

Viability-Guided Sim-to-Real Transfer for a Small Fixed-Wing Glider in Uncertain Indoor Updrafts

by
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Abstract

TBD.

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List of Symbols

Latin Letters

Symbol	Units	Description	Formula
$A_{\text{ring},f}$	m s^{-1}	annular strength parameter for fan f	
$a_{0,f}$	m s^{-1}	zeroth Fourier coefficient for fan f	
$a_{n,f}$	m s^{-1}	cosine Fourier coefficient of order n for fan f	
b	m	wing span	
b_H	m	horizontal tail span	
b_V	m	vertical tail height	
c	m	wing chord	
$\bar{c}$	m	mean aerodynamic chord	
c_H	m	horizontal tail chord	
c_V	m	vertical tail chord	
e_i	m s^{-1}	pointwise modelling error at sample i	$w_{\text{model}} - \bar{w}$
F		number of fans	
f		fan index	
f_i		index of the fan assigned to sample i	$\arg \min_f r_f(x_i, y_i)$
f_{scale}		robust loss scale parameter	
$\mathbf{K}$		kernel covariance matrix	
$\mathbf{K}_y$		effective training covariance	$\mathbf{K} + \mathbf{V}$
k		measurement-plane index	
$k(\cdot, \cdot)$		Gaussian process kernel function	
M		number of measurement samples	
m	g	aircraft mass	
N_f		Fourier order for fan f	
r_f	m	horizontal distance from a point to fan f	$\sqrt{(x - x_f)^2 + (y - y_f)^2}$
r_i	m	distance from sample i to its assigned fan	$r_{f_i}(x_i, y_i)$
$r_{\text{ring},f}$	m	annular radius for fan f	
s_d		feature standard deviation in dimension d	
S	m^2	wing area	

u_i		normalised residual used in robust fitting	$e_i/f_{\text{scale},f_i}\sigma_i$
$w_{0,f}$	m s^{-1}	background vertical velocity for fan f	
w_{model}	m s^{-1}	modelled mean vertical velocity	
$\bar{w}$	m s^{-1}	measured mean vertical velocity	
x_f	m	x-coordinate of fan centre f	
x_i	m	x-coordinate of sample i	
x_W	m	wing aerodynamic centre location	
x_H	m	horizontal tail aerodynamic centre location	
x_V	m	vertical tail aerodynamic centre location	
x_{CG}	m	longitudinal centre of gravity location	
y_f	m	y-coordinate of fan centre f	
y_i	m	y-coordinate of sample i	
y_{CG}	m	lateral centre of gravity location	
z_{CG}	m	vertical centre of gravity location	
z_{fan}	m	height above the fan outlet plane	
$z_{\text{fan},k}$	m	height of measurement plane k above the fan outlet plane	

Greek Letters

Symbol	Units	Description	Formula
$\mathcal{R}$		aspect ratio	b^2/S
$\delta_{\text{ring},f}$	m	annular thickness parameter for fan f	
η		harmonic penalty exponent	
Γ	$^\circ$	wing dihedral angle	
κ		harmonic cap ratio	
λ_{cap}		soft-cap regularisation weight	
λ_{ridge}		ridge regularisation weight	
ℓ_d		ARD length scale in dimension d	
μ_*	m s^{-1}	Gaussian process predictive mean	
μ_d		feature mean in dimension d	
ϕ_i		input feature vector	$[x_i, y_i, z_i]^\top$
$\tilde{\phi}_i$		standardised input feature vector	

ρ	kg m^{-3}	air density	
$\rho(\cdot)$		robust loss function	
σ_c	m s^{-1}	Gaussian process signal standard deviation	
σ_i	m s^{-1}	local fluctuation scale at sample i	
σ_w	m s^{-1}	Gaussian process white-noise standard deviation	
σ_*	m s^{-1}	Gaussian process predictive standard deviation	
θ_f	rad	azimuth angle about fan f	$\text{atan2}(y - y_f, x - x_f)$
$\boldsymbol{\theta}_{\text{ring},f}$		annular-Gaussian parameter vector for fan f	
$\boldsymbol{\Theta}_{\text{ring}}$		stacked annular-Gaussian parameter vector over all fans	$\left[\boldsymbol{\theta}_{\text{ring},1}^\top(z), \dots, \boldsymbol{\theta}_{\text{ring},F}^\top(z) \right]^\top$

Subscripts

Symbol	Description
$(\cdot)_{CG}$	quantity evaluated at the centre of gravity
$(\cdot)_{\text{ring}}$	annular component of the updraft model

Abbreviations

Symbol	Description
ARD	automatic relevance determination
CAD	computer-aided design
DoF	degrees of freedom
GP	Gaussian process
IPOPT	Interior Point OPTimiser
LDV	laser Doppler velocimetry
LQR	linear quadratic regulator
MDO	multidisciplinary optimisation
PCHIP	piecewise cubic Hermite interpolating polynomial
PIV	particle image velocimetry
PPM	pulse-position modulation
RBF	radial basis function

RC	radio control
RL	reinforcement learning
ROS	Robot Operating System
SAE	sum of absolute errors
UAV	unmanned aerial vehicle
WRMSE	weighted root mean square error
XPS	extruded polystyrene

1 Introduction

Small fixed-wing aircraft can gain energy from rising air, but the same flow can also become a disturbance when it is uncertain, spatially non-uniform, and encountered near safety limits. For a small glider in a confined arena, an updraft may increase altitude or local dwell capability, but it may also change angle of attack, roll moment, wall margin, actuator demand, and recovery margin. A controller that appears successful in simulation can therefore fail in flight because the aircraft, the implementation timing, or the surrounding air has been modelled imperfectly.

This thesis studies this problem as a controlled sim-to-real transfer task: can short feedback-stabilised fixed-wing manoeuvres developed in simulation remain useful when repeated on a physical glider in a measured indoor updraft, when selection is conditioned on the local flow context rather than on a fixed fan-layout script?

1.1 Motivation and Controlled Setting

The long-term motivation is low-altitude fixed-wing flight in local flow near terrain, slopes, obstacles, or built structures. Such environments may contain useful lift, but they are difficult to repeat with the precision needed for controlled sim-to-real evaluation. This thesis therefore uses a fan-generated indoor updraft as a measured vertical-flow forcing field. The aim is not to reproduce an outdoor thermal, ridge-lift region, or urban wind field. The aim is to create a repeatable aerodynamic mismatch that can be measured, simulated, randomised, and tested on hardware.

The experimental setting is deliberately narrow. A lightweight glider is launched in a motion-capture arena, flies through a mechanically generated updraft, and is evaluated through logged flight episodes. This makes the problem smaller than outdoor autonomous soaring, but more traceable: the measured launch state, flight volume, command path, vehicle model, measured flow field, termination condition, and later simulation replay inputs can all be recorded.

1.2 Reusable Indoor Sim-to-Real Workflow

A control result is not interpretable unless the experiment that produced it is also defined. In this thesis, the reusable part is therefore the experimental workflow: a tracked flight volume, measured updraft, documented glider, timed command path, safety-volume rule, and replayable episode log. These elements allow each physical launch to be compared with simulation using the same initial-state record, environment description, command timing, and outcome labels.

The workflow is implemented in Imperial College London’s Brahma Vasudevan Multi Terrain Aerial Robotics Arena². It is not tied to one controller or one aircraft. Later studies could change the glider, updraft layout, primitive catalogue, selector, or validation protocol while retaining the same evidence structure.

1.3 Primitive-Level Transfer Problem

Continuous path following is not the natural control unit in the Flight Arena. A fixed-wing glider needs space to turn, recover, and remain within speed, attitude, actuator, and wall limits. This thesis therefore uses short closed-loop manoeuvre primitives. A primitive is a short section of flight with entry checks, a finite duration, local feedback, exit checks, metrics, and failure labels.

The active library item is a primitive-controller variant. Each variant combines a primitive entry set, reference state and command, finite horizon, local time-invariant LQR stabiliser, exit checks, and transition labels. The LQR is not the high-level planner; it stabilises the primitive around its local

²To avoid repetition, this thesis uses the informal shortened form “Flight Arena” when referring to this facility.

reference. The high-level decision is to choose which already-supported primitive-controller variant to attempt from the current state and local flow context.

The evidence for a primitive is transition-based. A rollout is not judged only by whether it looks locally acceptable; it is judged by whether it starts from a supported entry class and reaches a compatible exit class without violating safety or recovery limits. Launch is one entry regime for the same primitive catalogue, not a separate controller family.

A rich set of primitive-controller variants is generated and selected in simulation, replayed under held-out randomised conditions, and compressed only after weak or unsupported transition cases have been identified. Online, a viability governor rejects candidates outside their supported entry regime, chain-compatible transition set, or safety envelope. Across repeated launches, a bounded spatial 3D updraft-utility belief can bias the score of candidates that have already passed these filters, but it cannot override state classification, transition compatibility, hard-failure risk, or safety filtering.

1.4 Research Questions and Contributions

The thesis addresses four research questions.

Research Question 1. How can a motion-capture flight arena be organised into a reusable indoor workflow for fixed-wing sim-to-real transfer under controlled aerodynamic forcing?

Research Question 2. How can a fan-generated updraft be measured and represented so that it provides repeatable local vertical-flow forcing without being treated as an exact model of outdoor lift?

Research Question 3. How can short LQR-stabilised manoeuvre primitives be generated, validated as entry-to-exit transition objects, and compressed into a compact real-time library without treating dense row count or local rollout success as sufficient evidence?

Research Question 4. Can environment-conditioned primitive selection, assisted by a bounded spatial flow-belief memory over repeated launches, improve lift capture and finite-horizon dwell while preserving safety through a viability governor?

The thesis contributes a reusable experimental workflow and a transition-aware primitive-selection method. The workflow links the tracked flight volume, measured updraft, command interface, latency model, glider description, safety volume, and episode log into a replayable evidence structure. The method builds LQR-stabilised primitive-controller variants, validates them as entry-to-exit transition objects, compresses the surviving library for real-time use, and evaluates the frozen selector using primitive outcomes, rejection labels, mission-facing lift and energy metrics, memory ablations, and matched sim-real replay where flight evidence is available.

The remainder of the thesis is organised as follows. Chapter 2 reviews sim-to-real transfer under aerodynamic forcing, controlled updraft modelling, small-glider modelling, manoeuvre primitives, safety filtering, and belief-guided action selection. Chapter 3 presents the indoor experimental workflow and timing architecture. Chapter 4 characterises the fan-generated updraft and defines the simulation role of the fitted surrogate models. Chapter 5 presents the glider design and manufacture. Chapter 6 formulates the environment-context encoding, primitive library, transition validation, viability governor, spatial flow-belief memory, and validation protocol. Chapter 7 evaluates the evidence chain from model checks to repeated-launch simulation, real-flight records, matched replay, and claim boundaries. Chapter 8 summarises the findings and limitations.

2 Background and Related Work

This chapter reviews the literature needed to formulate the thesis problem. It first defines sim-to-real transfer for fixed-wing flight under aerodynamic forcing, then narrows the discussion to controlled environment modelling, small glider design, manoeuvre primitive control, and the resulting research gap.

2.1 Sim-to-Real Transfer Under Aerodynamic Forcing

Sim-to-real³ transfer refers to using models, controllers, or policies developed in simulation on physical hardware [1]. The main difficulty of sim-to-real is the reality gap: the mismatch between simulated and real dynamics, sensing, actuation, timing, and environmental interaction [2], [3]. In robotics, this gap has been addressed through system identification [4], [5], domain and dynamics randomisation [6], [7], [8], adaptive randomisation [9], [10], and real-to-sim-to-real adaptation loops [11], [12], [13].

Aerial sim-to-real work has extended these methods to both multirotor and fixed-wing vehicles. Studies of agile multirotor racing have used large-scale simulation, randomisation, and systematic testing to transfer learned perception and control policies to reality [14], [15]. Fixed-wing examples are fewer, but recent work shows that high-fidelity modelling and parameter randomisation can improve transfer in pitch control and attitude regulation [16], [17]. These studies support simulation as a useful development tool and show that success of transfer should be assessed using measurable validation tasks.

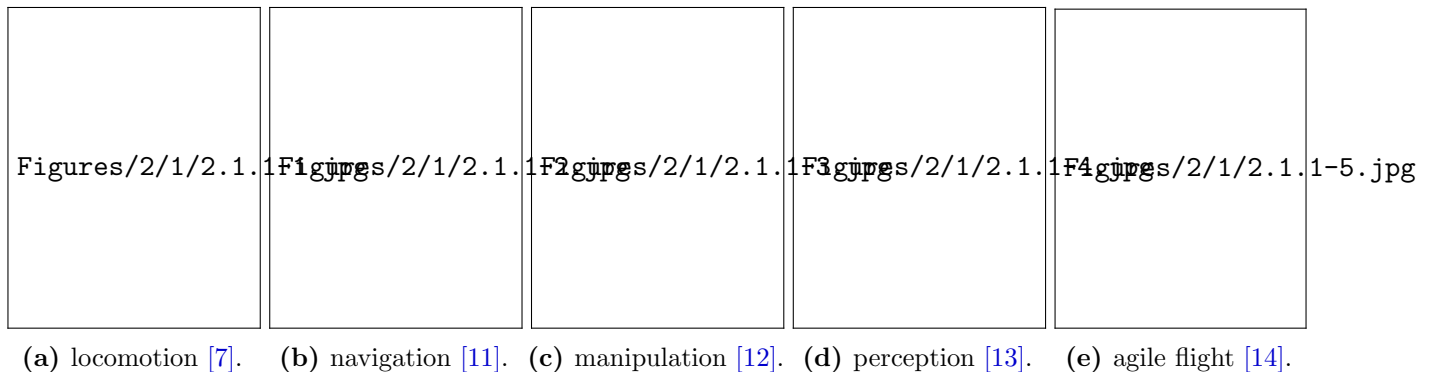
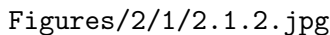


Fig. 2.1. Representative sim-to-real transfer examples in robotics and aerial autonomy.

When the surrounding flow is dynamically coupled to the vehicle, the reality gap should also include the aerodynamic forcing. Fixed-wing studies in wind and turbulence demonstrated that small aircraft are sensitive to non-uniform flow and gusts [18], [19]. Research on flow and dispersion in urban areas, including wind around buildings, further shows that structures can create shear layers, recirculation zones, and vertical motion [20], [21], [22]. Recent work on WindSeer also motivates sparse 3D wind prediction for small UAV operation in complex terrain [23]. For small fixed-wing vehicles, such spatially varying flow disturbances change the relative airflow at the lifting surfaces, affecting the vehicle’s energy state, flight path, control authority, and recoverability [24]. Hence, a simulator with an accurate aircraft model may still produce misleading closed-loop behaviour if the flow field is poorly represented.

Beyond treating wind as a disturbance to be predicted or tolerated, autonomous soaring provides the closest fixed-wing precedent for exploiting useful lift as a partially observable environmental resource [25]. Early flight-tested studies demonstrated that a small motorised sailplane could estimate and exploit thermal structure in flight [26], [27]. Subsequent work on atmospheric flow field modelling [28], biologically inspired soaring strategies [29], and thermalling control under limited flow observations [30] has advanced the representation, estimation, and use of uncertain lift fields. Recent orographic soaring experiments further show that fixed-wing MAVs can hover using local rising flow, while also revealing practical limitations in convergence to the lift region, circling behaviour, and roll stability [31], [32].

³Simulation-to-reality, sometimes also written as “sim2real” for brevity in paper titles, literature, and codebases.



Figures/2/1/2.1.2.jpg

Fig. 2.2. Representative simulation-to-flight progression in autonomous thermal soaring research, illustrated through Allen’s NASA Dryden studies as an early flight-tested example [26], [27], [33].

The present thesis does not attempt to reproduce nature thermals, orographic lift, or urban canyons directly. Instead, it uses indoor updrafts to study a narrower sim-to-real problem: can feedback-stabilised glider manoeuvre primitives be selected and transferred consistently across repeated launches when vehicle dynamics, implementation timing, and aerodynamic forcing differ between simulation and reality?

2.2 Environment Design, Characterisation, and Updraft Surrogates

A controlled sim-to-real study requires an environment that is repeatable enough for comparison, measurable enough for model construction, and variable enough across space to expose transfer degradation. Outdoor thermals [28], [34], terrain-induced lift [31], [32], and wind around buildings [20], [22] are relevant to low altitude fixed-wing operation and endurance extension, but they are difficult to reproduce and localise [35]. Heat-driven indoor thermals are also poorly matched to the present facility because their length scales and heat input are impractical for the available volume [36], [37]. Hence, mechanically generated updrafts are used in this thesis as controlled laboratory forcing fields.

Within this controlled setting, measurements are mainly required to build a useful updraft surrogate from finite samples. They must capture the dominant vertical velocity structure, resolve variations over the glider scale, and provide local fluctuation estimates [38]. Particle image velocimetry (PIV) [39] and laser Doppler velocimetry (LDV) [40] offer high-fidelity flow information, but their requirements for optical access, seeding, illumination, and setup effort can limit coverage in indoor test environments [41]. Pointwise anemometry with spatial scanning provides a lower-dimensional flow characterisation [42], but can still produce practical mean field maps and local fluctuation estimates when probe orientation, sampling duration, repeatability, and uncertainty are reported carefully [43], [44], [45].

To use measured samples in simulation, the finite flow data must be converted into updraft surrogates. This motivates the modelling framework developed in Chapter 4. Analytic Gaussian plume models provide compact descriptions for early guidance and energy exchange studies [46], [47]. Rotorcraft induced-flow and outwash theory offer physical intuition for induced velocity, source strength, and ground or boundary interaction effects [48], [49], [50]. Data-driven surrogate models, including Gaussian processes (GPs) [51], radial basis functions (RBFs) [52], and other response-surface methods [53], can interpolate measured data and, in some cases, provide uncertainty estimates. These models support validation and robustness studies, but should not be treated as exact descriptions of the physical flow, since finite measurements may miss temporal fluctuations, lateral velocities, or local wind shear.

2.3 Small Gliders and Control-Oriented Modelling

Small fixed-wing gliders are a natural benchmark for lift aware studies, since their sink rate, recoverability, and terminal energy depend directly on aerodynamic energy exchange [54]. In the present thesis, the vehicle is also part of the closed-loop system: geometry, wing loading, control surface authority, structural tolerance, and mass budget determine the attainable flight envelope, the available recovery margin, and the modelling assumptions that can be used for controller design [24].

This coupling between airframe design and control is especially restrictive in a confined indoor

arena with spatially varying inflow. Reducing wing loading and mass improves low-speed operation and reduces impact energy, but also increases sensitivity to inflow and limits structural margins. In contrast, increasing mass or wing loading can improve disturbance tolerance, but requires more space for turning and recovery. The representative platforms in Fig. 2.3 and Table 2.1 illustrate this compromise and support the lightweight foam and carbon construction adopted for the present glider.

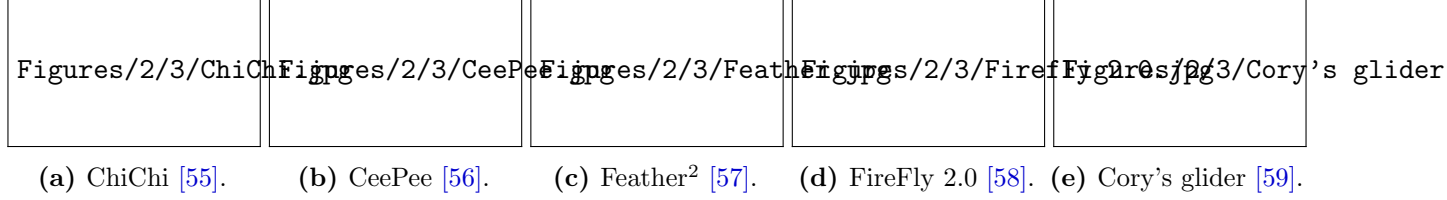


Fig. 2.3. Representative small fixed-wing gliders in related work and hobby literature.

Table 2.1 Key characteristics of the representative small fixed-wing gliders shown in Fig. 2.3.

Parameter	Symbol	Unit	ChiChi [55]	CeePee [56]	Feather² [57]	FireFly 2.0 [58]	Cory's glider [59]
Primary materials	-	-	balsa, plywood, heat-shrink film		balsa, CFRP tube, XPS foam	XPS foam, CFRP tube	Depron [60], CFRP tube
Control surface(s)	-	-	ailerons, ruddervators	ailerons, rudder, elevator	rudder, elevator	elevons	elevator
Mass	m	g	115–130	200	43–50	80–100	77
Wing span	b	m	0.600	0.750	0.736	0.812	0.813
Mean chord	$\bar{c}$	m	0.098	0.115	0.099	0.168	0.098
Wing area	S	m²	0.0585	0.0860	0.0729	0.1368	0.0797
Aspect ratio	$\mathcal{AR}$	-	6.15	6.54	7.43	4.82	8.30
Wing loading	$W S^{-1}$	N m⁻²	19.3–21.8	22.8	5.79–6.73	5.74–7.17	9.48

The dimensions and wing loadings in Table 2.1 also indicate the aerodynamic regime where such vehicles operate. For small fixed-wing gliders, the combination of short chord and low flight speed typically places the Reynolds number in the range between 10^4 and 10^5 . In this regime, lift, drag, and stall behaviour are highly sensitive to laminar separation, surface roughness, aerofoil thickness, and manufacturing error [61], [62], [63]. This discourages reliance on a precise aerofoil database unless the relevant aerofoil geometry and Reynolds number range have been experimentally validated.

Designing a small glider for this setting therefore involves coupled trade-offs among low-Reynolds-number aerodynamic efficiency, structural weight and stiffness, trim and stability, control authority, manufacturability, and mission performance [64]. These trade-offs may be addressed through iterative design built on exist configurations, optimisation within a multidisciplinary framework, or a combination of both [65]. In practice, prior aircraft layouts provide feasible starting points, and optimisation refines the geometry and sizing subject to the manufacturing and performance constraints.

Several open-source frameworks support multidisciplinary design optimisation (MDO), including OpenMDAO [66], SUAVE [67], and GPkit [68]. In this thesis, AeroSandbox [69] is selected as a lightweight differentiable framework for customised constraints, rapid iteration, and local sensitivity analysis, providing a transparent and repeatable glider design workflow presented in Chapter 5.

The same low-Reynolds-number sensitivity discussed above also shapes the vehicle model used later for control. Since a precise aerofoil database is not available for the as-built aircraft, the aerodynamic model must remain numerically robust over the expected operating envelope while supporting trim, local linearisation, and repeated closed-loop simulation [70], [71].

Quasi-steady flat plate aerodynamic models provide a practical solution to this requirement. For thin lifting surfaces, they give a compact representation that remains usable over a wider incidence range than conventional small-angle aerofoil models [72], [73]. Previous studies on aggressive small fixed-wing manoeuvres and perching have shown that averaged high-incidence lift and drag behaviour can be estimated by flat plate models, even though individual manoeuvre executions remain affected by unsteady vortex dynamics and repeatability limits [59], [74].

The indoor updrafts add a separate modelling demand through the aerodynamic loading imposed on the lifting surfaces. As discussed in Section 2.2, the forcing field is spatially non-uniform and is represented by finite measured surrogates in simulation. If the vertical velocity varies significantly over the wing span, representing the flow at the centre of gravity only may miss asymmetric changes in local relative airflow and the resulting roll, pitch, and yaw moments. This motivates the panel-wise aerodynamic model in Chapter 6, where each panel is evaluated using its own local relative velocity.

2.4 Guidance and Control with Manoeuvre Primitives

Conventional fixed-wing control architectures separate reference generation from stabilisation. Cascaded PID controllers [75], linear quadratic regulators (LQR) [76], robust control [77], and model predictive control (MPC) [78] are standard tools for stabilising aircraft around trim conditions or tracking feasible references [79]. Path-following methods then convert geometric errors into course, bank, altitude, or flight path angle commands [80], [81], [18]. This separation is effective when the aircraft has enough space to follow smooth references and when those references remain inside the attainable flight envelope.

For the present indoor problem, this separation is restrictive. The useful safety volume is small relative to the turn radius and recovery distances of a fixed-wing glider, and the glider may encounter strong local aerodynamic forcing while already close to the wall, height, speed, or actuator limits. Hence, the guidance problem is better posed as the selection of short, recoverable manoeuvres [82].

Manoeuvre-based methods have a long history in robotics and aerospace applications. Sequential composition [83], manoeuvre automata [84], LQR-Trees [85], and funnel libraries [86] decompose nonlinear control problems into locally valid manoeuvre primitives, each with admissible entry regions and expected exit conditions. Fixed-wing perching studies are particularly relevant because they combine reduced-order high-incidence aerodynamic modelling, time-varying feedback, and experimental validation of short aggressive manoeuvres [59], [74]. Therefore, a manoeuvre primitive is treated here as a short section of flight controlled by its own local feedback law, with defined entry conditions, a nominal reference and command profile, a finite time horizon, exit checks, and outcome metrics. In this setting, LQR is used as a local stabiliser for individual primitives, where each primitive has its own reference state, nominal command, linearisation, and performance objective [85].

Once control is framed as choosing between candidate manoeuvres, constraints must be checked before execution. Reference and command governors provide established methods for modifying or rejecting proposed commands before constraint violation occurs [87], [88]. Viability theory [89] and set invariance [90] treat safety as a state space property, identifying admissible states, recoverable regions, and trajectories that preserve constraints. In this thesis, the same logic is applied to manoeuvre primitives. A candidate primitive is rejected if its entry conditions are not satisfied, or if its predicted execution violates the safety volume, actuator limits, flight envelope, or recovery requirements.

The primitive library is the finite set of candidate manoeuvres made available to this governor. Its size creates a coverage problem: adding primitives can enlarge the reachable behaviour set, but each new primitive requires its own entry checks, uncertainty tests, exit criteria, and rejection conditions before it can be used [59], [85]. Related motion-planning work shows that richer primitive sets can improve coverage or motion quality, but also increase validation and computational burden [86], [91]. Instead of assuming that increasing the library size automatically improves sim-to-real transfer, this thesis treats library design as a trade-off between behavioural coverage and robustness of individual primitives.

With the primitive library and viability governor defined, one guidance question remains: which locally promising lift region should be targeted next, and which primitive objective should be proposed to reach it? This problem is partially observed because the aircraft cannot measure the full lift field during one flight. Partially Observable Markov Decision Processes (POMDPs) [92], [93], belief-space planning [94], [95], informative planning [98], [99], and reinforcement learning (RL) [100], [101], [102] provide general tools for action selection under incomplete information. In autonomous soaring research, related ideas appear in heuristic thermalling [103], online updraft estimation [26], stochastic decision making [104], and RL-based lift exploitation [105], [106], [107]. As discussed in Chapter 6, these ideas are implemented here through a spatial lift belief, which is updated after each launch and suggests the next lift-seeking target, viability governor then decides whether the candidate primitive can be executed safely. The resulting control architecture is evaluated in Chapter 7 through primitive acceptance, rejection, degradation, and improvement across repeated launches in the presence of the reality gap.

Primitive selection defines the guidance logic, but closed-loop behaviour also depends on the sensing, computation, and command path linking the aircraft and controller [108], [109], [110]. Indoor flight experiments commonly use motion capture systems for repeatable state estimation, but calibration, marker visibility, update timing, and communication delay must be managed carefully [111], [112], [113]. Networked and off-board control studies further show that delay and packet timing can degrade stability and tracking performance [114], [115]. These implementation effects are quantified in Chapter 3.

2.5 Research Gap and Thesis Positioning

Taken together, the reviewed literature places this thesis at the intersection of several established but only partly connected lines of research. Robotics sim-to-real studies provide mature tools for transfer evaluation [1], [3], system identification [4], [5], and randomisation [6], [8], but most examples do not treat external aerodynamic forcing as a dominant source of mismatch between simulation and reality. Research on fixed-wing wind-aware guidance [18], [19] and airflow around buildings [20], [22] explains why spatial wind variation around obstacles and structures should be represented in the control problem. Autonomous thermalling [25], [28] and orographic soaring [31], [32] show that fixed-wing aircraft can exploit useful lift even when the surrounding flow field is partially observable. Fixed-wing perching studies [59], [74], together with LQR-Trees [85] and funnel library methods [86], illustrate how short feedback-stabilised manoeuvres can be associated with admissible entry regions and safety constraints. What remains missing is a fixed-wing sim-to-real transfer setting that treats the uncertain flow field as part of the reality gap, instead of separating the vehicle, controller, experimental setup, and aerodynamic environment into independent problems.

This gap cannot be closed by a guidance and control method alone. Recent fixed-wing research platforms and autopilot software provide reusable aircraft designs, simulation tools, and controller interfaces, but not the full experimental setting required for sim-to-real transfer [116], [117]. Robotics testbed and benchmarking studies further show that experiments are easier to reproduce when they use common interfaces, recorded data, shared procedures, and clearly defined hardware requirements [118], [119], [120]. For fixed-wing sim-to-real transfer studies, the missing element is a reusable, reproducible, and extensible workflow for indoor aerial robotics laboratories that use motion capture, a setup increasingly common in universities and research institutions [121], [122], [123]. This workflow must specify the test volume, state measurement, sensing and command latency, aircraft design and hardware, test environment design and characterisation, and the data needed to rerun each flight in simulation.

Accordingly, this thesis addresses these gaps in two parts. It develops the required experimental workflow, and proposes a viability-guided primitive-selection algorithm for lift-aware guidance and control of a small fixed-wing glider in uncertain indoor updrafts. This algorithm is then evaluated through the workflow to test whether it can transfer successfully from simulation to reality when the reality gap comes from the combined mismatch in vehicle, sensing, command path, and flow field.

3 Experimental Platform and System Architecture

Building on the research gap identified in [Section 2.5](#), this chapter turns the required experimental workflow into a concrete infrastructure for an indoor aerial robotics arena instrumented with a motion capture system. It defines the physical test setup, computation and state measurement architecture, command interface, and sensing, command, and actuation latencies, establishing the experimental basis for later updraft modelling, glider design, controller development, and sim-to-real transfer evaluation.

3.1 Experimental Platform

The flight experiments were conducted in Imperial College London’s Flight Arena, an indoor test space with dimensions of $10.0 \times 6.2 \times 5.5$ m. The Flight Arena is equipped with 16 Vicon Vantage cameras connected through a Power over Ethernet network that supplies camera power and transfers data to the Vicon synchronisation hardware and host PC [\[124\]](#). Vicon Tracker software provides real-time 6-DoF rigid-body measurements with millimetre accuracy from at least three reflective markers fixed in a known geometry on each tracked object [\[111\]](#), [\[112\]](#).

Figures/3/1/3.1.1.JPG

Fig. 3.1. Flight Arena with Vicon motion capture system.

Within this facility, flight tests were further limited to an $8.0 \times 4.8 \times 3.5$ m volume protected by the recovery net and floor padding. The test vehicle is a small fixed-wing glider built for these indoor flight tests, its design and manufacture are presented in [Chapter 5](#). For controller development in [Chapter 6](#), a smaller operational region, $x_w \in [1.2, 6.6]$ m, $y_w \in [0.0, 4.4]$ m, $z_w \in [0.4, 3.5]$ m, is used so that the reflective markers mounted on the glider remain visible to at least 75% of the Vicon cameras.

3.1.1 Updraft Generation and Measurement Setup

To create repeatable indoor updraft environments for flow characterisation, modelling, and subsequent controller studies, experiments were carried out using either a single Clarke CAMAX30 axial drum fan [\[125\]](#) or four fans. The single-fan case provided an isolated baseline for updraft field characterisation and sensitivity tests, while the four-fan case produced a multi-source updraft for evaluating surrogate reconstruction quality and subsequent controller performance in a more spatially complex flow.

The updraft field was sampled on a locally refined Cartesian grid using a BENETECH GM8903 hot-wire anemometer mounted on a height-indexed rigid support [\[126\]](#). Ambient temperature was held between 18.5°C and 20.0°C to minimise temperature-induced sensor drift [\[127\]](#), while the probe orientation was kept fixed to reduce errors caused by lateral flow in shear or swirl regions [\[42\]](#). This setup provided a consistent basis for the dataset introduced in [Chapter 4](#).

Figures/3/1/Measurement grid.jpg

Figures/3/1/Measurement appar

(a) Measurement grid.

(b) Probe arrangement.

Fig. 3.2. Fan-generated updraft measurement setup.

3.2 Sensing, Computation, and Command Architecture

Sensing, computation, and command transmission determine how the state is measured, where the controller runs, what delays are introduced, and whether each experiment can be replayed in simulation. Since the full airframe design is introduced in [Chapter 5](#), this chapter treats the test vehicle only through four servo channels relevant to the actuation link: left aileron, right aileron, elevator, and rudder.

3.2.1 Computation and State Measurement Architecture

[Table 3.1](#) compares computation architectures by integration difficulty, real-time performance, onboard size, weight, and power (SWaP) burden, reliability, modularity, and extra cost relative to the available infrastructure. The ground computer is selected to keep the glider mechanically simple by handling control, logging, and safety supervision off-board.

[Table 3.2](#) scores state estimators by integration, accuracy, robustness, real-time performance, setup ease, and additional cost. Here, robustness refers to sensitivity to test conditions, and setup ease covers calibration and maintenance effort. Vicon motion capture system is chosen since it provides fast and accurate 6-DoF tracking without adding sensors, wiring, or processing load to glider, supporting the off-board control selected above.

Table 3.1 Computation architecture decision matrix.

Architecture	Integration	Real-time	Low SWaP	Reliability	Modularity	Low cost	Score
Weight	10%	20%	30%	20%	10%	10%	100%
Ground Computer	8	9	10	4	9	10	8.3
Onboard HW	5	7	3	6	5	5	5.0
Onboard LW	3	4	9	9	7	7	7.0
Split Compute	9	9	8	7	3	7	7.5

Table 3.2 State estimator decision matrix.

Estimator	Integration	Accuracy	Robustness	Real-time	Setup ease	Low cost	Score
Weight	10%	30%	20%	20%	10%	10%	100%
IMU	9	3	8	10	9	7	7.0
UWB	7	6	7	7	5	4	6.2
UWB + IMU	8	7	8	8	5	3	6.9
Vicon	9	10	9	9	4	10	8.9
Vicon + IMU	7	10	10	10	3	7	8.7
Camera + IMU	7	6	5	6	7	6	6.0
LiDAR + IMU	6	8	8	7	6	5	7.1
Optical Flow	7	3	5	8	8	6	5.6

3.2.2 Command Interface Selection

With the ground computer and Vicon motion capture system selected, a compatible command interface is needed to carry control commands to the onboard actuators. It should be inexpensive, lightweight, simple to integrate, stable, durable, reproducible, and extensible to future aircraft or actuator changes.

These requirements leave two practical routes. In direct servo command path ([Fig. 3.3a](#)), an onboard microcontroller receives commands over Wi-Fi and directly generates servo PWM signals⁴. In commercial off-the-shelf (COTS) radio control (RC) command chain ([Fig. 3.3b](#)), the microcontroller sends a PPM signal⁵ to a transmitter via its trainer port, the transmitter then relays the commands through the radio link to an onboard receiver, which outputs separate servo PWM signals.

⁴ Pulse-width modulation (PWM) represents a command by varying the width of repeated pulses. For standard RC servos, this pulse width sets the demanded position.

⁵ Pulse-position modulation (PPM) represents multiple channel commands by sending pulses sequentially within a repeated timing frame. In PPM-based RC interfaces, this allows several channels to share one signal line.

As discussed in [Section 2.4](#), previous studies have shown that implementation timing affects closed-loop control [\[109\]](#), [\[110\]](#), but stage-by-stage latency measurements of command paths are rarely reported. To address this gap, a repeatable electrical timing test ⁶ was designed for two candidate routes, each driven by a sequence of step changes in the four control surface commands. For a given run, the ground computer logged the scheduled command and dispatch time, while the Arduino Nano 33 IoT recorded the board receive and update times for the same command sequence. A logic analyser ⁷ was connected to the relevant pulse lines to capture when the electrical output first changed after each command step.

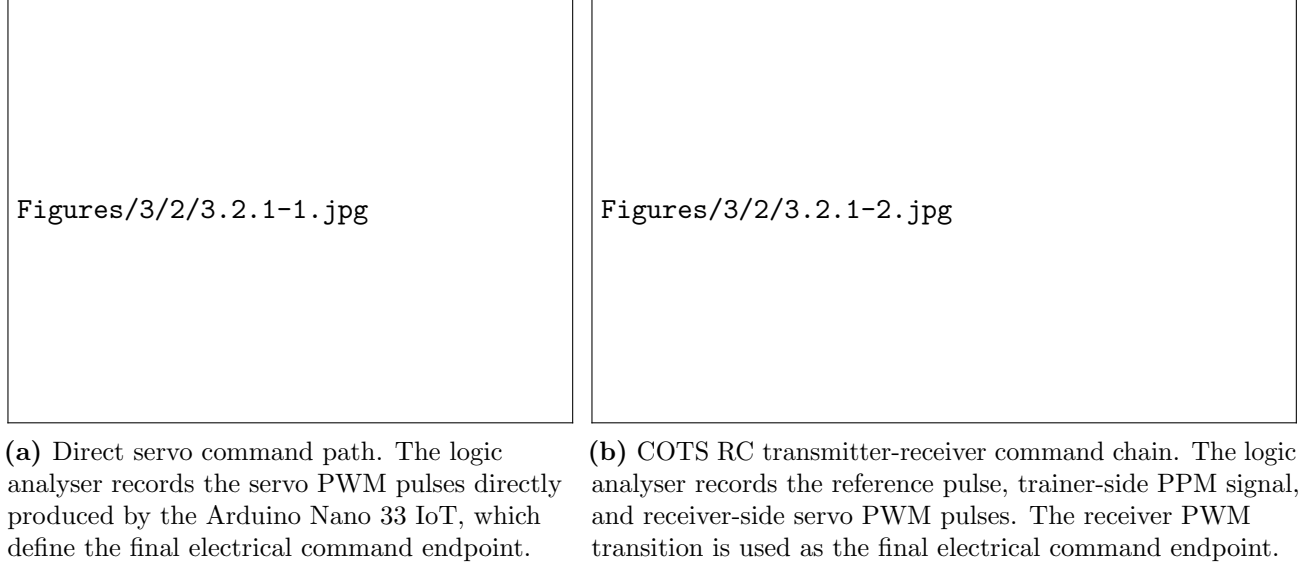


Fig. 3.3. Electrical command latency test setups for the two candidate interfaces.

Five timing runs were processed for each command interface. Each run lasted 59.0s and contained 2952 command updates for four control surfaces, equivalent to 11808 per-surface command records in total. The command logs and pulse traces were then aligned offline to compute the electrical delay at each stage of the command path and the final endpoint latency. The clock alignment and transition matching are detailed in [Appendix B.1](#), together with the timing tables for each run.

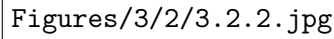
Table 3.3 Stage-by-stage electrical timing comparison for the two command interfaces.

Path segment	Unit	Direct servo				COTS RC			
		median	P_{95}	P_{99}	σ	median	P_{95}	P_{99}	σ
Ground command to dispatch	ms	0.25	0.99	4.03	0.89	0.16	0.51	1.17	0.34
Ground dispatch to board receive	ms	22.9	32.4	34.1	5.68	6.83	7.38	7.63	0.74
		over Wi-Fi				via micro USB			
Board receive to firmware update	ms	0.20	0.23	0.30	0.02	11.6	16.8	17.1	5.08
		servo PWM updated				trainer PPM updated			
Firmware update to electrical output	ms	16.2	28.0	30.5	6.99	3.01	20.1	20.1	7.83
		next servo PWM pulse				next trainer PPM frame			
Electrical output to receiver output	ms	-	-	-	-	6.89	33.6	45.6	12.8
						receiver PWM updated			
Ground command to final endpoint	ms	40.4	55.0	58.3	9.46	33.9	57.8	64.8	11.6

⁶ This test covers electrical command timing only. Mechanical control surface response is characterised in [Section 3.3](#).

⁷ A logic analyser records digital voltage signals over time. In these tests, it was sampled at 4 MHz, corresponding to a $0.25 \mu\text{s}$ interval, so the start, end, and width of millisecond-scale command pulses can be measured across several channels.

Table 3.3 above shows that the direct-servo command path has slightly higher median latency from ground command to final electrical endpoint, but is more consistent since its has smaller P_{95} and P_{99} latencies⁸, and smaller standard deviation, while the COTS RC path introduces additional receiver-side timing variation after the transmitter chain. These timing properties would favour the direct-servo route for a tightly coupled autonomous inner control loop. However, the final interface also had to support repeatable flight test, safe intervention, practical integration, and future extension. During the latency test, the COTS RC path ran for 25–35 min, compared with 7–15 min for the direct-servo setup, while preserving transmitter pilot override as a safety fallback, remaining compatible with standard RC servos without modification, and reducing custom onboard integration work. The direct-servo route also depends heavily on local Wi-Fi configuration, making exact reproduction across laboratories less reliable. Therefore, the COTS RC path is adopted as the experimental command interface used in this thesis.



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Fig. 3.4. Selected flight test sensing and command architecture.

3.3 State Processing and Closed-Loop Timing

Opening, do not touch it yet.

3.3.1 Vicon State Processing and Filtering

ABC why its in wrong page?

3.3.2 Control Surface Response and Implementation Timing

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⁸ P_{95} and P_{99} are the delay values below which 95% and 99% of matched events fall. They describe rare high-latency cases, which matter because the controller must remain stable beyond the typical delay.

The final implementation-delay model was identified from five accepted integrated command-path tests using the 20 Hz filtered Vicon surface-state signal. Each run contained 160 valid command events, giving 800 accepted events in total, with no rejected events in the final processed dataset. The timing reference for each event was the MATLAB serial write-start timestamp, denoted $t_{\text{write,start}}$, while the measured surface response time was computed using the Vicon-corrected capture timestamp. This separates host-side command issue time from Vicon frame-read time and avoids attributing Vicon communication delay to the actuator path.

The results show that the command-to-surface path has a repeatable effective delay. Across the accepted tests, the median response onset, defined by the 10% crossing, was approximately 73 ms. The nominal effective latency, defined by the 50% crossing, was approximately 108 ms. The near-final response, defined by the 90% crossing, was approximately 130 ms. A conservative upper bound based on the largest valid observed t_{50} was approximately 151 ms.

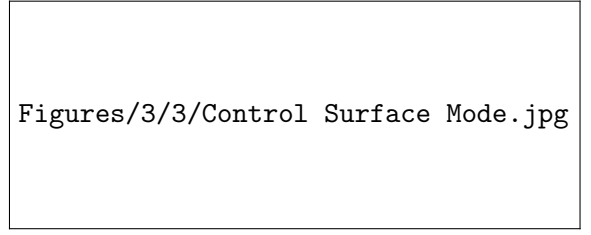
These values are not interpreted as Vicon latency. The Vicon measurement latency is logged separately by the Vicon system and was approximately 14.9 ms nominally, with a 95th percentile value of approximately 16.9 ms. When the selected 20 Hz one-pole filter is used for state feedback, an additional estimated filter delay of approximately 8.0 ms is included. Therefore, the simulation uses a nominal filtered state-feedback delay of approximately 22.9 ms, while the actuator command-response dynamics are modelled separately.

The simulation model therefore uses two distinct implementation-delay terms. First, the controller receives a delayed state estimate representing Vicon capture latency and filtering delay. Second, the control-surface command passes through the measured command-to-surface response model. The command path is represented as an onset delay followed by a first-order actuator response calibrated to the measured t_{50} . This preserves the physical interpretation that the state-measurement delay and actuator-response delay are different mechanisms.

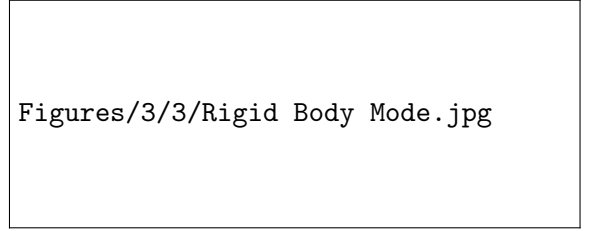
Table 3.4 Adopted latency values for the final simulation model.

Quantity	Unit	Value	Interpretation
Vicon nominal latency	ms	14.9	Measured Vicon capture and report delay
20 Hz filter delay	ms	8.0	Estimated delay introduced by the selected state filter
Nominal state-feedback delay	ms	22.9	Delayed state seen by the controller
Surface response onset, t_{10}	ms	73	First meaningful control-surface response
Nominal surface latency, t_{50}	ms	108	Primary actuator latency value used in simulation
Near-final response, t_{90}	ms	130	Large-response settling indicator
Conservative actuator bound	ms	151	Worst accepted valid t_{50} value

The main assumption is that the bang-bang surface tests are representative of the command-path timing relevant to the simulated manoeuvre primitives. The model does not assume that small-amplitude



(a) Individual control surface tracking.



(b) Airframe rigid-body tracking.

Fig. 3.5. Vicon tracking configurations.

servo response is perfectly linear, because the real mechanism showed deadband, backlash, and surface-dependent oscillation. Instead, the large-step latency test is used to identify a robust timing envelope for primitive-level sim-to-real validation. Surface deflection magnitude calibration is treated as less reliable than latency timing because the Vicon surface-angle estimates were sensitive to rigid-body setup and axis definition.

Accordingly, the final model is an empirical implementation-delay model rather than a complete servo physics model. It captures the dominant timing effects required for controller validation: delayed state feedback, command quantisation, command-to-surface onset delay, actuator lag, and asymmetric surface limits. The remaining uncertainty is mainly associated with surface-angle measurement quality, mechanical backlash, and possible amplitude dependence of very small servo commands. These uncertainties are handled by validating the controller over the measured latency envelope rather than relying on a single idealised latency value.

In the repeated-launch protocol, this timing model is applied to the measured initial state of each episode. Timing uncertainty is therefore evaluated together with launch-state variation, rather than under an identical-release assumption.

4 Updraft Characterisation and Modelling

In the present thesis, the dominant contributor to the reality gap is the indoor updraft, which strongly affects the glider’s vertical energy exchange. This chapter characterises the measured fan-generated flow and introduces the modelling framework used later to support controller development in simulation.

4.1 Updraft Measurement and Dataset

Using the artificial updraft and measurement setup introduced in [Section ??](#), the local vertical velocity $w(t)$ was characterised for both fan configurations, with the quasi-steady mean $\bar{w}$ used for modelling. For each configuration, $\bar{w}(x_i, y_i, z_{\text{fan},k})$ was recorded on seven horizontal planes at $z_{\text{fan}} \in \{0.20, 0.35, 0.50, 0.75, 1.10, 1.60, 2.20\}$ m above the fan outlet plane. Measurements were taken at 195 grid points on each plane, giving 1365 spatial locations per configuration across the seven planes.

Representative mean-field maps at $z_{\text{fan},2} = 0.35$ m and $z_{\text{fan},6} = 1.60$ m are shown in [Fig. 4.2](#) and [4.3](#), while annotated mean-field maps for all horizontal planes in both layouts are provided in [Appendix C.2](#).

Figures/4/1/Time-lapse.jpg

Fig. 4.1. Time-lapse composite of anemometer measurements of the fan-generated updraft.

Figures/4/1/z035_single_heatmap_main.png

(a) 0.35 m above fan outlet plane.

Figures/4/1/z160_single_heatmap_main.png

(b) 1.60 m above fan outlet plane.

Fig. 4.2. Spatial distribution of mean vertical velocity at representative heights (single fan).

Figures/4/1/z035_four_heatmap_main.png

(a) 0.35 m above fan outlet plane.

Figures/4/1/z160_four_heatmap_main.png

(b) 1.60 m above fan outlet plane.

Fig. 4.3. Spatial distribution of mean vertical velocity at representative heights (four fans).

In addition to the mean-field maps, temporal variability was assessed at six representative points spanning the near-outlet, off-outlet, and far-field regions. Each record consisted of 15 samples at 1 Hz, with duplicate runs to check repeatability. In the single-fan case, fan outlet height and speed were varied for sensitivity tests, while the four-fan case was measured only at the nominal operating condition.

4.2 Experimental Updraft Field Characterisation

The fan field shows strong height dependence and multi-source interaction. In single-fan case ([Fig. 4.2](#)), the peak occurs on an asymmetric annular ring, that spreads and weakens with height. The peak decreasing from 4.86 m/s at $z_{\text{fan},2} = 0.35$ m to 3.16 m s^{-1} at $z_{\text{fan},6} = 1.60$ m. In the four-fan case ([Fig. 4.3](#)),

four distinct asymmetric cores are observed at low altitude, which weaken and overlap with height. The local peaks reduce from $4.02 - 5.80 \text{ m s}^{-1}$ at $z_{\text{fan},2} = 0.35 \text{ m}$ to $2.04 - 2.60 \text{ m s}^{-1}$ at $z_{\text{fan},6} = 1.60 \text{ m}$, while the overlap raises the background lift in the central region.

Ground effect is then evaluated by varying the fan outlet height across low (0.245 m), baseline (0.330 m), and high (0.425 m) configurations. As shown in Fig. 4.4a, the effect is strongest near-core, where rising outlet height increases the lower-profile updraft and delays its decay with height, consistent with rotor-induced ground effect behaviour [49], [128]. The off-core response in Fig. 4.4b is weaker and less monotonic, showing local redistribution rather than a uniform change across the field. Fig. 4.4 further shows that the fluctuation level is spatially non-uniform: near-core point 1 exhibit $\sigma = 0.26 - 0.41 \text{ m s}^{-1}$ over $z_{\text{fan}} = 0.20 - 1.60 \text{ m}$, while off-core point 4 show smaller absolute fluctuations, about $0.09 - 0.30 \text{ m s}^{-1}$, but larger relative variability as the mean velocity is low⁹. Overall, the baseline outlet height was adopted as a compromise between updraft strength and reduced sensitivity to ground effects.

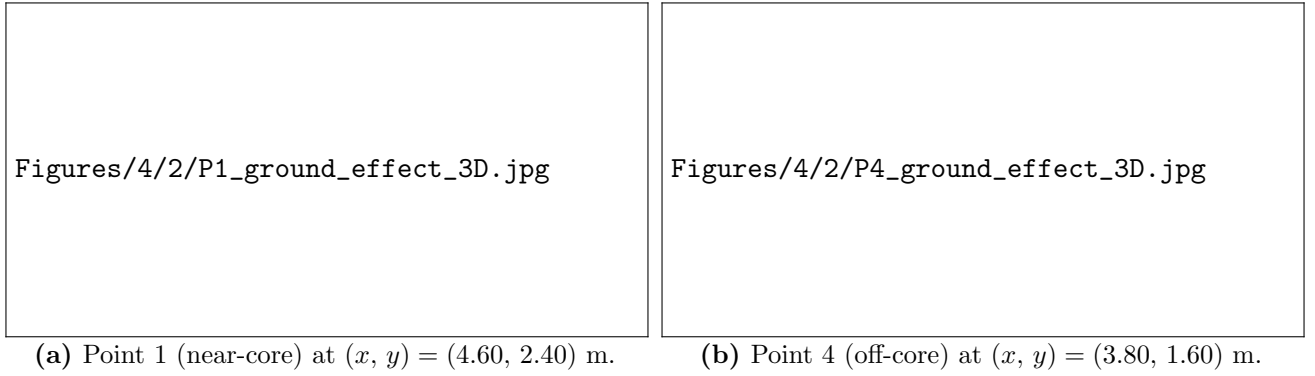


Fig. 4.4. Vertical velocity profiles at representative sampling points for three fan outlet plane heights (single fan). The solid line shows the mean, semi-transparent markers show the 15 samples (1 Hz over 15 s) at each measurement height, and shaded bands show $\pm 1\sigma$.

Fan speed sensitivity is also examined at 8800 rpm and 960 rpm. The higher speed was adopted, since the lower one led to occasionally blade-guard contact, likely due to insufficient centrifugal stiffening.

In the four-fan configuration, plume overlap produces non-zero mean velocity and measurable variance at points between the individual cores, as shown in Fig. 4.5. The largest absolute fluctuations occur in the near-core and interaction region. For example, at $z_{\text{fan},2} = 0.35 \text{ m}$, point 1 give $\bar{w} = 3.47 \text{ m s}^{-1}$ with $\sigma = 0.87 \text{ m s}^{-1}$, while the point 4 gives $\bar{w} = 0.69 \text{ m s}^{-1}$ with $\sigma = 0.15 \text{ m s}^{-1}$. These measurements clearly show that the uncertainty is highly non-uniform and varies with both height and position.

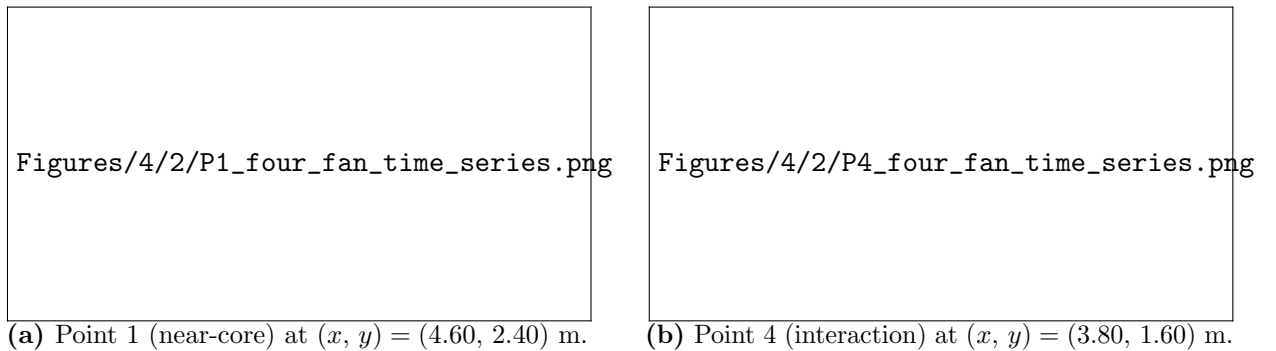


Fig. 4.5. Vertical velocity profiles at representative sampling points (four fans). The solid line shows the mean, semi-transparent markers show the 15 samples (1 Hz over 15 s) at each measurement height, and shaded bands show $\pm 1\sigma$.

The complete set of corresponding profiles for all six sampling points is provided in [Appendix C.3](#).

⁹ Since these records span only 15 s at 1 Hz, the reported σ only represent local short horizon fluctuation levels.

In conclusion, the measurements show that the updraft field is non-axisymmetric, height-dependent, and subject to spatially varying uncertainty. It is also sensitive to outlet height and fan speed, as observed from the single-fan sensitivity sweeps, while the four-fan case is additionally affected by plume overlap and redistribution under multi-source interaction.

4.3 Updraft Modelling Framework

These observed features motivate the simulation surrogates developed in this section, which are designed to preserve the measured field structure while retaining a representation of spatial uncertainty.

For a configuration with F fans, with fan centres (x_f, y_f) for $f = 1, \dots, F$, the horizontal distance and azimuth angle of an arbitrary point (x, y, z) relative to fan f are

$$r_f(x, y) = \sqrt{(x - x_f)^2 + (y - y_f)^2}, \quad \theta_f(x, y) = \text{atan2}(y - y_f, x - x_f). \quad (4.1)$$

Each sampling point (x_i, y_i) is then assigned to its nearest fan f , with relative radius r_i :

$$f_i = \arg \min_{f \in \{1, \dots, F\}} r_f(x_i, y_i), \quad r_i = r_{f_i}(x_i, y_i). \quad (4.2)$$

At measurement height $z_{\text{fan},k}$, the fluctuation scale is mapped from six representative time-series measurements in [Section 4.1](#). For each fan, these data define an empirical radial fluctuation profile, $\sigma_f(r_f, z_{\text{fan},k})$. The fluctuation assigned to sample i is then obtained by inverse-distance blending:

$$\sigma_i = \sum_{f=1}^F \omega_f \sigma_f(r_f, z_{\text{fan},k}), \quad \omega_f = \frac{(r_f + \epsilon)^{-p}}{\sum_{g=1}^F (r_g + \epsilon)^{-p}}. \quad (4.3)$$

The corresponding fluctuation field plots at all measurement heights are given in [Appendix D.1](#).

4.3.1 Error Metrics for Model Fitting

Model performance is quantified by comparing the fitted updraft field with the measured mean vertical velocity. For each sample (x_i, y_i) at height $z_{\text{fan},k}$, the pointwise error is defined as

$$e_i(z_{\text{fan},k}) = w_{\text{AG}}(x_i, y_i, z_{\text{fan},k}) - \bar{w}(x_i, y_i, z_{\text{fan},k}). \quad (4.4)$$

Two complementary error measures are used. To summarise the overall mismatch, the per-height sum of absolute errors (SAE) [\[129\]](#) and its aggregate over all heights are computed:

$$\text{SAE}(z_{\text{fan},k}) = \sum_i |e_i(z_{\text{fan},k})|, \quad \text{SAE}_{\text{total}} = \sum_k \sum_i |e_i(z_{\text{fan},k})|. \quad (4.5)$$

To account for measurement variability, weighted root mean square error (WRMSE) [\[129\]](#) is also used:

$$\text{WRMSE}(z_{\text{fan},k}) = \sqrt{\frac{\sum_i \left(\frac{e_i(z_{\text{fan},k})}{\sigma_i(z_{\text{fan},k)}} \right)^2}{\sum_i \left(\frac{1}{\sigma_i(z_{\text{fan},k)}} \right)^2}}, \quad \text{WRMSE}_{\text{total}} = \sqrt{\frac{\sum_k \sum_i \left(\frac{e_i(z_{\text{fan},k})}{\sigma_i(z_{\text{fan},k)}} \right)^2}{\sum_k \sum_i \left(\frac{1}{\sigma_i(z_{\text{fan},k)}} \right)^2}}. \quad (4.6)$$

Since the reference data are time-averaged, the reported SAE and WRMSE quantify only how closely each model reproduces the measured mean field, and do not represent the unsteady flow experienced by the real glider [\[130\]](#). Therefore, they are used mainly for relative comparison between models.

4.3.2 Harmonic Annular–Gaussian Surrogate

As discussed in [Section 4.2](#), the fan-generated updraft exhibits two dominant structural features: a ring-shaped peak and clear azimuthal asymmetry. A control-oriented surrogate should therefore preserve

the compact annular structure of the mean field while remaining flexible enough to represent the observed directional variation. Inspired by the harmonic inflow representations in rotorcraft induced-flow modelling [49] and motivated by the common use of Gaussian plume fits in soaring simulation [46], a harmonic annular-Gaussian model is designed to represent the measured field:

$$w_{\text{HAG}}(x, y; z) = \sum_{f=1}^F \left[w_{0,f}(z) + A_{\text{ring},f}(\theta_f(x, y); z) \exp\left(-\left(\frac{r_f(x, y) - r_{\text{ring},f}(z)}{\delta_{\text{ring},f}(z)}\right)^2\right) \right], \quad (4.7)$$

where w_0 is the background vertical velocity, r_{ring} and δ_{ring} are the ring radius and thickness, and A_{ring} is the azimuth-dependent ring strength, written as a constant term plus an N -th order Fourier series:

$$A_{\text{ring},f}(\theta_f(x, y); z) = a_{0,f}(z) + \sum_{n=1}^{N_f} \left(a_{n,f}(z) \cos(n\theta_f) + b_{n,f}(z) \sin(n\theta_f) \right). \quad (4.8)$$

For fan f , the corresponding parameter block of the model in (4.4) is given by

$$\boldsymbol{\theta}_{\text{ring},f}(z) = \left[w_{0,f}(z), r_{\text{ring},f}(z), \delta_{\text{ring},f}(z), a_{0,f}(z), \{a_{n,f}(z), b_{n,f}(z)\}_{n=1}^{N_f} \right]^\top, \quad (4.9)$$

and the full parameter vector used for fitting is obtained by stacking the parameter blocks for all fans:

$$\boldsymbol{\Theta}_{\text{ring}}(z) = \left[\boldsymbol{\theta}_{\text{ring},1}^\top(z), \dots, \boldsymbol{\theta}_{\text{ring},F}^\top(z) \right]^\top. \quad (4.10)$$

At each measurement height $z_{\text{fan},k}$, $\boldsymbol{\Theta}_{\text{ring}}(z_{\text{fan},k})$ is estimated by solving a least-squares optimisation problem with robust loss function $\rho(\cdot)$ [131]. To reduce sensitivity to local minima and improve the chance of reaching a good solution, the optimisation is initialised from multiple starting points [132]:

$$\begin{aligned} & \underset{\boldsymbol{\Theta}_{\text{ring}}(z_{\text{fan},k})}{\text{minimise}} \quad \sum_{i=1}^M \rho_{f_i}(u_i) + \mathcal{R}(\boldsymbol{\Theta}_{\text{ring}}(z_{\text{fan},k}), \boldsymbol{\gamma}), \\ & \text{subject to} \quad a_{0,f}(z_{\text{fan},k}) \geq 0, \quad \delta_{\text{ring},f}(z_{\text{fan},k}) \in [0.12, 1], \\ & \quad w_{0,f}(z_{\text{fan},k}) \in [\bar{w}_{\min}(z_{\text{fan},k}) - 1, \bar{w}_{\max}(z_{\text{fan},k}) + 1]. \end{aligned} \quad (4.11)$$

Here, M is the total number of measurements, u_i denotes the normalised residual, and $\mathcal{R}(\cdot, \cdot)$ is a regularisation term introduced to limit overfitting of the Fourier coefficients [133].

The normalised residual in (4.10) is constructed such that the denominator introduces variance-aware weighting, so samples with larger measured fluctuations have lower weight in the fit:

$$u_i = \frac{w_{\text{HAG}}(x_i, y_i; z_{\text{fan},k}, \boldsymbol{\Theta}_{\text{ring}}(z_{\text{fan},k})) - \bar{w}_i}{f_{\text{scale},f_i} \sigma_i(z_{\text{fan},k})}. \quad (4.12)$$

The regularisation term in (4.10) with hyperparameters $\boldsymbol{\gamma} = (\lambda_{\text{ridge}}, \lambda_{\text{cap}}, \kappa, \eta, N)$ is defined as

$$\begin{aligned} \mathcal{R}(\boldsymbol{\Theta}_{\text{ring}}(z_{\text{fan},k}), \boldsymbol{\gamma}) = & \sum_{f=1}^F \sum_{n=1}^{N_f} n^{\eta_f} \left\{ \lambda_{\text{ridge},f} \|[a_{n,f}, b_{n,f}]^\top\|_2^2 \right. \\ & \left. + \lambda_{\text{cap},f} \left[\max(0, \|[a_{n,f}, b_{n,f}]^\top\|_2 - \kappa_f a_{0,f}) \right]^2 \right\}, \end{aligned} \quad (4.13)$$

which combines: (i) an L_2 term that penalises large harmonic coefficients [133], and (ii) a soft ‘‘cap’’ that activates only when the harmonic magnitude exceeds a fixed fraction of a_0 [134]. Both penalties are weighted by n^η to filter out higher-order harmonics that are not clearly present in the measurements [135].

Accordingly, three aspects of the fitting setup are evaluated to select the configuration that minimises total SAE and WRMSE across all measurement heights: (i) the robust loss function $\rho(\cdot)$ in (4.10), including soft- L_1 [136], Huber [130], and Cauchy [137]; (ii) the corresponding scale parameter f_{scale} in

(4.11); and (iii) the regularisation hyperparameters γ in (4.12).

The selected configurations use the same robust loss, soft- L_1 , but different scale and regularisation settings. For single-fan fit, $f_{\text{scale}} = 1.5$ and $\gamma = (0.05, 0.20, 0.70, 1.5, 2)$. For four-fan case, the left pair of fans uses $f_{\text{scale}} = 1.5$ while the right pair uses $f_{\text{scale}} = 1$, with same $\gamma = (0.03, 0.10, 0.80, 1.5, 2)$ ¹⁰. The weaker harmonic penalties in four-fan fit allow stronger azimuthal correction, while the fan-dependent robust scale indicates that the right-side flow exhibits larger fluctuations.

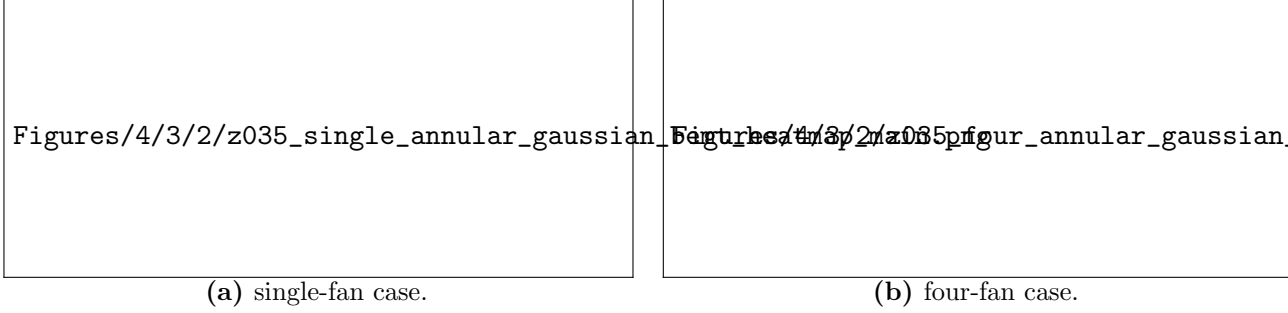


Fig. 4.6. Fitted mean updraft map of the harmonic annular-Gaussian updraft surrogate at a single measurement plane 0.35 m above the fan outlet plane.

Finally, to obtain an uncertainty-aware continuous model for simulation, $\{\Theta_{\text{ring}}(z_{\text{fan},k})\}_{k=1}^7$ and σ_i are interpolated over height, z , using shape-preserving piecewise cubic Hermite interpolation (PCHIP) [138]. Near and above the highest measured plane, extrapolation is blended to a separate upper-plane PCHIP fitted only through $z_{\text{fan},5}$, $z_{\text{fan},6}$, and $z_{\text{fan},7}$. Below the lowest measured plane, the parameters are fixed to their values at $z_{\text{fan},1}$ to reduce sensitivity to unsteadiness near the outlet [139].

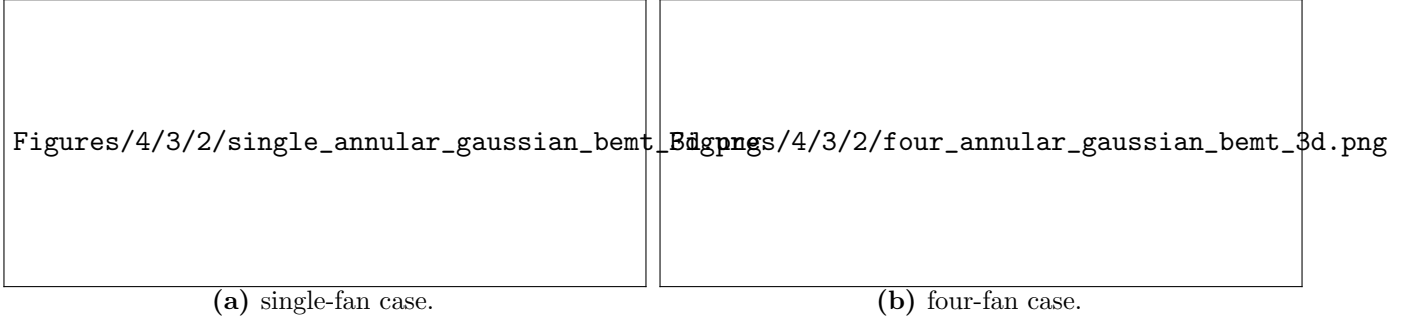


Fig. 4.7. 3D rendering of the harmonic annular-Gaussian updraft surrogate mean vertical velocity field.

4.3.3 Residual Correction Using Gaussian Process Regression

Although the harmonic annular-Gaussian model captures the dominant non-axisymmetric ring-shaped structure, small local discrepancies remain, especially where plume interaction and measurement noise are not fully captured by the parametric fit. To model these residual errors, a Gaussian process (GP) regressor is introduced as a residual corrector to refine the harmonic annular-Gaussian updraft representation [149], [140]. GP regression is a non-parametric Bayesian method that treats the unknown mapping $g(\cdot)$ as a random function whose values at any finite set of inputs are jointly Gaussian [51].

In the present case, the GP is trained on the residual dataset with M measurement points,

$$\mathcal{D} = \{(x_i, y_i, z_i, w_{\text{res},i}, \sigma_i)\}_{i=1}^M, \quad z_i \in \{z_{\text{fan},k}\}_{k=1}^7, \quad w_{\text{res},i} = \bar{w}_i - w_{\text{HAG},i}. \quad (4.14)$$

Using polar inputs, standardised by the sample mean μ_d and standard deviation s_d [140],

$$\phi_i = [r_i, \cos \theta_i, \sin \theta_i, z_i]^\top, \quad \tilde{\phi}_{i,d} = \frac{\phi_{i,d} - \mu_d}{s_d}, \quad d \in \{1, 2, 3, 4\}, \quad (4.15)$$

¹⁰ Since $N = 2$, both selected harmonic annular-Gaussian models include a second-order azimuthal Fourier correction.

the GP observation model for the residual field with sample-wise regularisation is then written as [141]

$$w_{\text{res},i} = g(\tilde{\phi}_i) + \varepsilon_i, \quad g(\cdot) \sim \mathcal{GP}(0, k(\cdot, \cdot)), \quad \varepsilon_i \sim \mathcal{N}(0, \alpha_i). \quad (4.16)$$

where $\alpha_i = \max(\alpha_{\text{scale}} \sigma_i^2, 10^{-8})$ is the sample-dependent observation noise variance with a minimum floor for numerical robustness. In the final selected model, the single-fan case uses $\alpha_{\text{scale}} = 1.2$, while the four-fan selects $\alpha_{\text{scale}} = 1.0$, indicating that the single-fan fit favours slightly stronger variance inflation.

The covariance between GP function values at two inputs is determined by the kernel function $k(\cdot, \cdot)$ [51]. For training inputs $\{\tilde{\phi}_i\}_{i=1}^M$, the effective training covariance is

$$\mathbf{K}_y = \mathbf{K} + \mathbf{V}, \quad K_{ij} = k(\tilde{\phi}_i, \tilde{\phi}_j), \quad \mathbf{V} = \text{diag}(\alpha_1, \dots, \alpha_M). \quad (4.17)$$

Kernel type, including radial basis function (RBF) [52], Matérn kernels with $\nu \in \{3/2, 5/2\}$ [142], and rational quadratic [141], is selected using grouped cross-validation [143], [144]: samples are split by measurement plane $z_{\text{fan},k}$, with three planes held out from training for validation, and the choice is made by minimising the validation WRMSE. This procedure selected a Matérn-3/2 kernel for single-fan and a Matérn-5/2 kernel for four-fan case [142], both with Automatic Relevance Determination (ARD) length scales [145] and an added white noise term to prevent overfitting by forcing exact interpolation [147]:

$$k(\tilde{\phi}, \tilde{\phi}') = \sigma_c^2 k_\nu(q) + \sigma_w^2 \delta_{\tilde{\phi}\tilde{\phi}'}, \quad k_\nu(q) = \begin{cases} (1 + \sqrt{3}q) \exp(-\sqrt{3}q), & \nu = 3/2, \\ (1 + \sqrt{5}q + 5q^2/3) \exp(-\sqrt{5}q), & \nu = 5/2, \end{cases} \quad (4.18)$$

where $q = \sqrt{\sum_{d=1}^4 (\tilde{\phi}_d - \tilde{\phi}'_d)^2 / \ell_d^2}$ is ARD-scaled distance, $\delta_{\tilde{\phi}\tilde{\phi}'}$ equals 1 when $\tilde{\phi} = \tilde{\phi}'$ and 0 otherwise.

The kernel hyperparameters, $\boldsymbol{\theta} = (\sigma_c, \ell_1, \ell_2, \ell_3, \ell_4, \sigma_w) \equiv (\sigma_c, \ell_r, \ell_{\cos\theta}, \ell_{\sin\theta}, \ell_z, \sigma_w)$, are estimated by maximising the GP log marginal likelihood [141],

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} \log p \left[-\frac{1}{2} \mathbf{w}_{\text{res}}^\top \mathbf{K}_y^{-1} \mathbf{w}_{\text{res}} - \frac{1}{2} \log |\mathbf{K}_y| - \frac{M}{2} \log(2\pi) \right], \quad (4.19)$$

with a multi-start procedure to reduce sensitivity to local minima [132].

After training, the GP surrogate is evaluated on each measured grid to produce maps of the predictive mean and standard deviation [141]. For a query input $\tilde{\phi}_*$, the predictive residual mean and variance are

$$\mu_{\text{res},*} = \mathbf{k}_*^\top \mathbf{K}_y^{-1} \mathbf{w}_{\text{res}}, \quad \sigma_{\text{res},*}^2 = k_{**} - \mathbf{k}_*^\top \mathbf{K}_y^{-1} \mathbf{k}_*, \quad (4.20)$$

where $\mathbf{w}_{\text{res}} = [w_{\text{res},1}, \dots, w_{\text{res},M}]^\top$, $\mathbf{k}_* = [k(\tilde{\phi}_*, \tilde{\phi}_1), \dots, k(\tilde{\phi}_*, \tilde{\phi}_M)]^\top$, and $k_{**} = k(\tilde{\phi}_*, \tilde{\phi}_*)$.

The final harmonic annular-Gaussian model with GP residual correction¹¹ is then obtained by

$$w_{\text{HAG-GP}}(x, y, z) = w_{\text{HAG}} + \mu_{\text{res}}, \quad \sigma_{\text{HAG-GP}}(x, y, z) = \sqrt{\sigma_1^2 + \sigma_{\text{res}}^2}. \quad (4.21)$$

The training procedure described above yields $\boldsymbol{\theta}^* = (2.29, 0.716, 0.659, 3.70, 0.371, 10^{-6})$ for the single-fan configuration and $\boldsymbol{\theta}^* = (0.789, 0.486, 100, 0.681, 1.96, 0.640)$ for the four-fan case. Relative to the single-fan case, the four-fan fit selects a smaller σ_c and a much larger white-noise level, σ_w , consistent with increased unresolved variability arising from multi-fan interactions [146]. The ARD length scales, $\{\ell_d\}_{d=1}^4$, further suggest that the residual field in the four-fan case is more radially localised, more anisotropic between the two angular feature directions, and shows little variation with height.

This observation is supported by the total fluctuation field shown in Fig. 4.9. The field is dominated by the empirical fluctuation term, σ_i , while a much smaller σ_{res} mainly reflects model uncertainty, and remains nearly height-invariant. In the single-fan case, $\sigma_{\text{HAG-GP}}$ remains concentrated around the fan-centred region, whereas in the four-fan case it is more spatially distributed and consistently higher.

¹¹ For simplicity, this model is called the residual annular Gaussian process surrogate in the remainder of the thesis.

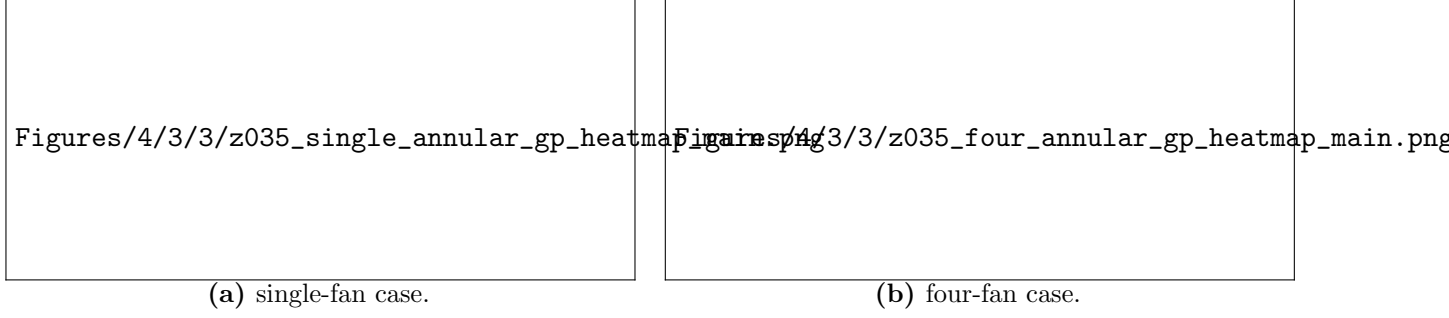


Fig. 4.8. Fitted mean updraft map of the residual annular Gaussian process surrogate at a single measurement plane 0.35 m above the fan outlet plane.

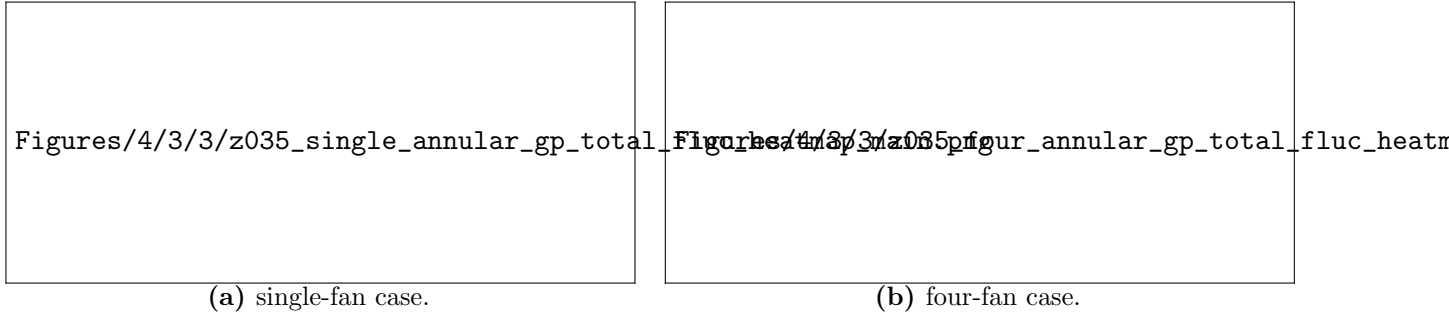


Fig. 4.9. Total fluctuation field, $\sigma_{\text{HAG-GP}}$, of the residual annular Gaussian process surrogate at a measurement plane 0.35 m above the fan outlet plane. The grey dotted rings indicate the annular regions used to assign the empirical fluctuation level, σ_i , as described at the beginning of [Section 4.3](#).

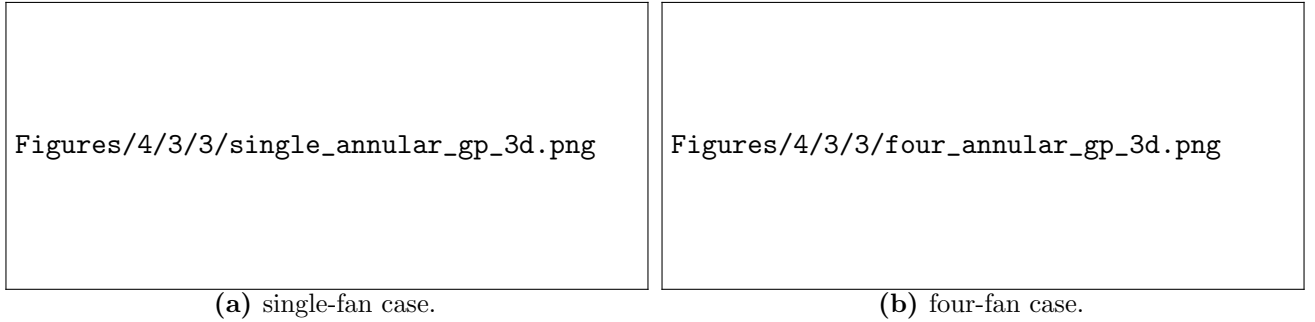


Fig. 4.10. 3D rendering of the residual annular Gaussian process surrogate mean vertical velocity field.

4.4 Model Comparison and Selection

Alongside the harmonic annular-Gaussian and residual annular GP surrogates, two simpler baselines are constructed: an axisymmetric annular-Gaussian model, obtained by removing the Fourier series terms from (4.7), and an axisymmetric Gaussian plume, formed by further removing the ring-radius term from (4.6). These four candidate updraft surrogates are evaluated using SAE and WRMSE metrics to compare their reconstruction quality and support the choice of surrogates used in controller development.

The fitted updraft maps for these two simpler baseline models are not shown in the main text for brevity. The complete set across all seven measurement planes is provided in [Appendix D](#).

For the single-fan configuration, [Table 4.1](#) and [Fig. 4.11](#) show a clear progression in fit quality from the Gaussian plume to the axisymmetric annular-Gaussian, harmonic annular-Gaussian, and residual annular GP surrogates, indicating that both the ring-shaped peak and azimuthal asymmetry are important features of the measured field. Once this dominant structure is captured, the remaining mismatch is weak, smooth, and spatially coherent, making it well-suited to residual GP correction.

For the four-fan case, the residual annular GP again yields the lowest SAE and WRMSE, but by a

much smaller margin, as shown in Table 4.1 and Fig. 4.12. This is consistent with the stronger and less coherent residual field left by plume interaction, fan-to-fan asymmetry, and height-dependent merging.

Table 4.1 Aggregate error metrics (SAE and WRMSE) for evaluated updraft models.

Setup	Error metric	Unit	Axisymmetric Gaussian plume	Axisymmetric annular-Gaussian	Harmonic annular-Gaussian	Residual annular Gaussian process
Single-fan	SAE	m s^{-1}	128	100	76.1	6.14
	WRMSE	m s^{-1}	0.210	0.152	0.120	0.00607
Four-fan	SAE	m s^{-1}	509	414	338	255
	WRMSE	m s^{-1}	0.419	0.367	0.321	0.241

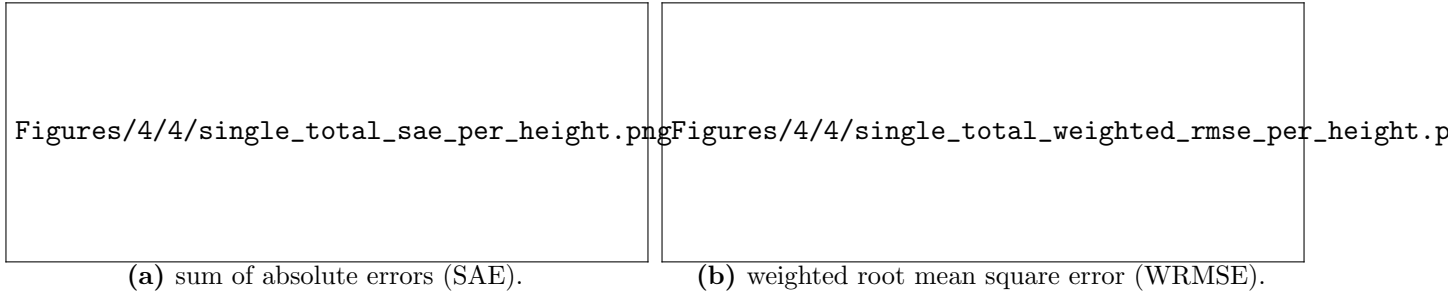


Fig. 4.11. Per-height error metrics for the evaluated updraft models (single fan).

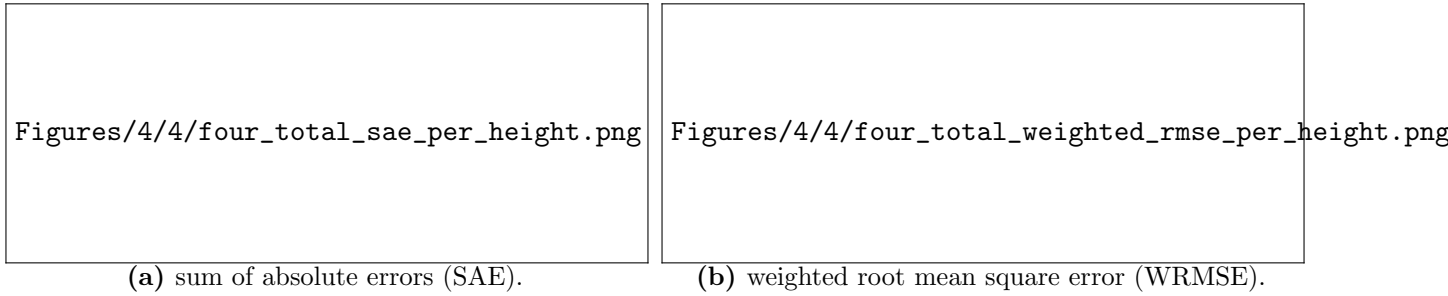


Fig. 4.12. Per-height error metrics for the evaluated updraft models (four fans).

Although the residual annular GP gives the most accurate reconstruction of the measured mean field, it remains an in-sample reconstruction rather than a complete description of the underlying unsteady flow. This mainly arises from two limitations. First, the mean field reconstruction is fitted to time-averaged measurements at a finite set of spatial points. The residual correction is flexible enough to reduce interpolation error at the measured locations, but an improved fit does not necessarily imply better physical generalisation, it may also reflect overfitting to the available data [141], [147]. Second, temporal fluctuation is sampled even more sparsely. The local fluctuation levels used in the model are estimated from short time-series records at only six representative locations, and then mapped across the rest of the field using the blending procedure in (4.3). Therefore, the resulting total fluctuation field in (4.21) is an approximate simulation uncertainty map, and may differ substantially from reality.

Instead of collecting additional data or adopting more accurate measurement methods to refine a single updraft model, the subsequent controller study in Chapter 6 adopts a more practical simulation strategy based on the principle of domain randomisation in robotics sim-to-real transfer [1], [3], [6]. The residual annular GP is selected as the most detailed measured surrogate, while its strength, spatial scale, centre location, number of active fans, and uncertainty level are randomised during simulation. According, controller design and validation are carried out across imperfect but physically related updraft families, reducing dependence on a single fitted reconstruction that may overfit the available measurements, while allowing reality to be treated as another variation of the simulated cases.

5 Glider Design and Optimisation

The glider is treated in this thesis not as a generic airframe, but as the physical control plant that must operate within the confined arena and the measured updraft environment. Its design therefore has to balance low-speed lift exploitation, manoeuvrability, structural simplicity, manufacturability, and practical avionics integration. This chapter formulates the vehicle design problem from that viewpoint and provides the aircraft that later defines the controller envelope studied in simulation and experiment.

5.1 Design Requirements and Low-Order Modelling

Write something.

In this thesis, AeroSandbox [69] is selected as a lightweight differentiable framework suited to highly customised constraints and objectives. The workflow uses CasADi [150] for automatic differentiation, and Interior Point OPTimiser (IPOPT) [151] to solve the resulting constrained nonlinear programmes efficiently.

The design target is not indefinite circulation inside the arena, but repeatable short episodes with sufficient controllability for lift capture, finite dwell, mild redirection, recovery, and safe termination.

5.1.1 Flight Arena Requirements

Write something.

5.1.2 Glider Model for Optimisation

Write something.

In this context, the flat-plate model is a sensible basis for a small indoor glider. It is not simply a crude approximation, but a suitable low-order model for thin lifting surfaces operating at low Reynolds number and potentially large incidence [59], [71]. However, a zero-thickness flat plate can be numerically problematic in optimisation frameworks that require aerofoil geometry or viscous-polar analysis [70]. In practice, it is often replaced by a very thin symmetric aerofoil, such as NACA 0002 or NACA 0004, to preserve near-flat-plate aerodynamic behaviour while improving numerical robustness [62], [152].

5.2 Optimisation Formulation and Design Selection

Write something.

5.2.1 Design Variables, Objective, and Constraints

Write something.

5.2.2 Multistart Search and Robust Evaluation

Write something.

5.2.3 Sensitivity Analysis

A local first-order sensitivity analysis was carried out about the selected optimum in order to identify the dominant drivers of the final design. Here, sensitivity refers to the total derivative of a scalar quantity of interest q with respect to a design parameter or requirement p ,

$$\bar{S}_{q,p} = \frac{p^*}{\max(|q^*|, \varepsilon_q)} \left. \frac{dq}{dp} \right|_{\star}, \quad (5.1)$$

evaluated at the final design $(\cdot)_*$. In practical terms, this measures how the local optimum changes when a single parameter or constraint is perturbed while the remaining active conditions are kept tight.

Since the optimisation was solved using a gradient-based primal dual method, these first order sensitivities were obtained directly from the converged solution with negligible additional cost. Selected entries were additionally checked by finite differences for verification. The purpose of this analysis is interpretive rather than exhaustive: it is used to distinguish quantities that are truly design-driving from those that are merely present in the formulation.

The step-size of the sensitivity analysis has been checked to ensure that it's neither too large to introduce truncation error nor too small to have the round-off error of the problem. The final step-size is listed in [Table 5.1](#), while the illustration of the error related to the step-size can be found in [Appendix E.2](#).

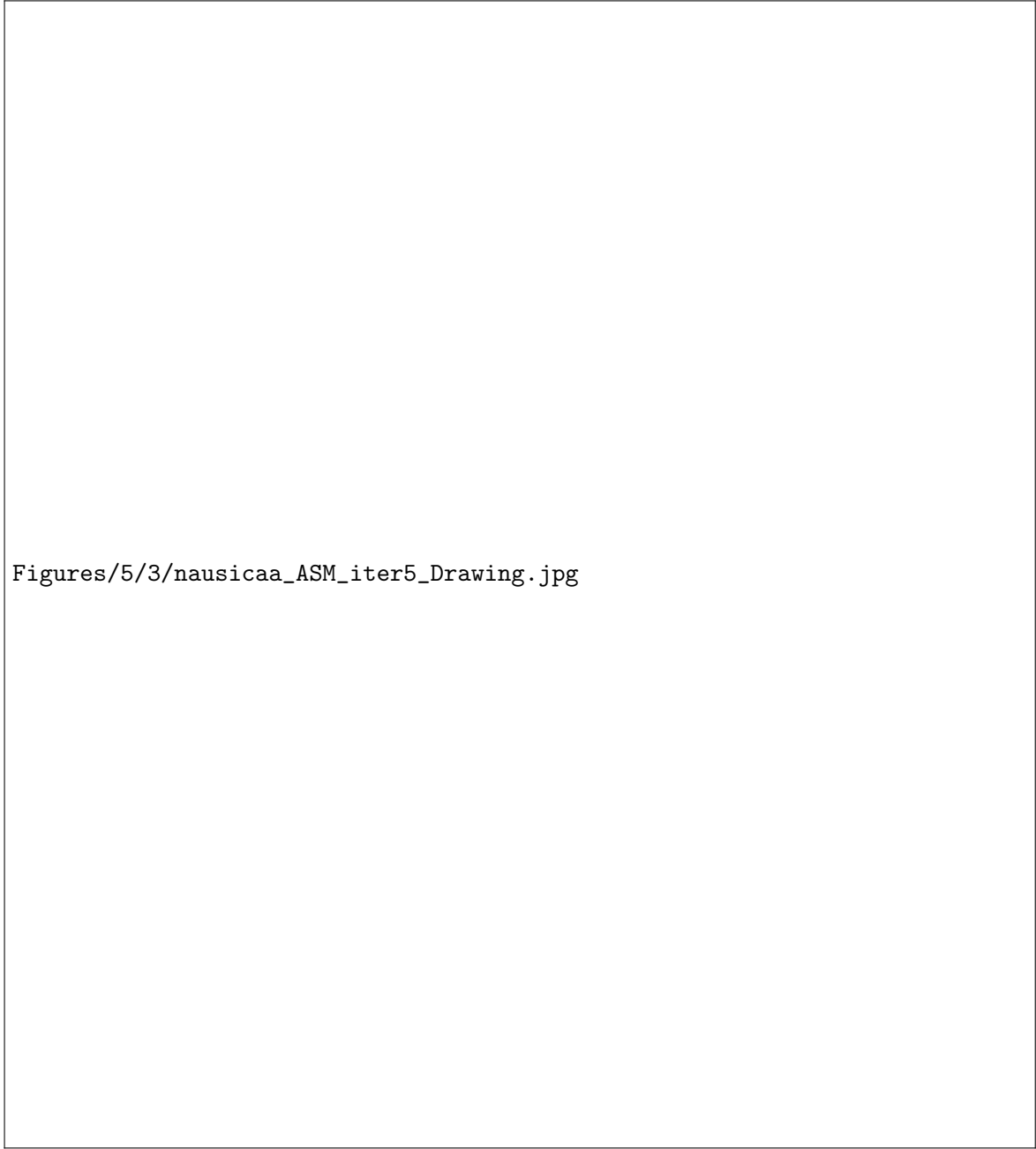
Table 5.1 Selected local first-order sensitivities about the final design.

Perturbed quantity	Symbol	Unit	Metric	Unit	$\bar{S}_{q,p}$	Interpretation
Bank entry time target	t_{entry}	s	J	-	[]	Larger magnitude indicates stronger sensitivity of the optimum to the agility requirement
Wall clearance	c_{wall}	m	J	-	[]	Quantifies the local cost of tightening the turning-space requirement
Static margin lower bound	SM_{min}	%MAC	J	-	[]	Indicates whether the longitudinal stability margin is weakly or strongly design driving
Nominal lift cap limit	$C_{L,\text{max}}$	-	J	-	[]	Shows the local benefit of relaxing or tightening the aerodynamic cap
Wing span	b	m	$\dot{h}_{\text{sink}}$	m s^{-1}	[]	Measures how strongly sink performance changes with span at the selected point
Tail arm	l_t	m	τ_ϕ	s	[]	Indicates the local effect of tail-arm changes on the roll-agility proxy
Horizontal tail span	b_h	m	$\delta_{e,\text{util}}$	-	[]	Useful for identifying whether trim authority is locally tail-size limited
Vertical tail height	h_v	m	r_{lat}	-	[]	Indicates the local influence on the lateral-trim or directional-residual metric

Accordingly, the reported sensitivities in [Table 5.1](#) are used to support two forms of interpretation. At the design-space level, they indicate which requirements most strongly shape the local optimum. At the engineering level, they indicate where additional effort, such as model improvement, uncertainty reduction, or requirement relaxation, is most likely to yield a useful performance gain.

5.3 Detailed Design and Manufacture

Write something.



Figures/5/3/nausicaa_ASM_iter5_Drawing.jpg

Fig. 5.1. General arrangement and key dimensions of the fifth iteration glider assembly, showing the Depron lifting surfaces, carbon-fibre boom and spar, and 3D-printed mounting and skid components. Avionics, wiring, and tape reinforcements are omitted for clarity.

Write something.

Table 5.2 As-designed and as-built parameters of the glider.

Parameter	Symbol	Unit	Designed	Measured	$\Delta\%$
Mass	m	g	117.53	133.56	13.6%
Wing span	b	m	0.762	0.764	0.262%
Wing chord	c	m	0.165	0.165	0%
Dihedral	Γ	deg	10.0°	9.28°	7.20%
Fuselage length	l	m	0.564	0.568	0.709%
Horizontal tail span	b_H	m	0.366	0.364	- 0.546%
Horizontal tail chord	c_H	m	0.091	0.091	0%
Vertical tail height	b_V	m	0.119	0.119	0%
Vertical tail chord	c_V	m	0.059	0.059	0%
Wing area	S	m ²	0.125	0.126	0.848%
Aspect ratio	$\mathcal{AR}$	-	4.63	4.63	0%
Wing loading	$W S^{-1}$	N m ⁻²	9.20	TBD	TBD%
Wing AC	x_W	m	0.0412	0.0413	0.243
Horizontal tail AC	x_H	m	0.423	0.426	0.709%
Vertical tail AC	x_V	m	0.424	0.433	2.12%
Longitudinal CG	x_{CG}	m	0.114	0.126	TBD%
		%MAC	69.1	76.4	10.6%
Lateral CG	y_{CG}	m	0	≈ 0	TBD%
Vertical CG	z_{CG}	m	0.0158	TBD	TBD%

Such a systematic optimisation method seems like maybe a bit 'overkill' for the current project, but it does provide a very good basis for the further integration of the powerplant, unique wing design, aeroelasticity and even morphing or flapping wing...

6 Viability-Guided Manoeuvre Primitive Selection

TBD.

6.1 Problem Formulation

TBD.

6.2 Control-Oriented Glider Model

TBD.

7 Sim-to-Real Evaluation and Results

TBD.

8 Conclusions and Future Work

Write something.

8.1 Summary of Contributions and Main Findings

Write something.

8.2 Recommendations for Future Work

Although the present thesis deliberately adopts a simpler quasi-steady and largely rigid-body optimisation model, the broader literature suggests a clear route for future extension toward coupled aerostructural and mission-aware design. Peter Sharpe’s Doctoral thesis [153]. In particular, Fasel and co-workers showed that concurrent aerodynamic, structural, and mission-level optimisation can be beneficial when the wing properties are allowed to adapt across operating conditions [154], [155].

While the present thesis does not require a full aeroelastic controller, this literature supports the broader principle that control-oriented reduced-order models are often the appropriate intermediate fidelity between rigid-body simplicity and high-order simulation [156].

We can integrate the co-design into this project, or the morphing wing. For co-design, we can reference Paul Master’s thesis and Adrian’s paper [73]. For morphing, a good example is *Flight through Narrow Gaps with Morphing-Wing Drones* [72], and research on Dr Fasel’s side. Also mentioned the wire control (interesting).

A natural next step is to move from the present sequential workflow toward a more integrated co-design formulation in which vehicle geometry, control authority, and mission or guidance objectives are optimised together rather than in isolation.

1) Can the model become much more precise? Aeroelastic? Real CFD? 2) Can we extend this to combine with the co-design problem? (Buda’s work) So it has been one of the first co-designs with an indoor fixed-wing test. 3) Can we extend this to a morphing wing? The flapping wing, etc. 4) A different control method is also interesting. How about wire control? In such a case, we even can extend the problem to integrate the control system into the manipulation of a multi-fingered dexterous hand, hence the area of robotics. 5) The ultimate goal is to achieve full autonomy, see the proposal of Wen-hua Chen and other researchers.

This project gives a very good platform to extend the research on all the above questions with minor modifications of the current apparatus and experimental design, and can be widely reproduced in many research institutions and universities.

Humans, or all intelligent cultures, rely on past experience to adapt to new things, but the biggest difference between humans themselves and other intelligences is that human use the different methods, such as education, training, or documentation, to share the experience over the different identity, and the result of the accumulation of the experience, is so called engineering institution in engineer perspective.

But it is actually reproducible and can be systematically transferred; in one day, I believe we can build a truly intelligent system based on that. We do not need to redesign everything or reproduce everything; we just need to make the intelligence itself an engineer.

References

- [1] Muratore F, Ramos F, Turk G, Yu W, Gienger M, Peters J. Robot Learning From Randomized Simulations: A Review. *Frontiers in Robotics and AI*. 2022;9. <https://doi.org/10.3389/frobt.2022.799893>.
- [2] Jakobi N, Husbands P, Harvey I. Noise and the reality gap: The use of simulation in evolutionary robotics. *European Conference on Artificial Life*. 1995. p. 704–720. https://doi.org/10.1007/3-540-59496-5_337.
- [3] Aljalbout E, Xing J, Romero A, Akinola I, Garrett CR, Heiden E, *et al.* The Reality Gap in Robotics: Challenges, Solutions, and Best Practices. *Annual Review of Control Robotics and Autonomous Systems*. 2025; <https://doi.org/10.1146/annurev-control-031924-100130>.
- [4] Allevato AD, Schaertl Short E, Pryor M, Thomaz AL. Iterative residual tuning for system identification and sim-to-real robot learning. *Autonomous Robots*. 2020;44(7): 1167–1182. <https://doi.org/10.1007/s10514-020-09925-w>.
- [5] Du Y, Watkins O, Darrell T, Abbeel P, Pathak D. Auto-Tuned Sim-to-Real Transfer. *2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE; 2021. <https://doi.org/10.1109/icra48506.2021.9562091>.
- [6] Tobin J, Fong R, Ray A, Schneider J, Zaremba W, Abbeel P. Domain randomization for transferring deep neural networks from simulation to the real world. *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE; 2017. p. 23–30. <https://doi.org/10.1109/IROS.2017.8202133>.
- [7] Schaff C, Sedal A, Ni S, Walter MR. Sim-to-real transfer of co-optimized soft robot crawlers. *Autonomous Robots*. 2023;47(8): 1195–1211. <https://doi.org/10.1007/s10514-023-10130-8>.
- [8] Peng XB, Andrychowicz M, Zaremba W, Abbeel P. Sim-to-Real Transfer of Robotic Control with Dynamics Randomization. *2018 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE; 2018. p. 3803–3810. <https://doi.org/10.1109/ICRA.2018.8460528>.
- [9] Tiboni G, Arndt K, Ville Kyrki. DROPO: Sim-to-real transfer with offline domain randomization. *Robotics and Autonomous Systems*. 2023;166: 104432. <https://doi.org/10.1016/j.robot.2023.104432>.
- [10] Belkhale S, Li R, Kahn G, McAllister R, Calandra R, Levine S. Model-Based Meta-Reinforcement Learning for Flight With Suspended Payloads. *IEEE Robotics and Automation Letters*. 2021;6(2): 1471–1478. <https://doi.org/10.1109/lra.2021.3057046>.
- [11] Zhu S, Mou L, Li D, Ye B, Huang R, Zhao H. VR-Robo: A Real-to-Sim-to-Real Framework for Visual Robot Navigation and Locomotion. *IEEE Robotics and Automation Letters*. 2025;10(8): 7875–7882. <https://doi.org/10.1109/lra.2025.3575648>.
- [12] Chebotar Y, Handa A, Makoviychuk V, Macklin M, Issac J, Ratliff N, *et al.* Closing the Sim-to-Real Loop: Adapting Simulation Randomization with Real World Experience. *2019 International Conference on Robotics and Automation (ICRA)*. IEEE; 2019. p. 8973–8979. <https://doi.org/10.1109/ICRA.2019.8793789>.
- [13] Bousmalis K, Irpan A, Wohlhart P, Bai Y, Kelcey M, Kalakrishnan M, *et al.* Using Simulation and Domain Adaptation to Improve Efficiency of Deep Robotic Grasping. *2018 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE; 2018. p. 4243–4250. <https://doi.org/10.1109/ICRA.2018.8460875>.
- [14] Kaufmann E, Bauersfeld L, Loquercio A, Müller M, Koltun V, Scaramuzza D. Champion-level drone racing using deep reinforcement learning. *Nature*. 2023;620(7976): 982–987. <https://doi.org/10.1038/s41586-023-06419-4>.
- [15] Loquercio A, Kaufmann E, Ranftl R, Dosovitskiy A, Koltun V, Scaramuzza D. Deep Drone Racing: From Simulation to Reality With Domain Randomization. *IEEE Transactions on Robotics*. 2020;36(1): 1–14. <https://doi.org/10.1109/tro.2019.2942989>.
- [16] Wada D, Araujo-Estrada S, Windsor S. Sim-to-Real Transfer for Fixed-Wing Uncrewed Aerial Vehicle: Pitch Control by High-Fidelity Modelling and Domain Randomization. *IEEE Robotics and Automation Letters*. 2022;7(4): 11735–11742. <https://doi.org/10.1109/lra.2022.3205442>.

- [17] Bøhn E, Coates EM, Reinhardt D, Johansen TA. Data-Efficient Deep Reinforcement Learning for Attitude Control of Fixed-Wing UAVs: Field Experiments. *IEEE Transactions on Neural Networks and Learning Systems*. 2023;35(3): 3168–3180. <https://doi.org/10.1109/tnnls.2023.3263430>.
- [18] Beard RW, Ferrin J, Humpherys J. Fixed Wing UAV Path Following in Wind With Input Constraints. *IEEE Transactions on Control Systems Technology*. 2014;22(6): 2103–2117. <https://doi.org/10.1109/tcst.2014.2303787>.
- [19] Mohamed A, Massey K, Watkins S, Clothier R. The attitude control of fixed-wing MAVS in turbulent environments. *Progress in Aerospace Sciences*. 2014;66: 37–48. <https://doi.org/10.1016/j.paerosci.2013.12.003>.
- [20] Britter R, Hanna SR. Flow and Dispersion in Urban Areas. *Annual Review of Fluid Mechanics*. 2003;35(1): 469–496. <https://doi.org/10.1146/annurev.fluid.35.101101.161147>.
- [21] Tominaga Y, Mochida A, Yoshie R, Kataoka H, Nozu T, Yoshikawa M, *et al.* AIJ guidelines for practical applications of CFD to pedestrian wind environment around buildings. *Journal of Wind Engineering and Industrial Aerodynamics*. 2008;96(10-11): 1749–1761. <https://doi.org/10.1016/j.jweia.2008.02.058>.
- [22] Blocken B. Computational Fluid Dynamics for urban physics: Importance, scales, possibilities, limitations and ten tips and tricks towards accurate and reliable simulations. *Building and Environment*. 2015;91: 219–245. <https://doi.org/10.1016/j.buildenv.2015.02.015>.
- [23] Achermann F, Stastny T, Danciu B, Kolobov A, Chung JJ, Siegwart R, *et al.* WindSeer: real-time volumetric wind prediction over complex terrain aboard a small uncrewed aerial vehicle. *Nature Communications*. 2024;15(1): 3507. <https://doi.org/10.1038/s41467-024-47778-4>.
- [24] McCormick BW. *Aerodynamics, Aeronautics, and Flight Mechanics*. Hoboken, NJ: John Wiley & Sons; 1995.
- [25] Gavrinev VO, Volkova MA, Romanov AM. Autonomous Thermal Soaring Methods: A Systematic Review. *Unmanned Systems*. 2025; 1–25. <https://doi.org/10.1142/s2301385027300010>.
- [26] Allen MJ. Autonomous Soaring for Improved Endurance of a Small Uninhabited Air Vehicle. *43rd AIAA Aerospace Sciences Meeting and Exhibit*. 2005. <https://doi.org/10.2514/6.2005-1025>.
- [27] Allen MJ, Lin V. Guidance and Control of an Autonomous Soaring Vehicle with Flight Test Results. *45th AIAA Aerospace Sciences Meeting and Exhibit*. 2007. <https://doi.org/10.2514/6.2007-867>.
- [28] Bencatel R, Tasso de Sousa J, Girard A. Atmospheric flow field models applicable for aircraft endurance extension. *Progress in Aerospace Sciences*. 2013;61: 1–25. <https://doi.org/10.1016/j.paerosci.2013.03.001>.
- [29] Ákos Z, Nagy M, Leven S, Vicsek T. Thermal soaring flight of birds and unmanned aerial vehicles. *Bioinspiration & Biomimetics*. 2010;5(4): 045003. <https://doi.org/10.1088/1748-3182/5/4/045003>.
- [30] Reddy G, Celani A, Sejnowski TJ, Massimo Vergassola. Learning to soar in turbulent environments. *Proceedings of the National Academy of Sciences*. 2016;113(33): E4877–E4884. <https://doi.org/10.1073/pnas.1606075113>.
- [31] Suys T, Hwang S, De Croon GCHE, Remes BDW. Autonomous Control for Orographic Soaring of Fixed-Wing UAVs. *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE; 2023. p. 5338–5344. <https://doi.org/10.1109/icra48891.2023.10161578>.
- [32] Hwang S, Remes BDW, De Croon GC. AOSoar: Autonomous Orographic Soaring of a Micro Air Vehicle. *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE; 2023. p. 5003–5010. <https://doi.org/10.1109/iros55552.2023.10341699>.
- [33] Thomas C. NASA Dryden aerospace engineer Michael Allen hand-launches a model motorized sailplane during a study validating the use of heat thermals to extend flight time. NASA Image and Video Library. 2005. <https://images.nasa.gov/details/EC05-0198-11> [Accessed 7th March 2026].

- [34] Wharington J, Israel H. Control of a High Endurance Unmanned Air Vehicle. *21st Congress of International Council of the Aeronautical Sciences*. 1998. https://www.icas.org/icas_archive/ICAS1998/PAPERS/371.
- [35] Lenschow DH, Stephens PL. The role of thermals in the convective boundary layer. *Boundary-Layer Meteorology*. 1980;19(4): 509–532. <https://doi.org/10.1007/bf00122351>.
- [36] Turner JS. *Buoyancy Effects in Fluids*. Cambridge: Cambridge University Press; 2009.
- [37] Morto BR, Taylor GI, John Stewart T. Turbulent gravitational convection from maintained and instantaneous sources. *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*. 1956;234(1196): 1–23. <https://doi.org/10.1098/rspa.1956.0011>.
- [38] Bower G, Flanzer T, Naiman A, Saripalli S. Dynamic Environment Mapping for Autonomous Thermal Soaring. *AIAA Guidance, Navigation, and Control Conference*. 2010. <https://doi.org/10.2514/6.2010-8031>.
- [39] Raffel M, Willert CE, Scarano F, Kähler CJ, Wereley ST, Kompenhans J. *Particle Image Velocimetry: A Practical Guide*. Cham: Springer; 2018.
- [40] Zagaglia D, Giuni M, Green RB. Investigation of the Rotor–Obstacle Aerodynamic Interaction in Hovering Flight. *Journal of the American Helicopter Society*. 2018;63(3): 1–12. <https://doi.org/10.4050/jahs.63.032007>.
- [41] Ramasamy M, Leishman JG. Benchmarking Particle Image Velocimetry with Laser Doppler Velocimetry for Rotor Wake Measurements. *AIAA Journal*. 2007;45(11): 2622–2633. <https://doi.org/10.2514/1.28130>.
- [42] Jørgensen FE. *How to Measure Turbulence with Hot-wire Anemometers*. Dantec Dynamics; 2005.
- [43] Barlow JB, Rae WH, Pope A. *Low-speed wind tunnel testing*. 3rd ed. Hoboken, NJ: John Wiley & Sons; 1999.
- [44] Bendat JS, Piersol AG. *Random Data: Analysis and Measurement Procedure*. 4th ed. Hoboken, NJ: John Wiley & Sons; 2011.
- [45] Coleman HW, Steele WG. *Experimentation, Validation, and Uncertainty Analysis for Engineers*. Hoboken, NJ: John Wiley & Sons; 2009.
- [46] Allen MJ. Updraft Model for Development of Autonomous Soaring Uninhabited Air Vehicles. *44th AIAA Aerospace Sciences Meeting and Exhibit*. 2006. <https://doi.org/10.2514/6.2006-1510>.
- [47] Gedeon J. Dynamic analysis of dolphin-style thermal cross-country flight. *Technical Soaring*. 1974;3(1): 9–19. <https://journals.sfu.ca/ts/index.php/ts/article/view/1114>.
- [48] Coleman RP, Feingold AM, Stempin CW. *Evaluation of the induced-velocity field of an idealized helicopter rotor*. National Advisory Committee for Aeronautics. 1945. <https://ntrs.nasa.gov/citations/19930093086>.
- [49] Leishman JG. *Principles of Helicopter Aerodynamics*. Cambridge: Cambridge University Press; 2008.
- [50] Tanner PE, Overmeyer AD, Jenkins LN, Yao CS, Bartram SM. Experimental Investigation of Rotorcraft Outwash in Ground Effect. *AHS International Annual Forum & Technology Display*. No. NF1676L-21054; 2015. <https://ntrs.nasa.gov/citations/20160006428>.
- [51] Seeger M. Gaussian Processes for Machine Learning. *International Journal of Neural Systems*. 2004;14(02): 69–106. <https://doi.org/10.1142/s0129065704001899>.
- [52] Buhmann MD. *Radial Basis Functions*. Cambridge: Cambridge University Press; 2003.
- [53] Sacks J, Welch WJ, Mitchell TJ, Wynn HP. Design and Analysis of Computer Experiments. *Statistical Science*. 1989;4(4): 409–423. <https://doi.org/10.1214/ss/1177012413>.
- [54] Simons M. *Model Aircraft Aerodynamics*. Fox Chapel Publishing; 2015.

- [55] Coussot JL. *On ne vas pas faire de chichis... Mais on peut faire un "CHICHI"!* JLC Aviation. <https://www.jlc-aviation.fr/a%C3%A9romod%C3%A9lisme/essais/silence-model-chichi> [Accessed 3rd March 2026].
- [56] Silence Model. *CeePee 750 mm*. <https://silencemodel.fr/kit-de-0-a-1990-mm/549-ceepee-750-mm.html> [Accessed 3rd March 2026].
- [57] MicroBirds. *Feather² Squared*. <https://microbirds.com/rc-plane-radio-control-kit-glider-feather> [Accessed 22nd January 2026].
- [58] MicroBirds. *FireFly 2.0*. <https://microbirds.com/radio-control-rc-gliders-firefly2> [Accessed 9th March 2026].
- [59] Cory RE. *Supermaneuverable perching*. [Doctoral Thesis] [Massachusetts Institute of Technology]; 2010. <http://hdl.handle.net/1721.1/60142>.
- [60] Depron. *Depron Insulation board: Technical Data Sheet*. 2013. https://www.depron-daemmplatte.de/site/assets/files/1044/pdb-depron-d.mmplatte_gb.20160301.pdf [Accessed 23rd February 2026].
- [61] Lissaman PBS. Low-Reynolds-Number Airfoils. *Annual Review of Fluid Mechanics*. 1983;15(1): 223–239. <https://doi.org/10.1146/annurev.fl.15.010183.001255>.
- [62] Mateescu D, Abdo M. Analysis of flows past airfoils at very low Reynolds numbers. *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*. 2010;224(7): 757–775. <https://doi.org/10.1243/09544100jaero715>.
- [63] Mueller TJ, DeLaurier JD. Aerodynamics of small vehicles. *Annual Review of Fluid Mechanics*. 2003;35(1): 89–111. <https://doi.org/10.1146/annurev.fluid.35.101101.161102>.
- [64] Anderson JD. *Aircraft Performance and Design*. Boston, MA: Mcgraw-Hill Higher Education; 2012.
- [65] Raymer DP. *Aircraft Design: A Conceptual Approach*. Reston, VA: American Institute of Aeronautics And Astronautics; 2012.
- [66] Gray JS, Hwang JT, Martins JRRA, Moore KT, Naylor BA. OpenMDAO: an open-source framework for multidisciplinary design, analysis, and optimization. *Structural and Multidisciplinary Optimization*. 2019;59(4): 1075–1104. <https://doi.org/10.1007/s00158-019-02211-z>.
- [67] Lukaczyk TW, Wendorff AD, Colonna M, Economou TD, Alonso JJ, Orra TH, *et al*. SUAVE: An Open-Source Environment for Multi-Fidelity Conceptual Vehicle Design. *16th AIAA/ISSMO Multidisciplinary Analysis and Optimization Conference*. 2015. <https://doi.org/10.2514/6.2015-3087>.
- [68] Burnell E, Damen NB, Hoburg W. GPkit: A Human-Centered Approach to Convex Optimization in Engineering Design. *CHI '20: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 2020. <https://doi.org/10.1145/3313831.3376412>.
- [69] Sharpe P. *AeroSandbox: A Differentiable Framework for Aircraft Design Optimization*. [Master Thesis] [Massachusetts Institute of Technology]; 2021. <https://hdl.handle.net/1721.1/140023>.
- [70] Drela M. XFOIL: An Analysis and Design System for Low Reynolds Number Airfoils. In: Mueller TJ (ed.) *Low Reynolds Number Aerodynamics*. Berlin, Heidelberg: Springer; 1989. p. 1–12. https://link.springer.com/chapter/10.1007/978-3-642-84010-4_1.
- [71] Khan W. *Dynamics Modeling of Agile Fixed-Wing Unmanned Aerial Vehicles*. [Doctoral Thesis] [McGill University]; 2016. <https://escholarship.mcgill.ca/concern/theses/4m90dz005>.
- [72] Wanner J, Phan HV, Toumeh C, Floreano D. Flight through Narrow Gaps with Morphing-Wing Drones. *ArXiv [Preprint]*. Version 1. 2026. <https://doi.org/10.48550/arXiv.2603.12059>.
- [73] Buda AA, Chen X, Botteghi N, Fasel U. Robust Co-design Optimisation for Agile Fixed-Wing UAVs. *ArXiv [Preprint]*. Version 1. 2025. <https://doi.org/10.48550/arXiv.2603.11130>.

- [74] Moore J. *Robust Post-Stall Perching with a Fixed-Wing UAV*. [Doctoral Thesis] [Massachusetts Institute of Technology]; 2014. <http://hdl.handle.net/1721.1/93861>.
- [75] Stevens BL, Lewis FL, Johnson EN. *Aircraft Control and Simulation: dynamics, Controls design, and Autonomous Systems*. Hoboken, NJ: John Wiley & Sons; 2016.
- [76] Anderson BDO, Moore JB. *Optimal control: linear quadratic methods*. Mineola, NY: Dover Publications; 2007.
- [77] López J, Dormido R, Dormido S, Gómez JP. A Robust H_∞ Controller for an UAV Flight Control System. *The Scientific World Journal*. 2015;2015(1). <https://doi.org/10.1155/2015/403236>.
- [78] Mathisen S, Gryte K, Gros S, Johansen TA. Precision Deep-Stall Landing of Fixed-Wing UAVs Using Nonlinear Model Predictive Control. *Journal of Intelligent & Robotic Systems*. 2020;101(1). <https://doi.org/10.1007/s10846-020-01264-3>.
- [79] Vachtsevanos G, Tang L, Drozeski G, Gutierrez L. From mission planning to flight control of unmanned aerial vehicles: Strategies and implementation tools. *Annual Reviews in Control*. 2005;29(1): 101–115. <https://doi.org/10.1016/j.arcontrol.2004.11.002>.
- [80] Sujit PB, Saripalli S, Sousa JB. Unmanned Aerial Vehicle Path Following: A Survey and Analysis of Algorithms for Fixed-Wing Unmanned Aerial Vehicles. *IEEE Control Systems Magazine*. 2014;34(1): 42–59. <https://doi.org/10.1109/mcs.2013.2287568>.
- [81] Nelson DR, Barber DB, McLain TW, Beard RW. Vector Field Path Following for Miniature Air Vehicles. *IEEE Transactions on Robotics*. 2007;23(3): 519–529. <https://doi.org/10.1109/tro.2007.898976>.
- [82] Frazzoli E, Dahleh MA, Feron E. Maneuver-based motion planning for nonlinear systems with symmetries. *IEEE Transactions on Robotics*. 2005;21(6): 1077–1091. <https://doi.org/10.1109/tro.2005.852260>.
- [83] Burridge RR, Rizzi AA, Koditschek DE. Sequential Composition of Dynamically Dexterous Robot Behaviors. *The International Journal of Robotics Research*. 1999;18(6): 534–555. <https://doi.org/10.1177/0278364992066385>.
- [84] Frazzoli E, Dahleh MA, Feron E. Real-Time Motion Planning for Agile Autonomous Vehicles. *Journal of Guidance, Control, and Dynamics*. 2002;25(1): 116–129. <https://doi.org/10.2514/2.4856>.
- [85] Tedrake R, Manchester IR, Tobenkin M, Roberts JW. LQR-trees: Feedback Motion Planning via Sums-of-Squares Verification. *The International Journal of Robotics Research*. 2010;29(8): 1038–1052. <https://doi.org/10.1177/0278364910369189>.
- [86] Tobenkin M, Manchester IR, Tedrake R. Invariant Funnels around Trajectories using Sum-of-Squares Programming. *IFAC Proceedings Volumes*. 2011;44(1): 9218–9223. <https://doi.org/10.3182/20110828-6-it-1002.03098>.
- [87] Garone E, Di Cairano S, Kolmanovsky I. Reference and command governors for systems with constraints: A survey on theory and applications. *Automatica*. 2017;75: 306–328. <https://doi.org/10.1016/j.automatica.2016.08.013>.
- [88] Bemporad A. Reference governor for constrained nonlinear systems. *IEEE Transactions on Automatic Control*. 1998;43(3): 415–419. <https://doi.org/10.1109/9.661611>.
- [89] Aubin JP, Bayen AM, Saint-Pierre P. *Viability theory: new directions*. Springer; 2013.
- [90] Blanchini F. *Set invariance in control*. *Automatica*. 1999;35(11): 1747–1767. [https://doi.org/10.1016/S0005-1098\(99\)00113-2](https://doi.org/10.1016/S0005-1098(99)00113-2).
- [91] Botros A, Smith SL. Spatio-Temporal Lattice Planning Using Optimal Motion Primitives. *IEEE Transactions on Intelligent Transportation Systems*. 2023;24(11): 11950–11962. <https://doi.org/10.1109/tits.2023.3297068>.
- [92] Monahan GE. State of the Art—A Survey of Partially Observable Markov Decision Processes: Theory, Models, and Algorithms. *Management Science*. 1982;28(1): 1–16. <https://doi.org/10.1287/mnsc.28.1.1>.

- [93] Kaelbling LP, Littman ML, Cassandra AR. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*. 1998;101(1-2): 99–134. [https://doi.org/10.1016/s0004-3702\(98\)00023-x](https://doi.org/10.1016/s0004-3702(98)00023-x).
- [94] Prentice S, Roy N. The Belief Roadmap: Efficient Planning in Belief Space by Factoring the Covariance. *The International Journal of Robotics Research*. 2009;28(11-12): 1448–1465. <https://doi.org/10.1177/0278364909341659>.
- [95] van den Berg J, Patil S, Alterovitz R. Motion planning under uncertainty using iterative local optimization in belief space. *The International Journal of Robotics Research*. 2012;31(11): 1263–1278. <https://doi.org/10.1177/0278364912456319>.
- [96] Indelman V, Carlone L, Dellaert F. Planning in the continuous domain: A generalized belief space approach for autonomous navigation in unknown environments. *The International Journal of Robotics Research*. 2015;34(7): 849–882. <https://doi.org/10.1177/0278364914561102>.
- [97] Kondo K, Tordesillas J, How JP. SANDO: Safe Autonomous Trajectory Planning for Dynamic Unknown Environments. ArXiv [Preprint]. Version 1. 2026. <https://arxiv.org/abs/2604.07599>.
- [98] Brian Y. A Frontier-Based Approach for Autonomous Exploration. *Proceedings 1997 IEEE International Symposium on Computational Intelligence in Robotics and Automation CIRA'97. 'Towards New Computational Principles for Robotics and Automation'*. IEEE; 1997. <https://doi.org/10.1109/CIRA.1997.613851>
- [99] Placed JA, Strader J, Carrillo H, Atanasov N, Indelman V, Carlone L, *et al.* A Survey on Active Simultaneous Localization and Mapping: State of the Art and New Frontiers. *IEEE Transactions on Robotics*. 2023;39(3): 1686–1705. <https://doi.org/10.1109/tro.2023.3248510>.
- [100] Kober J, Bagnell JA, Peters J. Reinforcement learning in robotics: A survey. *The International Journal of Robotics Research*. 2013;32(11): 1238–1274. <https://doi.org/10.1177/0278364913495721>.
- [101] Moerland TM, Broekens J, Plaat A, Jonker CM. Model-based Reinforcement Learning: A Survey. *Foundations and Trends in Machine Learning*. 2023;16(1): 1–118. <https://doi.org/10.1561/22000000086>.
- [102] Brunke L, Greeff M, Hall AW, Yuan Z, Zhou S, Panerati J, *et al.* Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning. *Annual Review of Control, Robotics, and Autonomous Systems*. 2022;5(1). <https://doi.org/10.1146/annurev-control-042920-020211>.
- [103] Tabor S, Guilliard I, Kolobov A. ArduSoar: An Open-Source Thermalling Controller for Resource-Constrained Autopilots. *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE; 2018. p. 6255–6262. <https://doi.org/10.1109/iros.2018.8593510>.
- [104] Guilliard I, Rogahn R, Piavi J, Kolobov A. Autonomous Thermalling as a Partially Observable Markov Decision Process. *14th Robotics: Science and Systems, RSS 2018*. MIT Press Journals; 2018. <https://doi.org/10.15607/rss.2018.xiv.068>.
- [105] Reddy G, Wong-Ng J, Celani A, Sejnowski TJ, Vergassola M. Glider soaring via reinforcement learning in the field. *Nature*. 2018;562(7726): 236–239. <https://doi.org/10.1038/s41586-018-0533-0>.
- [106] Cui Y, Yan D, Wan Z. Study on the Glider Soaring Strategy in Random Location Thermal Updraft via Reinforcement Learning. *Aerospace*. 2023;10(10): 834. <https://doi.org/10.3390/aerospace10100834>.
- [107] Zhao J, Li J, Zeng L. Energy-Harvesting Strategy Investigation for Glider Autonomous Soaring Using Reinforcement Learning. *Aerospace*. 2023;10(10): 895–895. <https://doi.org/10.3390/aerospace10100895>.
- [108] Kumar V, Michael N. Opportunities and challenges with autonomous micro aerial vehicles. *The International Journal of Robotics Research*. 2012;31(11): 1279–1291. <https://doi.org/10.1177/0278364912455954>.
- [109] Coletti C, Williams K. Real-Time Control Interface for Research and Development Using Commercial-Off-The-Shelf (COTS) Mobile Robotics and UAVs. *AIAA SCITECH 2024 Forum*. Reston, Virginia: American Institute of Aeronautics and Astronautics; 2024. <https://doi.org/10.2514/6.2024-0235>.

- [110] Cheng Z, West R, Einstein C. End-to-End Analysis and Design of a Drone Flight Controller. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*. 2018;37(11): 2404–2415. <https://doi.org/10.1109/tcad.2018.2857399>.
- [111] Windolf M, Götzen N, Morlock M. Systematic accuracy and precision analysis of video motion capturing systems—exemplified on the Vicon-460 system. *Journal of Biomechanics*. 2008;41(12): 2776–2780. <https://doi.org/10.1016/j.jbiomech.2008.06.024>.
- [112] Merriaux P, Dupuis Y, Boutteau R, Vasseur P, Savatier X. A Study of Vicon System Positioning Performance. *Sensors*. 2017;17(7): 1591. <https://doi.org/10.3390/s17071591>.
- [113] Mellinger D, Kumar V. Minimum snap trajectory generation and control for quadrotors. *2011 IEEE International Conference on Robotics and Automation*. IEEE; 2011. <https://doi.org/10.1109/icra.2011.5980409>.
- [114] Schoellig AP, Mueller FL, D’Andrea R. Optimization-based iterative learning for precise quadcopter trajectory tracking. *Autonomous Robots*. 2012;33(1-2): 103–127. <https://doi.org/10.1007/s10514-012-9283-2>.
- [115] Zhang W, Branicky MS, Phillips SM. Stability of networked control systems. *IEEE Control Systems Magazine*. 2001;21(1): 84–99. <https://doi.org/10.1109/37.898794>.
- [116] Miao Y, Shen W, Cui H, Mitra S. FalconWing: An Ultra-Light Indoor Fixed-Wing UAV Platform for Vision-Based Autonomy. ArXiv [Preprint]. Version 3. 2025. <https://doi.org/10.48550/arXiv.2505.01383>.
- [117] Reid I, Ritchie J, Moore J, Sutherland B, Snow G, Tokumaru P, *et al.* ROSplane 2.0: A Fixed-Wing Autopilot for Research. ArXiv [Preprint]. Version 2. 2026; <https://doi.org/10.48550/arXiv.2510.01041>.
- [118] Pickem D, Glotfelter P, Wang L, Mote M, Ames A, Feron E, *et al.* The Robotarium: A remotely accessible swarm robotics research testbed. *2017 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE; 2017. p. 1699–1706. <https://doi.org/10.1109/ICRA.2017.7989200>.
- [119] Tani J, Daniele AF, Bernasconi G, Camus A, Petrov A, Courchesne A, *et al.* Integrated Benchmarking and Design for Reproducible and Accessible Evaluation of Robotic Agents. *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE; 2020. p. 6229–6236. <https://doi.org/10.1109/iros45743.2020.9341677>.
- [120] Dasari S, Wang J, Hong J, Bahl S, Lin Y, Wang A, *et al.* RB2: Robotic Manipulation Benchmarking with a Twist. ArXiv [Preprint]. Version 2. 2022. <https://doi.org/10.48550/arXiv.2203.08098>.
- [121] Lupashin S, Schöllig A, Hehn M, D’Andrea R. The Flying Machine Arena as of 2010. *2011 IEEE International Conference on Robotics and Automation*. IEEE; 2011. <https://doi.org/10.1109/icra.2011.5980308>.
- [122] Srivastava K, Kulkarni R, Velmurugan M, Sanker NJ. VizFlyt: Perception-centric Pedagogical Framework For Autonomous Aerial Robots. *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE; 2025. p. 9920–9927. <https://doi.org/10.1109/icra55743.2025.11128463>.
- [123] Kaya YF, Häusermann D, Nguyen PH, Orr L, Kovač M. The DroneHub: a multi-environment testbed for sustainability robotics. *npj Robotics*. 2026;4(1). <https://doi.org/10.1038/s44182-025-00054-z>.
- [124] Vicon Motion Systems Limited. *Vicon Tracker User Guide*. <https://help.vicon.com/space/Tracker44/376309877/PDF+downloads+for+Vicon+Tracker> [Accessed 29th November 2025].
- [125] Clarke International. *Clarke CAMAX30 30” Extra High Output Drum Fan*. <https://www.clarkeinternational.com/p/clarke-camax30-extra-high-output-drum-fan> [Accessed 1st December 2025].
- [126] BENETECH. *Hot Wire Anemometer GM8903*. <http://www.benetechno.net/en/products/gm8903.html> [Accessed 20th January 2026].
- [127] Hultmark M, Smits AJ. Temperature corrections for constant temperature and constant current hot-wire anemometers. *Measurement Science and Technology*. 2010;21(10): 105404. <https://doi.org/10.1088/0957-0233/21/10/105404>.

- [128] Yang Y, Sciacchitano A, Veldhuis LLM, Eitelberg G. Analysis of propeller-induced ground vortices by particle image velocimetry. *Journal of Visualization*. 2017;21(1): 39–55. <https://doi.org/10.1007/s12650-017-0439-1>.
- [129] Hodson TO. Root-mean-square error (RMSE) or mean absolute error (MAE): when to use them or not. *Geoscientific Model Development*. 2022;15(14): 5481–5487. <https://doi.org/10.5194/gmd-15-5481-2022>.
- [130] Huber PJ, Ronchetti EM. *Robust Statistics*. Hoboken, NJ: John Wiley & Sons; 2011.
- [131] Virtanen P, Gommers R, Oliphant TE, Haberland M, Reddy T, Cournapeau D, *et al*. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*. 2020;17(3): 261–272. <https://www.nature.com/articles/s41592-019-0686-2>.
- [132] Ugray Z, Lasdon L, Plummer J, Glover F, Kelly J, Martí R. Scatter Search and Local NLP Solvers: A Multistart Framework for Global Optimization. *INFORMS Journal on Computing*. 2007;19(3): 328–340. <https://doi.org/10.1287/ijoc.1060.0175>.
- [133] Hoerl AE, Kennard RW. Ridge Regression: Biased Estimation for Nonorthogonal Problems. *Technometrics*. 1970;12(1): 55–67. <https://doi.org/10.1080/00401706.1970.10488634>.
- [134] Bertsekas DP. On Penalty and Multiplier Methods for Constrained Minimization. *SIAM Journal on Control and Optimization*. 1976;14(2): 216–235. <https://doi.org/10.1137/0314017>.
- [135] Vandevein H. Family of spectral filters for discontinuous problems. *Journal of Scientific Computing*. 1991;6(2): 159–192. <https://doi.org/10.1007/bf01062118>.
- [136] Charbonnier P, Blanc-Feraud L, Aubert G, Barlaud M. Two deterministic half-quadratic regularization algorithms for computed imaging. *Proceedings of 1st International Conference on Image Processing*. 1994. <https://doi.org/10.1109/icip.1994.413553>.
- [137] Black MJ, Anandan P. The Robust Estimation of Multiple Motions: Parametric and Piecewise-Smooth Flow Fields. *Computer Vision and Image Understanding*. 1996;63(1): 75–104. <https://doi.org/10.1006/cviu.1996.0006>.
- [138] Fritsch FN, Carlson RW. Monotone Piecewise Cubic Interpolation. *SIAM Journal on Numerical Analysis*. 1980;17(2): 238–246. <https://doi.org/10.1137/0717021>.
- [139] Transtrum MK, Machta BB, Sethna JP. Geometry of nonlinear least squares with applications to sloppy models and optimization. *Physical Review E*. 2011;83(3). <https://doi.org/10.1103/physreve.83.036701>.
- [140] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, *et al*. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*. 2011;12. <https://doi.org/10.5555/1953048.2078195>.
- [141] Bousquet O, Von Luxburg U, Rätsch G. *Advanced Lectures on Machine Learning*. Berlin Heidelberg: Springer; 2004.
- [142] Bishop CM. *Pattern Recognition and Machine Learning*. New York, NY: Springer; 2006.
- [143] Roberts DR, Bahn V, Ciuti S, Boyce MS, Elith J, Guillera-Aroita G, *et al*. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*. 2017;40(8): 913–929. <https://doi.org/10.1111/ecog.02881>.
- [144] Arlot S, Celisse A. A survey of cross-validation procedures for model selection. *Statistics Surveys*. 2010;4(0): 40–79. <https://doi.org/10.1214/09-ss054>.
- [145] Wipf D, Nagarajan S. A new view of automatic relevance determination. *Advances in Neural Information Processing Systems 20 (NIPS 2007)*. 2007. https://proceedings.neurips.cc/paper_files/paper/2007/file/9c01802ddb981e6bcfbec0f0516b8e35-Paper.pdf.
- [146] Adrian L, Joseph L. Dynamic interaction of multiple buoyant jets. *Journal of Fluid Mechanics*. 2012;708: 539–575. <https://doi.org/10.1017/jfm.2012.332>.

- [147] Andrianakis I, Challenor PG. The effect of the nugget on Gaussian process emulators of computer models. *Computational Statistics & Data Analysis*. 2012;56(12): 4215–4228. <https://doi.org/10.1016/j.csda.2012.04.020>.
- [148] Fernández-Godino MG. Review of Multi-fidelity Models. ArXiv [Preprint]. Version 6. 2023. <https://doi.org/10.20944/preprints202304.1264.v1>.
- [149] Forrester AIJ, Keane AJ. Recent advances in surrogate-based optimization. *Progress in Aerospace Sciences*. 2009;45(1-3): 50–79. <https://doi.org/10.1016/j.paerosci.2008.11.001>.
- [150] Andersson JAE, Gillis J, Horn G, Rawlings JB, Diehl M. CasADi: a software framework for nonlinear optimization and optimal control. *Mathematical Programming Computation*. 2018;11(1): 1–36. <https://doi.org/10.1007/s12532-018-0139-4>.
- [151] Wächter A, Biegler LT. On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming. *Mathematical Programming*. 2006;106(1): 25–57. <https://doi.org/10.1007/s10107-004-0559-y>.
- [152] Kurtulus DF. Critical Angle and Fundamental Frequency of Symmetric Airfoils at Low Reynolds Numbers. *Journal of Applied Fluid Mechanics*. 2022;15(3): 723–735. <https://doi.org/10.47176/jafm.15.03.33099>.
- [153] Sharpe P. *Accelerating Practical Engineering Design Optimization with Computational Graph Transformations*. [Doctoral Thesis] [Massachusetts Institute of Technology]; 2024. <https://hdl.handle.net/1721.1/140023>.
- [154] Fasel U, Keidel D, Molinari G, Ermanni P. Aerostructural optimization of a morphing wing for airborne wind energy applications. *Smart Materials and Structures*. 2017;26(9): 095043. <https://doi.org/10.1088/1361-665x/aa7c87>.
- [155] Fasel U, Tiso P, Keidel D, Ermanni P. Concurrent Design and Flight Mission Optimization of Morphing Airborne Wind Energy Wings. *AIAA Journal*. 2021;59(4): 1254–1268. <https://doi.org/10.2514/1.j059621>.
- [156] Wang Y, Wynn A, Palacios R. Nonlinear Aeroelastic Control of Very Flexible Aircraft Using Model Updating. *Journal of Aircraft*. 2018;55(4): 1551–1563. <https://doi.org/10.2514/1.c034684>.

A Reproducibility and Research Integrity

A.1 Reproducibility Statement

This thesis is accompanied by a reproducibility package including: (i) the complete source code for simulation, modelling, and analysis; (ii) the configuration files used to generate the reported results; (iii) the processed datasets underlying the figures in the main text and appendices; and (iv) the editable source files for author-designed illustrative and schematic figures. The package is hosted in a public repository at <https://github.com/GH-X-ST/Nausicaa>.

All computational results reported in this thesis can be reproduced by executing the provided scripts in a clean environment, following the instructions in `README.md`. The software environment is specified in `environment.yml` and was tested on Windows 11 25H2 with Python 3.12.11, MATLAB R2026a and Arduino IDE 2.3.8. For stochastic components, fixed random seeds are provided; minor numerical differences across platforms are expected within standard floating-point tolerances.

Experimental results reported in [Appendices B, C and F](#) depend on physical measurements in the Flight Arena and therefore cannot be regenerated deterministically without the same hardware and setup. To support verification, the repository includes the raw measurement logs, preprocessing scripts, and plotting code used to produce the reported statistics and heat maps.

Author-designed figures, including conceptual illustrations, workflow diagrams, and schematic figures, were prepared using standard graphical software, including Microsoft PowerPoint, Adobe Photoshop, Adobe Illustrator and Union. Their preparation drew on the author’s prior graphic design experience without any external assistance.

A.2 Declaration of Generative AI Use

Generative AI tools were used in a limited and documented manner to support the preparation of this thesis, consistent with Imperial College London guidance on responsible use ¹².

Specifically, generative AI was used for: (i) proofreading, polishing, and grammar refinement of draft text written by the author, (ii) restructuring of $\text{\LaTeX}$ formatting, and (iii) programming assistance for non-novel tasks (e.g. refactoring and debugging). All technical decisions, modelling assumptions, experimental procedures, results, and interpretations remain the author’s work.

Accordingly, generative AI was not used to: (i) fabricate, alter, or misrepresent experimental measurements, simulation outputs, or numerical results, (ii) generate figures or tables without author verification and reproduction from the project codebase, (iii) create citations, or (iv) produce assessment content in a manner that would replace the author’s original analysis and understanding.

All AI-assisted content was reviewed and edited by the author. Code suggestions were adopted only after inspection, execution, and validation against expected behaviour. Textual suggestions were checked for technical accuracy and consistency with the project record. No confidential, personal, or restricted information (e.g. unpublished third-party materials) was entered into generative AI tools.

¹² Available at <https://www.imperial.ac.uk/about/education/resources/ai-education-hub/generative-ai-principles>.

B Supplementary State Processing and Timing Characterisation

B.1 Stage-by-Stage Command Interface Timing

B.1.1 Direct Servo Command Path Latency Test

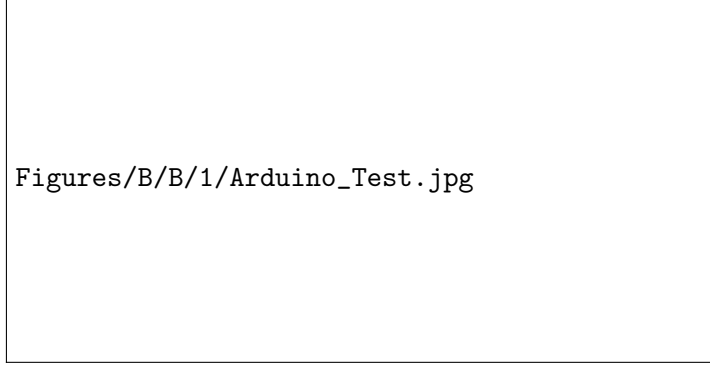


Fig. B.1. Hardware implementation of the direct servo command path latency test setup.

Algorithm B.1 Direct servo command path latency extraction

Input: Direct servo test firmware F_D , command schedule $\mathcal{U}$, capture set $\mathcal{Y}_D$, matching tolerances τ_D , and run mode r .

Output: Accepted transition set $\mathcal{E}_D$, latency segment table Λ_D , and rejection audit A_D .

```

1 Upload  $F_D$  and verify communication with the direct servo controller
2  $\mathcal{E}_D \leftarrow \emptyset$ ,  $A_D \leftarrow \emptyset$ 
3 if  $r = \text{new}$  then
4    $\mathcal{U} \leftarrow$  generate a repeatable surface transition schedule
5    $(\mathcal{G}_D, \mathcal{B}_D) \leftarrow$  execute  $\mathcal{U}$  through the direct servo command path
6    $\mathcal{Y}_D \leftarrow$  capture reference and servo output pulse trains
7 else
8    $(\mathcal{G}_D, \mathcal{B}_D, \mathcal{Y}_D, \mathcal{U}) \leftarrow$  load the selected recorded run
9  $\phi_D \leftarrow$  fit ground computer, controller, and capture clock maps from  $\mathcal{G}_D$ ,  $\mathcal{B}_D$ , and  $\mathcal{Y}_D$ 
10  $\mathcal{C}_D \leftarrow$  decode command transitions and servo output transitions under  $\phi_D$ 
11 for each command transition  $c_i \in \mathcal{C}_D$  do
12    $a_i \leftarrow$  unique controller apply event associated with  $c_i$ 
13    $y_i \leftarrow$  first admissible servo output transition after  $a_i$  within  $\tau_D$ 
14   if  $a_i$  and  $y_i$  satisfy uniqueness, ordering, and pulse consistency tests then
15      $\lambda_i \leftarrow \{t_{\text{rx}} - t_{\text{cmd}}, t_{\text{apply}} - t_{\text{rx}}, t_{\text{out}} - t_{\text{apply}}, t_{\text{out}} - t_{\text{sched}}\}$ 
16     Add  $(c_i, \lambda_i)$  to  $\mathcal{E}_D$ 
17   else
18     Add  $(c_i, \text{reason})$  to  $A_D$ 
19  $\Lambda_D \leftarrow$  aggregate latency segments from  $\mathcal{E}_D$  by surface and over all surfaces
20 return  $\mathcal{E}_D$ ,  $\Lambda_D$ , and  $A_D$ 

```

Table B.1 Per run stage-by-stage electrical timing for the direct servo command path. The n_{match} column denotes accepted matched transition events.

Run	n_{match}	Path segment	Unit	median	P_{95}	P_{99}	σ
1	914	Ground command to dispatch	ms	0.26	1.14	4.76	0.84
		Ground dispatch to controller receive	ms	21.6	24.0	24.4	1.33
		Controller receive to PWM update	ms	0.20	0.21	0.22	0.01
		PWM update to next servo pulse	ms	19.9	29.3	30.4	5.93
		Ground command to servo pulse	ms	42.3	51.7	52.7	5.90
2	940	Ground command to dispatch	ms	0.20	0.41	0.94	0.15
		Ground dispatch to controller receive	ms	21.7	34.1	35.6	8.14
		Controller receive to PWM update	ms	0.20	0.21	0.22	0.01
		PWM update to next servo pulse	ms	12.2	21.8	23.1	6.02
		Ground command to servo pulse	ms	34.2	51.7	56.8	10.3
3	910	Ground command to dispatch	ms	0.27	0.90	3.36	0.97
		Ground dispatch to controller receive	ms	22.6	32.3	33.2	5.32
		Controller receive to PWM update	ms	0.21	0.30	0.31	0.04
		PWM update to next servo pulse	ms	12.5	21.8	23.0	5.97
		Ground command to servo pulse	ms	36.5	50.7	54.3	8.14
4	956	Ground command to dispatch	ms	0.20	0.64	1.96	0.85
		Ground dispatch to controller receive	ms	27.5	28.8	29.1	1.15
		Controller receive to PWM update	ms	0.20	0.21	0.22	0.01
		PWM update to next servo pulse	ms	20.5	30.2	31.5	5.92
		Ground command to servo pulse	ms	48.6	58.1	58.9	5.86
5	932	Ground command to dispatch	ms	0.34	2.24	7.38	1.22
		Ground dispatch to controller receive	ms	21.5	33.1	33.5	6.20
		Controller receive to PWM update	ms	0.20	0.21	0.22	0.01
		PWM update to next servo pulse	ms	14.7	24.5	25.3	6.03
		Ground command to servo pulse	ms	38.1	52.0	57.3	8.25

B.1.2 COTS RC Transmitter-Receiver Command Chain Latency Test

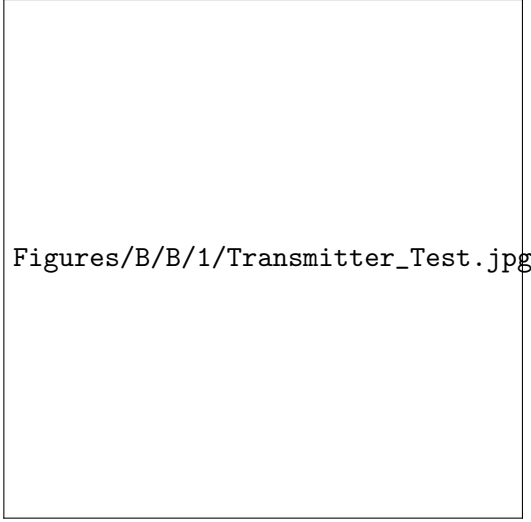


Fig. B.2. Hardware implementation of the COTS RC command chain latency test setup¹³.

Algorithm B.2 COTS RC transmitter-receiver command chain latency extraction

Input: COTS RC bridge firmware F_C , command schedule $\mathcal{U}$, trainer and receiver captures $\mathcal{Y}_C$, clock repair model ϕ_C , matching tolerances τ_C , and run mode r .

Output: Accepted COTS RC transition set $\mathcal{E}_C$, command chain latency table Λ_C , and audit A_C .

```

1 Upload  $F_C$  and verify the ground computer command link, trainer output, and receiver output
2  $\mathcal{E}_C \leftarrow \emptyset$ ,  $A_C \leftarrow \emptyset$ 
3 if  $r = \text{new}$  then
4    $\mathcal{U} \leftarrow$  generate a repeatable COTS RC transition schedule
5    $(\mathcal{G}_C, \mathcal{B}_C) \leftarrow$  execute  $\mathcal{U}$  through the RC command path
6    $\mathcal{Y}_C \leftarrow$  capture reference, trainer, and receiver pulse trains
7 else
8    $(\mathcal{G}_C, \mathcal{B}_C, \mathcal{Y}_C, \mathcal{U}) \leftarrow$  load the selected recorded run
9  $\phi_C \leftarrow$  repair the controller clock from ground computer dispatch, board receive, and board commit events
10  $\mathcal{C}_C \leftarrow$  decode scheduled, trainer output, and receiver output transitions under  $\phi_C$ 
11 for each scheduled transition  $c_i \in \mathcal{U}$  do
12    $b_i \leftarrow$  associated board receive and commit pair
13    $p_i \leftarrow$  associated trainer output transition
14    $q_i \leftarrow$  associated receiver output transition
15   if  $b_i$ ,  $p_i$ , and  $q_i$  satisfy uniqueness and  $\tau_C$  then
16      $\lambda_i \leftarrow \{t_{\text{rx}} - t_{\text{cmd}}, t_{\text{commit}} - t_{\text{rx}}, t_{\text{trainer}} - t_{\text{commit}}, t_{\text{receiver}} - t_{\text{trainer}}\}$ 
17     Add  $(c_i, \lambda_i)$  to  $\mathcal{E}_C$ 
18   else
19     Add  $(c_i, \text{reason})$  to  $A_C$ 
20  $\Lambda_C \leftarrow$  aggregate command chain latency segments from  $\mathcal{E}_C$  by surface and over all surfaces
21 Add clock repair residuals and rejected event counts to  $A_C$ 
22 return  $\mathcal{E}_C$ ,  $\Lambda_C$ , and  $A_C$ 

```

¹³ The transmitter was configured using the factory trainer port settings. For ACCESS-based FrSky transmitters running OpenTX, firmware version 2.3.15 or later is recommended for this setup, since the 2.3 branch includes a fix for trainer port detection on selected FrSky ACCESS radios, including the X7 Access and X9 Lite. The related OpenTX issue and fix are documented at <https://github.com/opentx/opentx/issues/8670> and <https://github.com/opentx/opentx/pull/8811>.

Table B.2 Per run stage-by-stage electrical timing for the COTS RC transmitter-receiver command chain. The n_{match} column denotes accepted matched transition events.

Run	n_{match}	Path segment	Unit	median	P_{95}	P_{99}	σ
1	919	Ground command to dispatch	ms	0.16	0.44	0.83	0.17
		Ground dispatch to Nano receive	ms	5.28	5.74	6.07	0.36
		Nano receive to PPM update	ms	16.5	17.1	17.2	0.43
		PPM update to next trainer frame	ms	4.39	20.0	20.1	8.10
		Trainer PPM to receiver PWM	ms	12.9	34.2	34.4	11.1
		Ground command to receiver PWM	ms	37.8	57.8	57.8	7.17
2	947	Ground command to dispatch	ms	0.16	0.47	1.15	0.17
		Ground dispatch to Nano receive	ms	6.64	7.07	7.26	0.26
		Nano receive to PPM update	ms	3.55	3.97	4.07	0.34
		PPM update to next trainer frame	ms	4.44	20.1	20.1	8.25
		Trainer PPM to receiver PWM	ms	6.98	28.5	48.6	12.9
		Ground command to receiver PWM	ms	20.5	40.6	60.6	9.28
3	912	Ground command to dispatch	ms	0.17	0.57	1.43	0.41
		Ground dispatch to Nano receive	ms	6.96	7.53	7.71	0.26
		Nano receive to PPM update	ms	11.6	11.9	12.0	0.52
		PPM update to next trainer frame	ms	4.44	20.1	20.2	8.23
		Trainer PPM to receiver PWM	ms	12.1	33.6	33.7	11.0
		Ground command to receiver PWM	ms	34.0	54.0	54.0	6.97
4	953	Ground command to dispatch	ms	0.17	0.72	1.57	0.58
		Ground dispatch to Nano receive	ms	7.07	7.49	7.68	0.22
		Nano receive to PPM update	ms	5.50	5.83	5.90	0.69
		PPM update to next trainer frame	ms	4.38	20.0	20.1	8.26
		Trainer PPM to receiver PWM	ms	5.39	43.8	46.9	15.7
		Ground command to receiver PWM	ms	21.1	61.0	61.2	12.8
5	941	Ground command to dispatch	ms	0.16	0.37	0.80	0.11
		Ground dispatch to Nano receive	ms	6.91	7.31	7.52	0.26
		Nano receive to PPM update	ms	14.6	15.0	15.1	0.30
		PPM update to next trainer frame	ms	1.48	4.48	4.52	1.63
		Trainer PPM to receiver PWM	ms	2.41	40.3	42.7	11.9
		Ground command to receiver PWM	ms	25.2	64.7	65.4	11.8

B.2 Vicon State Processing and Filtering

B.2.1 Vicon Rigid-Body Tracking Characterisation

Algorithm B.3 Rigid-body tracking quality and filter selection

Input: Rigid-body subject q , phase schedule $\mathcal{P}$, sample time Δt , candidate filters $\mathcal{H}$, and score weights w .

Output: Processed trajectory $\mathcal{X}_q$, phase statistics M_q , filter scores J_H , selected filter H^* , and tracking audit A_q .

```

1  $\mathcal{X}_q \leftarrow \emptyset, M_q \leftarrow \emptyset, J_H \leftarrow \emptyset, A_q \leftarrow \emptyset$ 
2 Connect the ground computer to the Vicon host PC through the Vicon DataStream SDK over TCP/IP
  and initialise the stream for subject  $q$ 
3 for each phase  $p_j \in \mathcal{P}$  do
4   for each sample time in phase  $p_j$  do
5      $x_k \leftarrow$  latest non-occluded pose of  $q$ 
6     if  $x_k$  is valid then
7       | Add  $(x_k, p_j)$  to  $\mathcal{X}_q$ 
8     else
9       | Add an occlusion record for  $p_j$  to  $A_q$ 
10 Close the Vicon DataStream connection and reject the run if  $\mathcal{X}_q = \emptyset$ 
11 Compute derivatives and high frequency residuals of  $\mathcal{X}_q$ 
12 for each phase  $p_j \in \mathcal{P}$  do
13   | Add sample-quality, noise, residual, velocity, and angular-rate metrics for  $p_j$  to  $M_q$ 
14 for each candidate filter  $H_i \in \mathcal{H}$  do
15   |  $\hat{\mathcal{X}}_i \leftarrow H_i(\mathcal{X}_q)$ 
16   |  $J_i \leftarrow$  weighted score of delay, static noise reduction, and dynamic motion preservation using  $w$ 
17   | Add  $J_i$  to  $J_H$ 
18  $H^* \leftarrow \arg \min_{H_i \in \mathcal{H}} J_i$  subject to the tracking quality and latency limits
19 return  $\mathcal{X}_q, M_q, J_H, H^*$ , and  $A_q$ 

```

B.3 Control Surface Response and Closed-Loop Timing

B.3.1 Command-to-Surface Latency Test

Algorithm B.4 Command-to-surface response latency analysis

Input: Surface command schedule $\mathcal{B}$, sample time Δt , command interface $\mathcal{I}$, Vicon measurement stream $\mathcal{Y}_S$, state filter H , and crossing fractions Γ .

Output: Accepted response event set $\mathcal{E}_S$, latency table Λ_S , and rejection audit A_S .

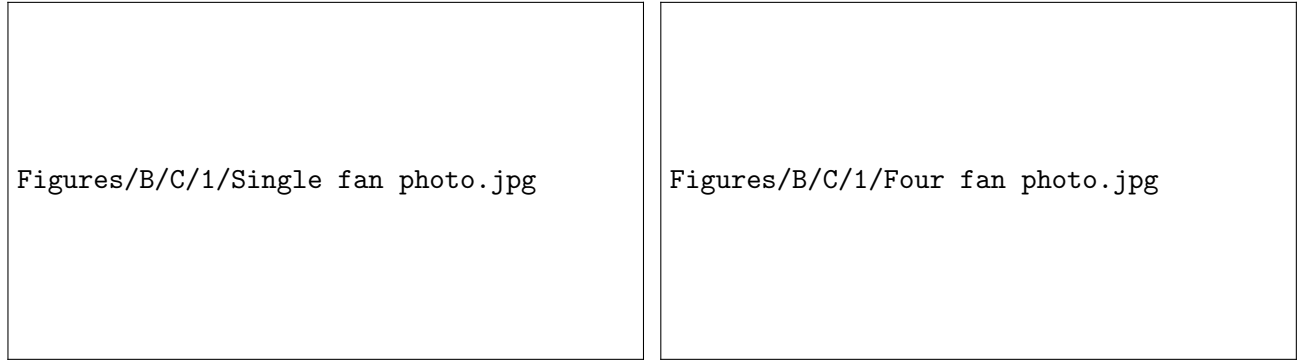
```

1  $\mathcal{X} \leftarrow \emptyset, \mathcal{U}_S \leftarrow \emptyset, \mathcal{E}_S \leftarrow \emptyset, A_S \leftarrow \emptyset$ 
2 Initialise  $\mathcal{I}$ , set all surfaces to neutral, connect the ground computer to the Vicon host PC through the
  Vicon DataStream SDK over TCP/IP, and estimate the neutral state  $x_0$ 
3 for each sample time  $t_k = 0, \Delta t, \dots, T$  do
4    $y_k \leftarrow$  latest Vicon rigid-body and surface measurements from  $\mathcal{Y}_S$ 
5    $x_k \leftarrow H(y_k)$ 
6   Add  $x_k$  to  $\mathcal{X}$ 
7   if an event  $b_i \in \mathcal{B}$  is scheduled at  $t_k$  then
8      $u_i \leftarrow$  command vector for event  $b_i$ 
9     Write  $u_i$  through  $\mathcal{I}$  and add  $(b_i, u_i, t_k, t_{\text{write}})$  to  $\mathcal{U}_S$ 
10 Set all surfaces to neutral and close the command interface and Vicon DataStream connection
11 for each logged command event  $(b_i, u_i, t_i) \in \mathcal{U}_S$  do
12    $W_0 \leftarrow$  pre-command baseline window from  $\mathcal{X}$ 
13    $W_1 \leftarrow$  response and endpoint windows from  $\mathcal{X}$ 
14   if  $W_0$  and  $W_1$  are valid and the measured response is large enough then
15      $t_\gamma \leftarrow$  first crossing time for each response fraction  $\gamma \in \Gamma$ 
16     Add  $(b_i, u_i, t_\gamma - t_i)$  to  $\mathcal{E}_S$ 
17   else
18     Add  $(b_i, \text{reason})$  to  $A_S$ 
19  $\Lambda_S \leftarrow$  aggregate command-to-surface crossing latencies from  $\mathcal{E}_S$ 
20 return  $\mathcal{E}_S, \Lambda_S$ , and  $A_S$ 

```

C Experimental Characterisation of the Fan-Generated Updraft

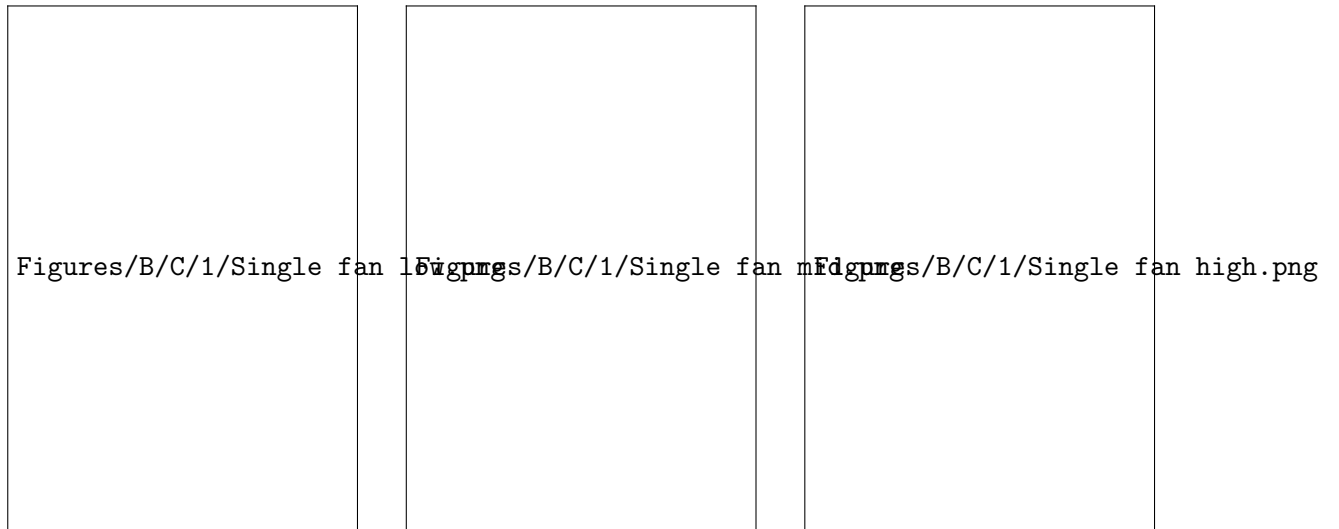
C.1 Test Facility and Measurement Configurations



(a) with a single axial drum fan.

(b) with four axial drum fans.

Fig. C.1. Experimental layout for indoor updraft measurements.



(a) outlet height: 0.245 m,
support offset: 0.040 m.

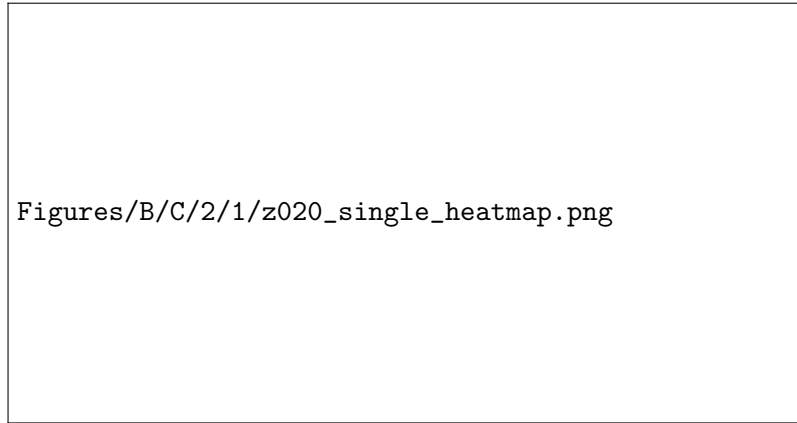
(b) outlet height: 0.330 m,
support offset: 0.125 m.

(c) outlet height: 0.425 m,
support offset: 0.220 m.

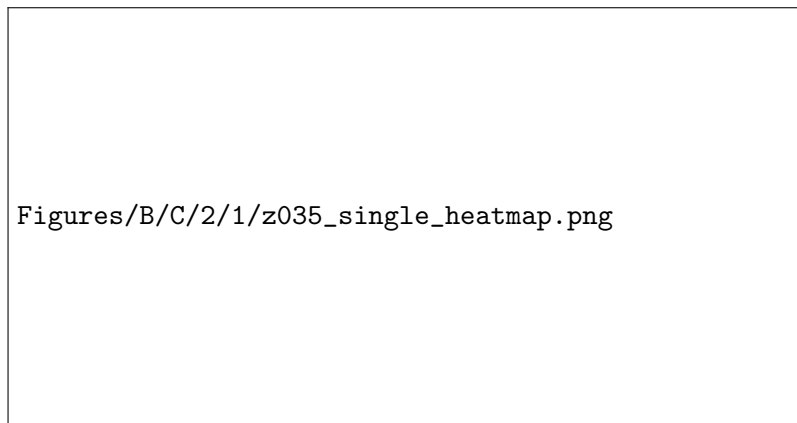
Fig. C.2. Single fan configuration with varying outlet heights for ground-effect assessment.

C.2 Spatial Field Measurements

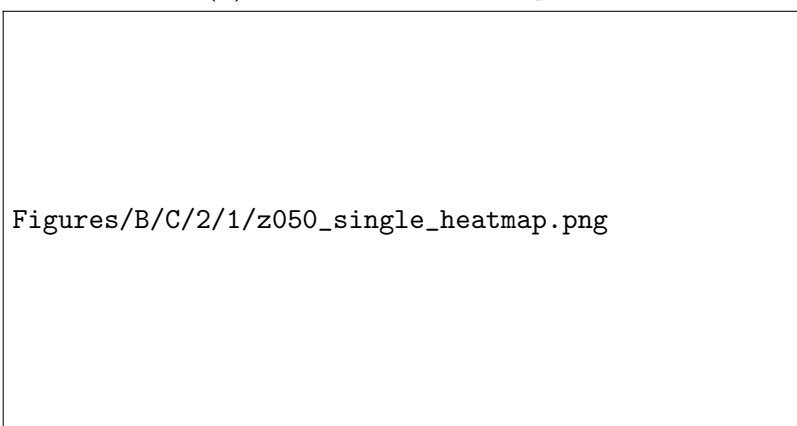
C.2.1 Single-Fan Case



(a) 0.20 m above fan outlet plane.



(b) 0.35 m above fan outlet plane.



(c) 0.50 m above fan outlet plane.

Fig. C.3. Spatial distribution of mean vertical velocity at lower measurement heights (single fan).

Figures/B/C/2/1/z075_single_heatmap.png

(a) 0.75 m above fan outlet plane.

Figures/B/C/2/1/z110_single_heatmap.png

(b) 1.10 m above fan outlet plane.

Figures/B/C/2/1/z160_single_heatmap.png

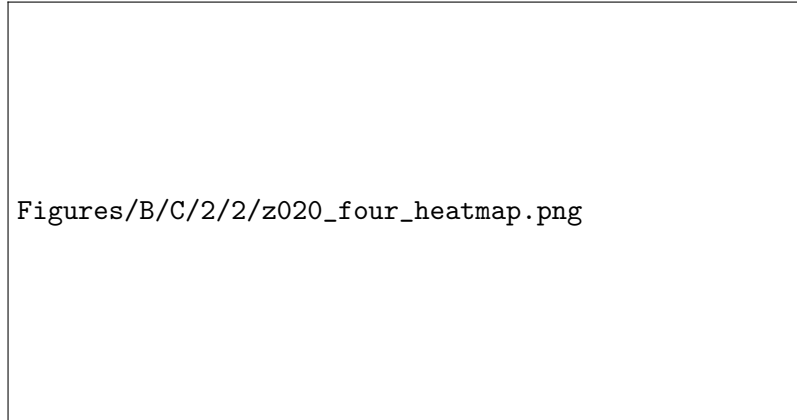
(c) 1.60 m above fan outlet plane.

Figures/B/C/2/1/z220_single_heatmap.png

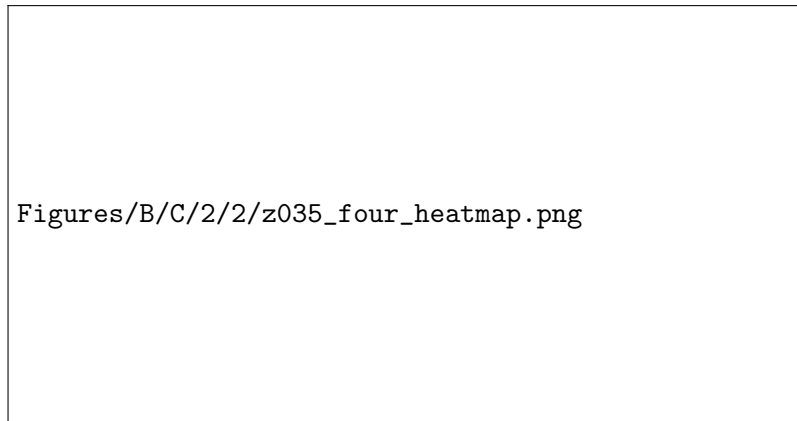
(d) 2.20 m above fan outlet plane.

Fig. C.4. Spatial distribution of mean vertical velocity at higher measurement heights (single fan).

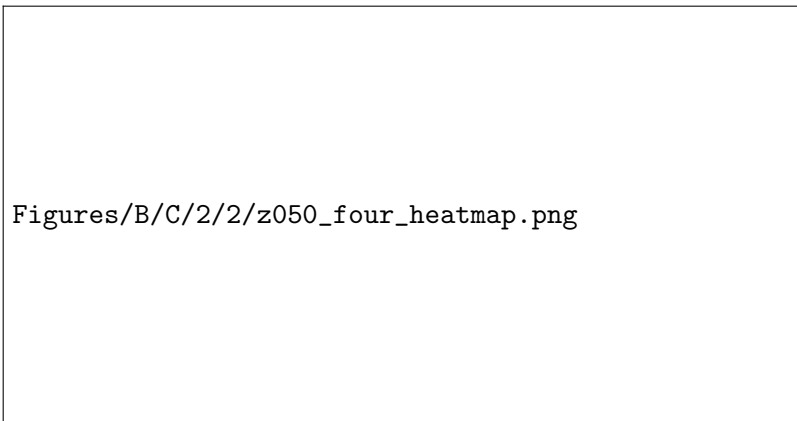
C.2.2 Four-Fan Case



(a) 0.20 m above fan outlet plane.

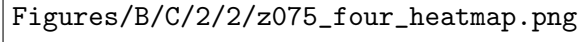


(b) 0.35 m above fan outlet plane.



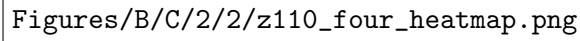
(c) 0.50 m above fan outlet plane.

Fig. C.5. Spatial distribution of mean vertical velocity at lower measurement heights (four fans).



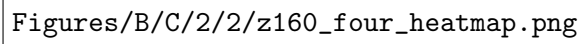
Figures/B/C/2/2/z075_four_heatmap.png

(a) 0.75 m above fan outlet plane.



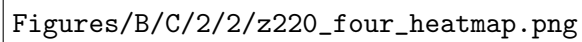
Figures/B/C/2/2/z110_four_heatmap.png

(b) 1.10 m above fan outlet plane.



Figures/B/C/2/2/z160_four_heatmap.png

(c) 1.60 m above fan outlet plane.



Figures/B/C/2/2/z220_four_heatmap.png

(d) 2.20 m above fan outlet plane.

Fig. C.6. Spatial distribution of mean vertical velocity at higher measurement heights (four fans).

C.3 Supplementary Measurements at Sampling Points

C.3.1 Single-Fan Case

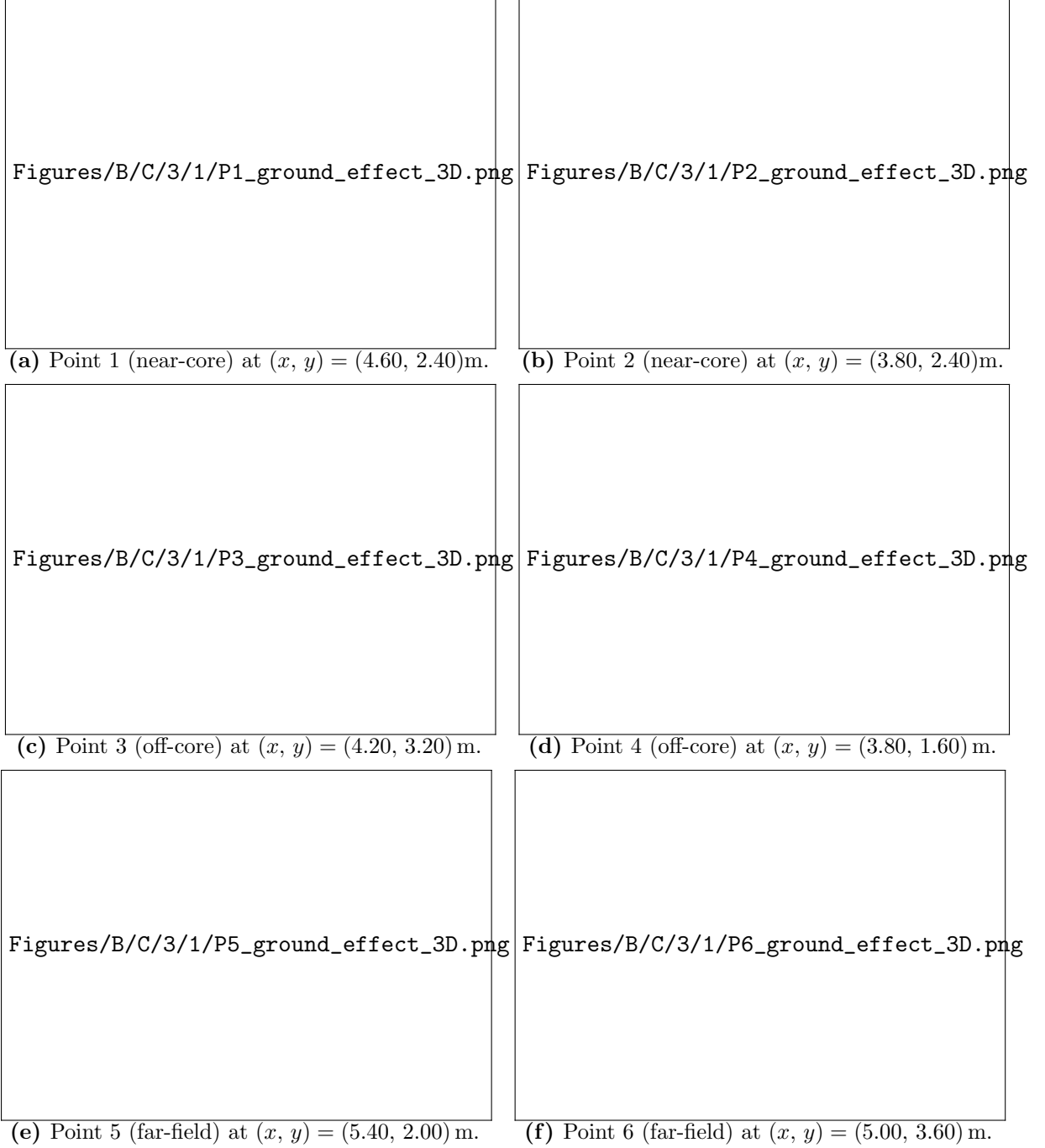
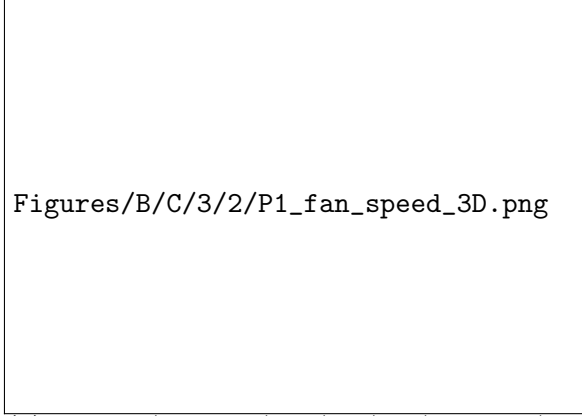
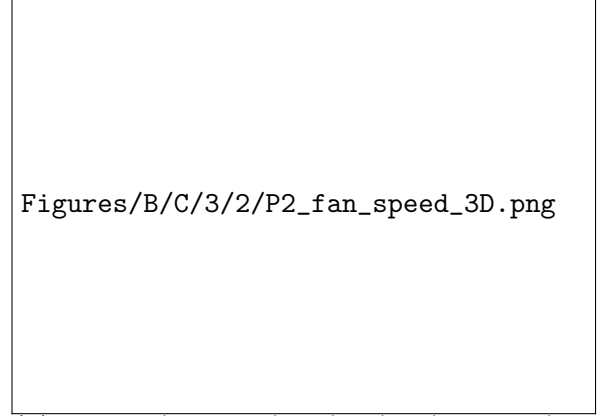


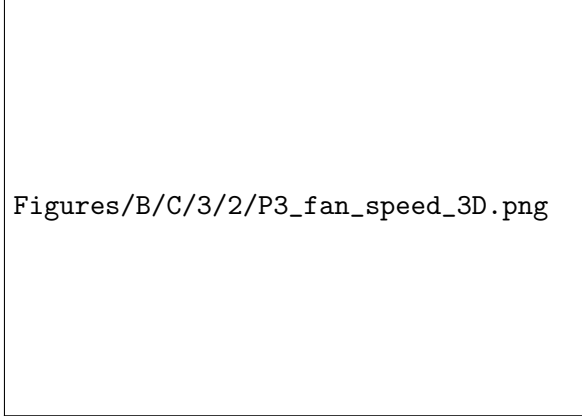
Fig. C.7. Vertical velocity profiles at six sampling points for three fan outlet plane heights (single fan). The solid line shows the mean, semi-transparent markers show the 15 samples (1 Hz over 15 s) at each measurement height, and shaded bands show $\pm 1\sigma$.



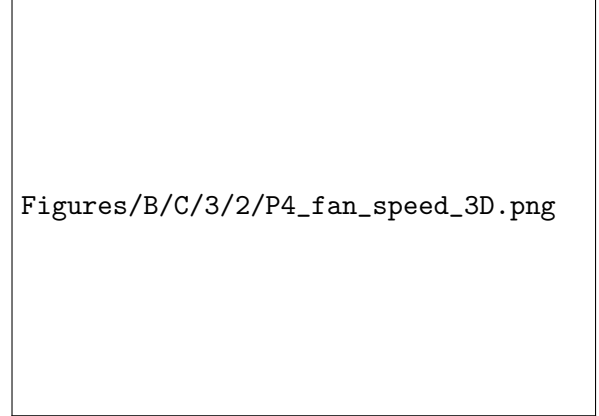
(a) Point 1 (near-core) at $(x, y) = (4.60, 2.40)$ m.



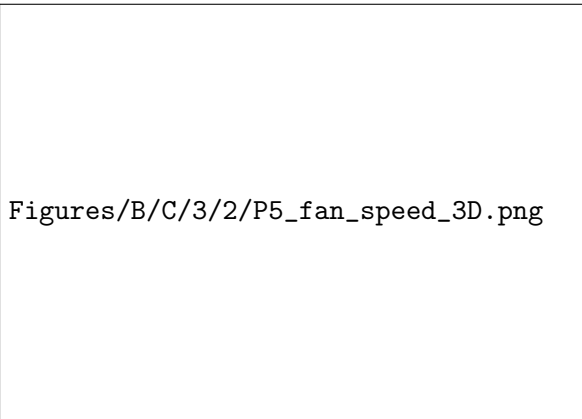
(b) Point 2 (near-core) at $(x, y) = (3.80, 2.40)$ m.



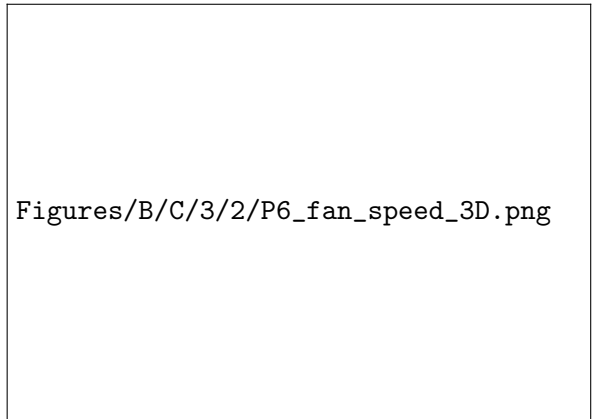
(c) Point 3 (off-core) at $(x, y) = (4.20, 3.20)$ m.



(d) Point 4 (off-core) at $(x, y) = (3.80, 1.60)$ m.



(e) Point 5 (far-field) at $(x, y) = (5.40, 2.00)$ m.



(f) Point 6 (far-field) at $(x, y) = (5.00, 3.60)$ m.

Fig. C.8. Vertical velocity profiles at six sampling points for two fan speed settings (single fan). The solid line shows the mean, semi-transparent markers show the 15 samples (1 Hz over 15 s) at each measurement height, and shaded bands show $\pm 1\sigma$.

C.3.2 Four-Fan Case

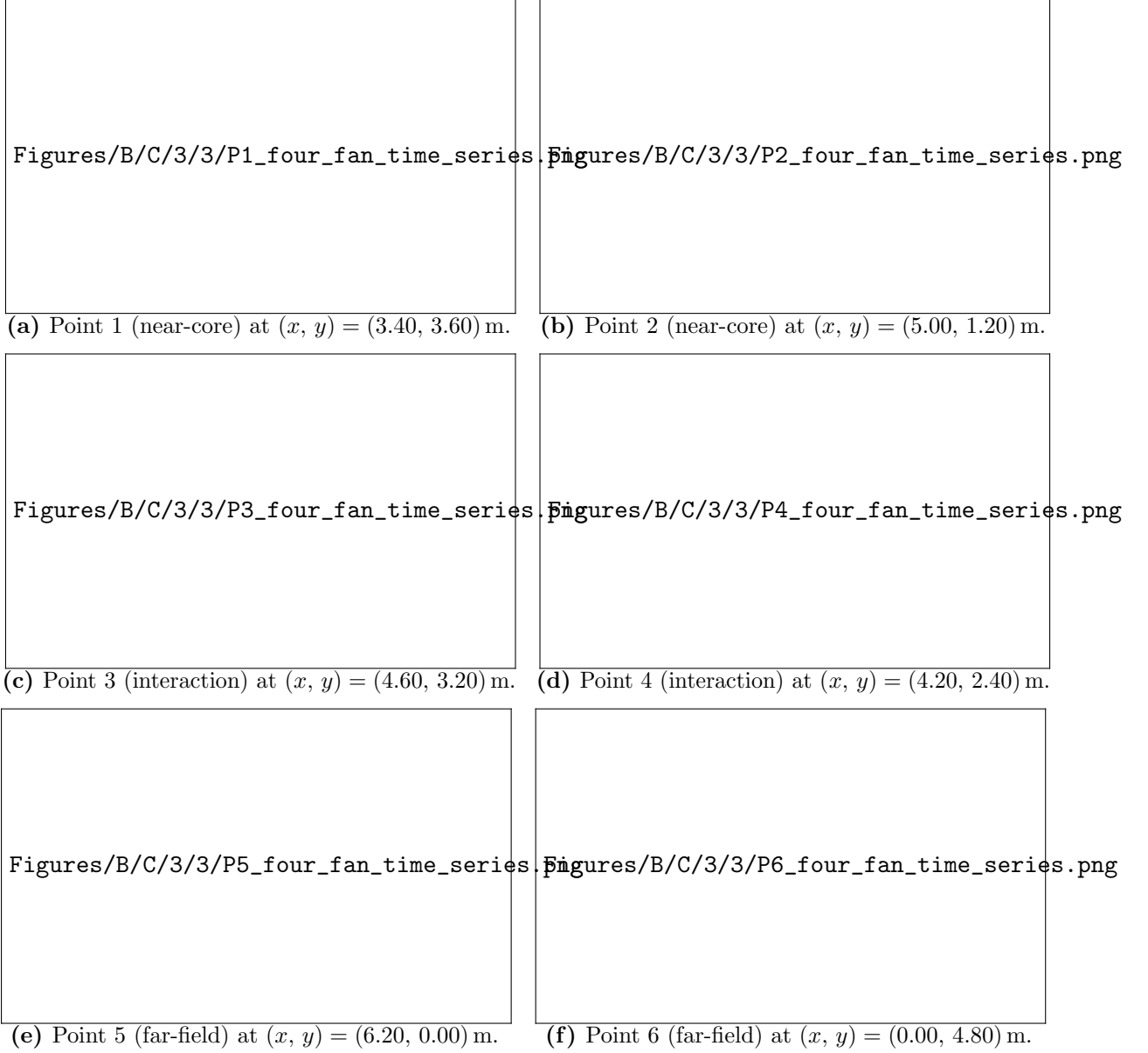


Fig. C.9. Vertical velocity profiles at six sampling points (four fans). The solid line shows the mean, semi-transparent markers show the 15 samples (1 Hz over 15 s) at each measurement height, and shaded bands show $\pm 1\sigma$.

D Supplementary Updraft Surrogate Fits at Measurement Planes

D.1 Empirical Fluctuation Fields Used for Variance Assignment

D.1.1 Single-Fan Empirical Fluctuation Field

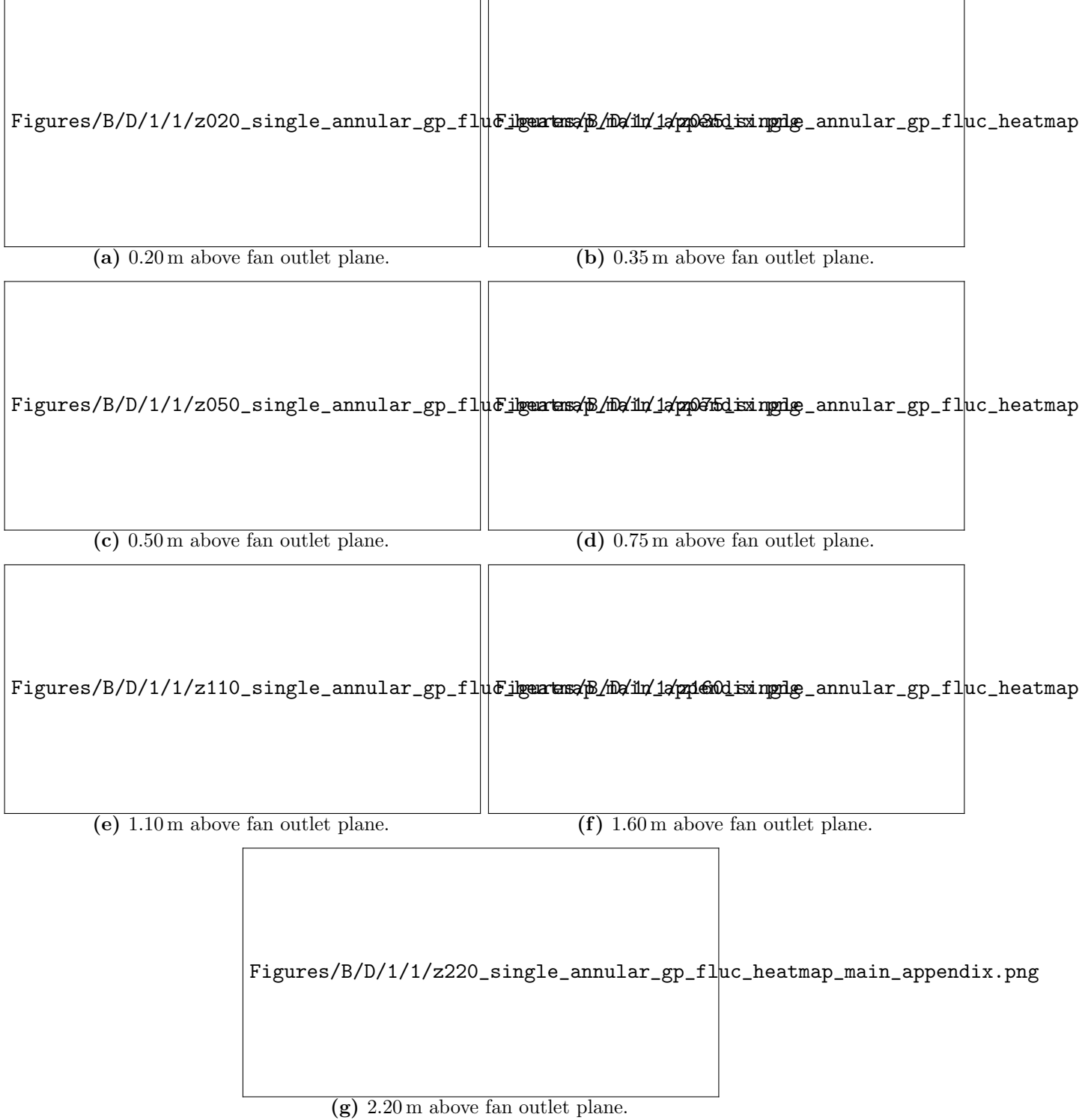


Fig. D.1. Supplementary empirical fluctuation field, σ_i (single fan). The grey dotted rings indicate the annular regions used to assign the empirical fluctuation level, σ_i , as described at the beginning of [Section 4.3](#).

D.1.2 Four-Fan Empirical Fluctuation Field

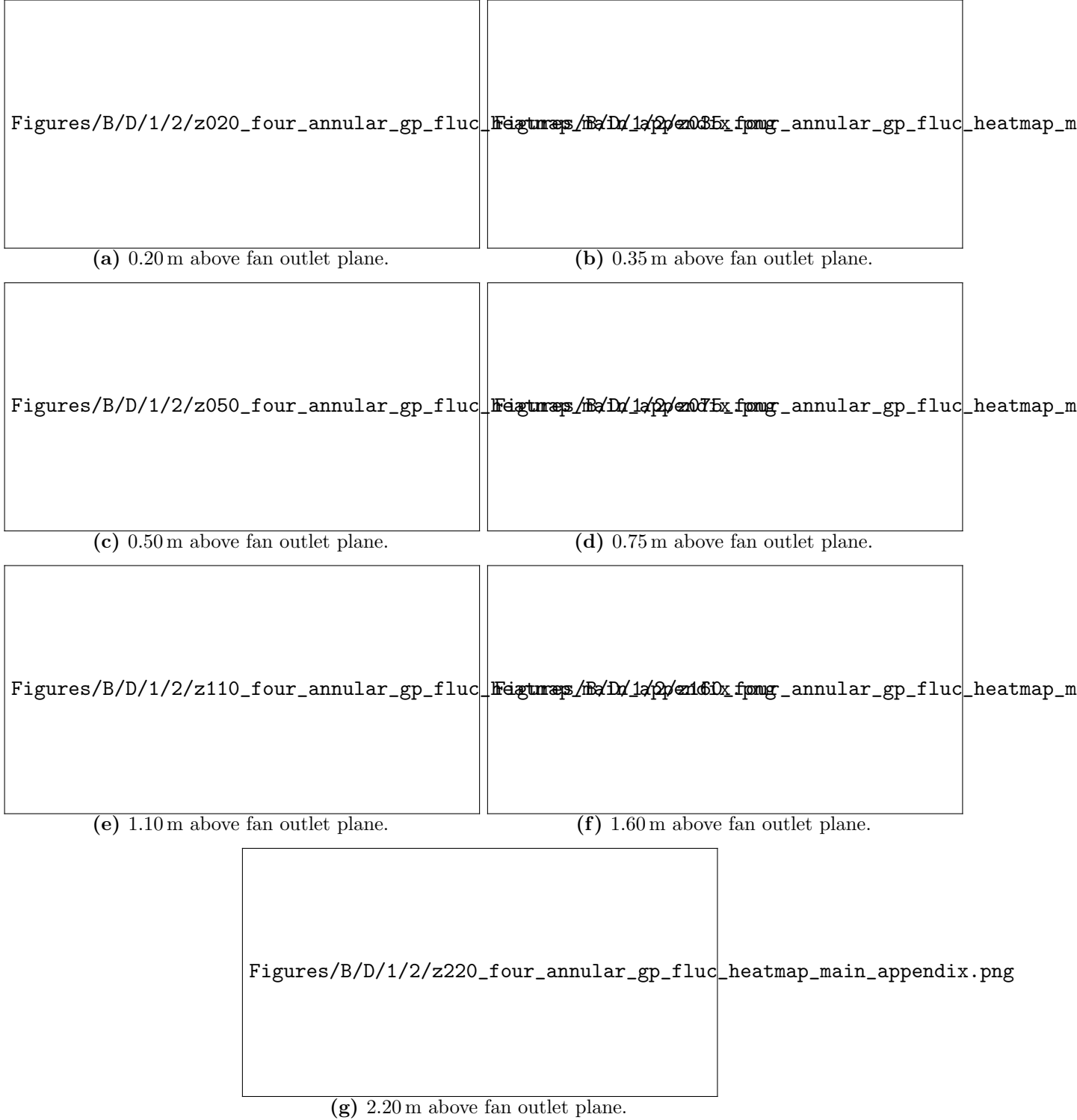


Fig. D.2. Supplementary empirical fluctuation field, σ_i (four fans). The grey dotted rings indicate the annular regions used to assign the empirical fluctuation level, σ_i , as described at the beginning of [Section 4.3](#).

D.2 Axisymmetric Gaussian Plume Baseline Fits

D.2.1 Single-Fan Axisymmetric Gaussian Plume Baseline



Fig. D.3. Supplementary spatial distribution of fitted mean vertical velocity from the Gaussian plume baseline model (single fan).

D.2.2 Four-Fan Axisymmetric Gaussian Plume Baseline



Fig. D.4. Supplementary spatial distribution of fitted mean vertical velocity from the Gaussian plume baseline model (four fans).

D.3 Axisymmetric Annular–Gaussian Surrogate Fits

D.3.1 Single-Fan Axisymmetric Annular–Gaussian Surrogate



Fig. D.5. Supplementary spatial distribution of fitted mean vertical velocity from the axisymmetric annular–Gaussian surrogate (single fan).

D.3.2 Four-Fan Axisymmetric Annular–Gaussian Surrogate



Fig. D.6. Supplementary spatial distribution of fitted mean vertical velocity from the axisymmetric annular–Gaussian surrogate (four fans).

D.4 Harmonic Annular–Gaussian Surrogate Fits

D.4.1 Single-Fan Harmonic Annular–Gaussian Surrogate



Fig. D.7. Supplementary spatial distribution of fitted mean vertical velocity from the harmonic annular–Gaussian surrogate (single fan).

D.4.2 Four-Fan Harmonic Annular–Gaussian Surrogate



Fig. D.8. Supplementary spatial distribution of fitted mean vertical velocity from the harmonic annular–Gaussian surrogate (four fans).

D.5 Residual Annular Gaussian Process Surrogate Predictions

D.5.1 Single-Fan Residual Annular Gaussian Process Surrogate



Fig. D.9. Supplementary spatial distribution of fitted mean vertical velocity from the residual annular Gaussian process surrogate (single fan).

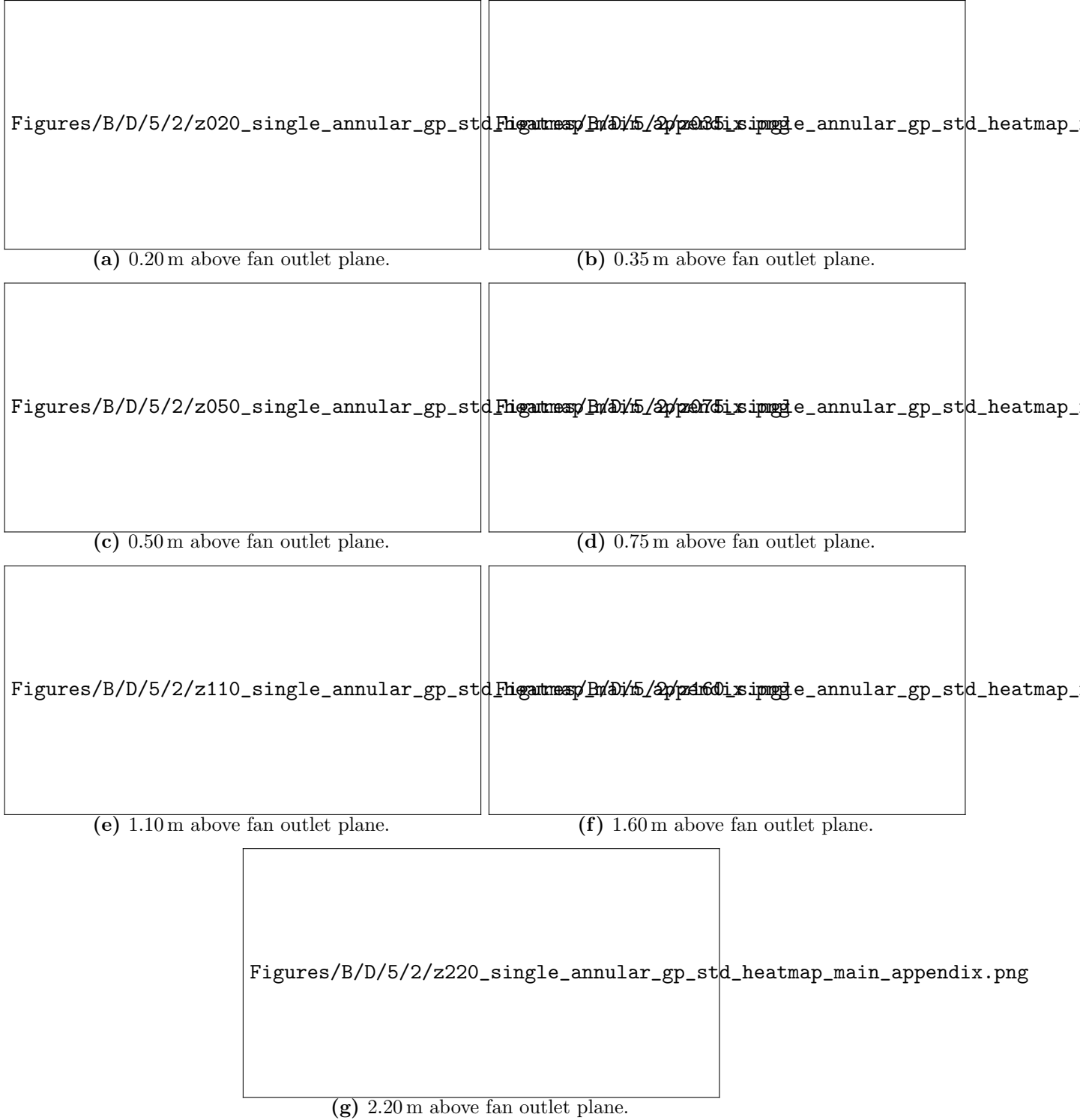


Fig. D.10. Supplementary predictive standard deviation field, σ_{res} , of the residual annular Gaussian process surrogate (single fan).



Fig. D.11. Supplementary total fluctuation field, $\sigma_{\text{HAG-GP}}$, of the residual annular Gaussian process surrogate (single fan). The grey dotted rings indicate the annular regions used to assign the empirical fluctuation level, σ_i , as described at the beginning of [Section 4.3](#).

D.5.2 Four-Fan Residual Annular Gaussian Process Surrogate



Fig. D.12. Supplementary spatial distribution of fitted mean vertical velocity from the residual annular Gaussian process surrogate (four fans).



Fig. D.13. Supplementary predictive standard deviation field, σ_{res} , of the residual annular Gaussian process surrogate (four fans).



Fig. D.14. Supplementary total fluctuation field, $\sigma_{\text{HAG-GP}}$, of the residual annular Gaussian process surrogate (four fans). The grey dotted rings indicate the annular regions used to assign the empirical fluctuation level, σ_i , as described at the beginning of [Section 4.3](#).

E Glider Design and Manufacturing Details

E.1 Glider Optimisation Workflow

E.1.1 Pseudocode for Optimisation Workflow

Algorithm E.1 Nominal AeroSandbox [69] glider design optimisation

Input: Design bounds $\mathcal{B}$, fixed parameters θ , mission condition ξ_0 , objective J , and requirements $\mathcal{C}$.

Output: Feasible design d^* , solved decision vector z^* , and feasibility audit A^* .

```

1 Define  $z = \{\alpha, \delta_a, \delta_e, \delta_r, b, c, l_t, b_H, b_V\}$  with  $z \in \mathcal{B}$ 
2 For a candidate  $z$ , construct  $G(z; \theta)$ ,  $M(z; \theta)$ , and the trim model at  $\xi_0$ 
3 Define  $J(z)$  and  $\mathcal{C}(z)$  using the objective terms and requirements in Tables E.2 and E.3
4  $z^\dagger \leftarrow \arg \min_{z \in \mathcal{B}} J(z)$  subject to  $\mathcal{C}(z) \leq 0$ 
5  $A^\dagger \leftarrow$  post-solve feasibility audit of  $z^\dagger$ 
6 if the solver converges and  $A^\dagger$  passes then
7   |  $z^* \leftarrow z^\dagger$ ,  $d^* \leftarrow d(z^*)$ , and  $A^* \leftarrow A^\dagger$ 
8 else
9   |  $z^* \leftarrow \emptyset$ ,  $d^* \leftarrow \emptyset$ , and  $A^* \leftarrow A^\dagger$ 
10 return  $d^*$ ,  $z^*$ , and  $A^*$ , or an infeasibility flag

```

Algorithm E.2 Multistart candidate selection and robust postcheck

Input: Initial guesses $\{z_0^{(i)}\}_{i=1}^{N_s}$, retained count K , uncertainty scenarios $\Omega = \{\omega_j\}_{j=1}^{N_\omega}$, and Algorithm E.1.

Output: Robust selected design d^* , selected candidate index i^* , and robust audit R^* .

```

1  $\mathcal{F} \leftarrow \emptyset$ 
2 for each initial guess  $z_0^{(i)}$  do
3   | Solve Algorithm E.1 from  $z_0^{(i)}$  to obtain  $(d_i, z_i, A_i)$ 
4   | if  $d_i \neq \emptyset$  and  $A_i$  passes then
5     | | Add  $(d_i, z_i, J_i, A_i)$  to  $\mathcal{F}$ 
6 if  $\mathcal{F} = \emptyset$  then
7   | return a failure flag
8 Remove duplicate geometries from  $\mathcal{F}$ , sort by nominal objective, and retain the best  $K$  candidates
9 for each retained candidate  $d_i \in \mathcal{F}$  do
10  |  $\mathcal{M}_i \leftarrow \emptyset$ 
11  | for each uncertainty scenario  $\omega_j \in \Omega$  do
12    | | Apply  $\omega_j$  to  $d_i$ , re-solve trim, and compute scenario feasibility and performance metrics
13    | | Add the scenario result to  $\mathcal{M}_i$ 
14  |  $R_i \leftarrow$  robust audit from  $\mathcal{M}_i$ 
15  $i^* \leftarrow \arg \min_i R_i$  using success rate, tail sink metric, and nominal objective as ordered ranking criteria
16 return  $d^* = d_{i^*}$ ,  $i^*$ , and  $R^* = R_{i^*}$ 

```

Algorithm E.3 Finite difference sensitivity analysis

Input: Selected design d^* , parameter set $\mathcal{P}$, output metrics $\mathcal{Q}$, step ladder $\mathcal{H}$, and stability tolerances.

Output: Sensitivity matrix $\mathcal{S}$, selected step table H^* , and stability audit A_S .

```

1  $\mathcal{S} \leftarrow \emptyset$ ,  $H^* \leftarrow \emptyset$ ,  $A_S \leftarrow \emptyset$ 
2 Evaluate the baseline model for  $d^*$  to obtain  $q_j^0$  for all  $q_j \in \mathcal{Q}$ 
3 for each parameter  $p_i \in \mathcal{P}$  do
4   Select a central-difference scheme, or a forward scheme if a lower bound is active
5   Build the admissible step set  $\mathcal{H}_i \subseteq \mathcal{H}$ 
6   for each step  $h_{i,\ell} \in \mathcal{H}_i$  do
7     Evaluate feasible perturbed designs at  $p_i \pm h_{i,\ell}$ , or  $p_i + h_{i,\ell}$  for a forward scheme
8     Store the perturbed metrics and feasibility flags
9   for each output metric  $q_j \in \mathcal{Q}$  do
10     $D_{ij} \leftarrow$  derivative candidates from feasible perturbation pairs
11     $e_{ij} \leftarrow$  truncation and round-off error estimates for  $D_{ij}$ 
12    if at least one non-round-off step is defensible then
13       $h_{ij}^* \leftarrow \arg \min_{h_{i,\ell}} e_{ij}$ 
14       $\mathcal{S}_{ij} \leftarrow (p_i / \max(|q_j^0|, q_{j,\min})) \partial q_j / \partial p_i$  at  $h_{ij}^*$ 
15      Add  $(p_i, q_j, h_{ij}^*, \text{stable})$  to  $A_S$ 
16    else
17       $\mathcal{S}_{ij} \leftarrow \text{NaN}$ 
18      Add  $(p_i, q_j, \emptyset, \text{unresolved})$  to  $A_S$ 
19 return  $\mathcal{S}$ ,  $H^*$ , and  $A_S$ 

```

E.1.2 Optimisation Inputs and Robustness Settings

Table E.1 Design variables and bounds used in the nominal optimisation.

Quantity	Symbol	Unit	Bounds
Trim angle of attack	α_{trim}	deg	$[-4.0, 8.0]$
Aileron deflection	δ_a	deg	$[-30, 30]$
Elevator deflection	δ_e	deg	$[-30, 30]$
Rudder deflection	δ_r	deg	$[-30, 30]$
Wing span	b	m	$[0.30, 0.75]$
Wing chord	c	m	$[0.10, 0.30]$
Tail arm	l_t	m	$[0.40, 0.80]$
Horizontal tail span	b_H	m	$[0.20, 0.50]$
Vertical tail height	b_V	m	$[0.10, 0.30]$
Tail boom length	l_b	m	$[0.35, 0.70]$

Table E.2 Principal feasibility limits enforced in the optimiser.

Quantity	Symbol	Unit	Limit
Pitching moment tolerance	$ C_m $	-	1×10^{-6}
Maximum nominal lift coefficient	$C_{L,\max}$	-	1.2
Minimum glide ratio	L/D	-	6.0
Minimum wing Reynolds number	Re_c	-	2×10^4
Wing loading interval	W/S	N m ⁻²	$[2.0, 20.0]$
Static margin interval	SM	MAC	$[0.05, 0.10]$
Horizontal tail volume	V_H	-	$[0.50, 1.00]$
Vertical tail volume	V_V	-	$[0.03, 0.08]$
Roll damping magnitude	$ C_{\ell_p} $	-	$\geq 1 \times 10^{-4}$
Servo torque rating	T_s	N m	0.009

Table E.3 Objective terms and weights used in the nominal optimisation.

Objective	Symbol	Scale	Weight
Sink speed	J_{sink}	1.0 m s^{-1}	1.000
Mass	J_m	0.10 kg	0.04699
Trim effort	J_{trim}	10°	0.01326
Wing deflection	J_{δ_w}	–	2.152
Horizontal tail deflection	J_{δ_h}	–	0.07831
Roll time constant	J_{τ_p}	0.45 s	0.2471
Mass included in penalty		all aircraft mass	

Table E.4 Solver and candidate-selection settings used by the workflow.

Setting	Value
Solver	IPOPT
Maximum iterations	1000
Hessian treatment	limited memory
Multistart solves	30
Retained candidates	10
Multistart seed	20
Robust postcheck scenarios	100
Scenario seed	30
Geometry deduplication	[0.010, 0.005, 0.010] m
Active set tolerance	1.0×10^{-3}

Table E.5 Uncertainty intervals used in the robust postcheck.

Perturbation	Unit	Interval
Mass scale	–	[1.00, 1.10]
Longitudinal CG shift	MAC	[−0.06, 0.06]
Incidence bias	deg	[−2.0, 2.0]
Control effectiveness	–	[0.85, 1.00]
Control neutral bias	deg	[−3.0, 3.0]
Principal inertia scale	–	[0.90, 1.10]
Surface stiffness scale	–	[0.80, 1.30]
Nominal vertical gust	m s^{-1}	[−0.30, 0.30]
Turn vertical gust	m s^{-1}	[−0.20, 0.20]
Drag scale	–	[1.00, 1.25]

Table E.6 Robust ranking and rate-check policy used after the nominal solve.

Setting	Value
Selection order	success rate, tail risk, objective
Tail-risk fraction	20%
Default robust-in-loop cases	8
Robust-in-loop case interval	4–8
Scenario families	mild, harsh, gust, compound
Nominal trim time	1.5 s
Bank entry time	0.20 s
Rate utilisation allowance	0.80
Gust time constant	0.50 s
Robust gust flag	on

E.2 Optimised Glider Design Iterations

E.2.1 First Iteration

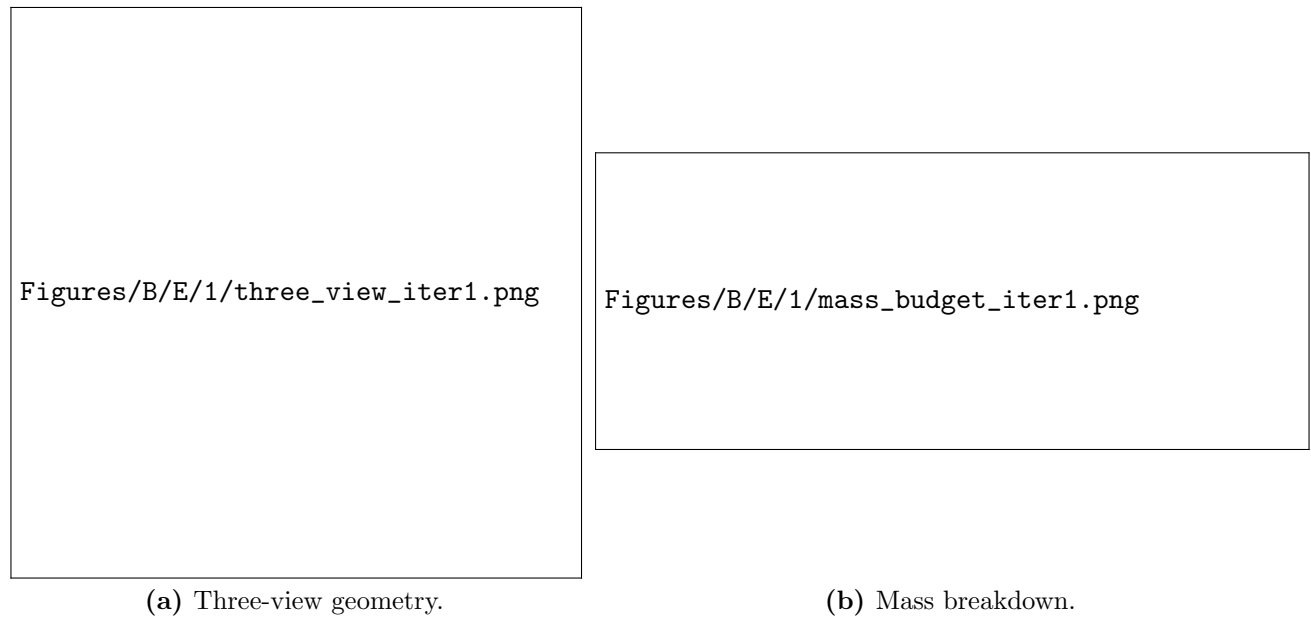


Fig. E.1. First iteration optimised glider configuration.

E.2.2 Second Iteration

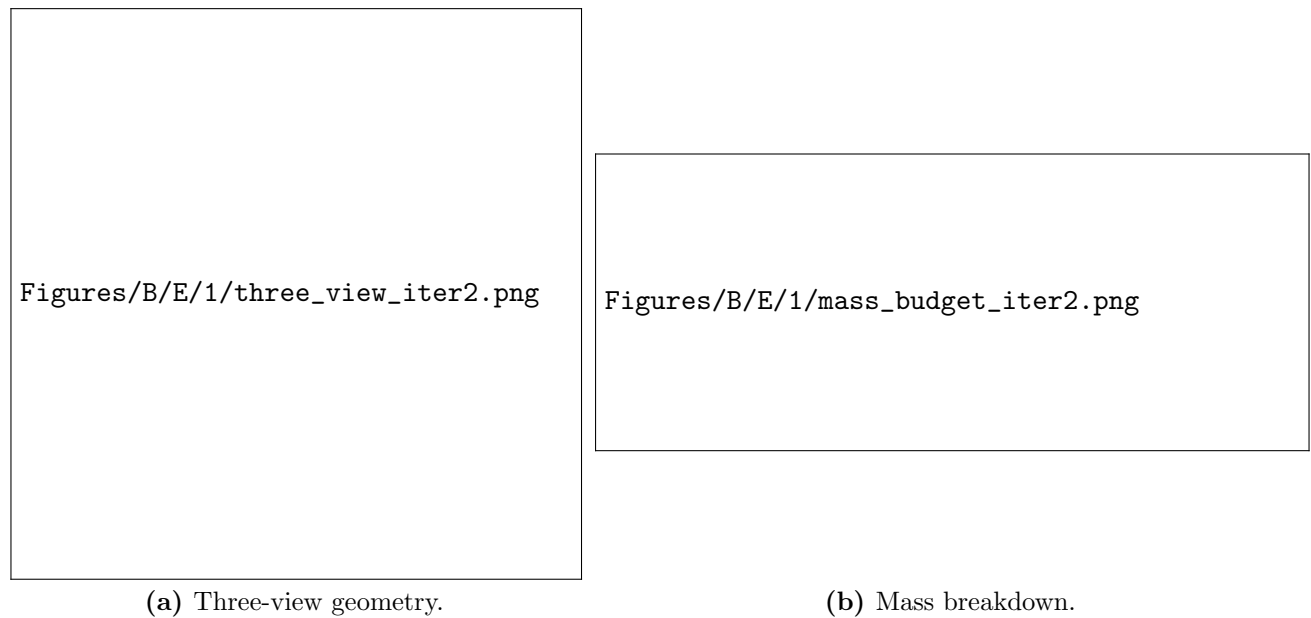
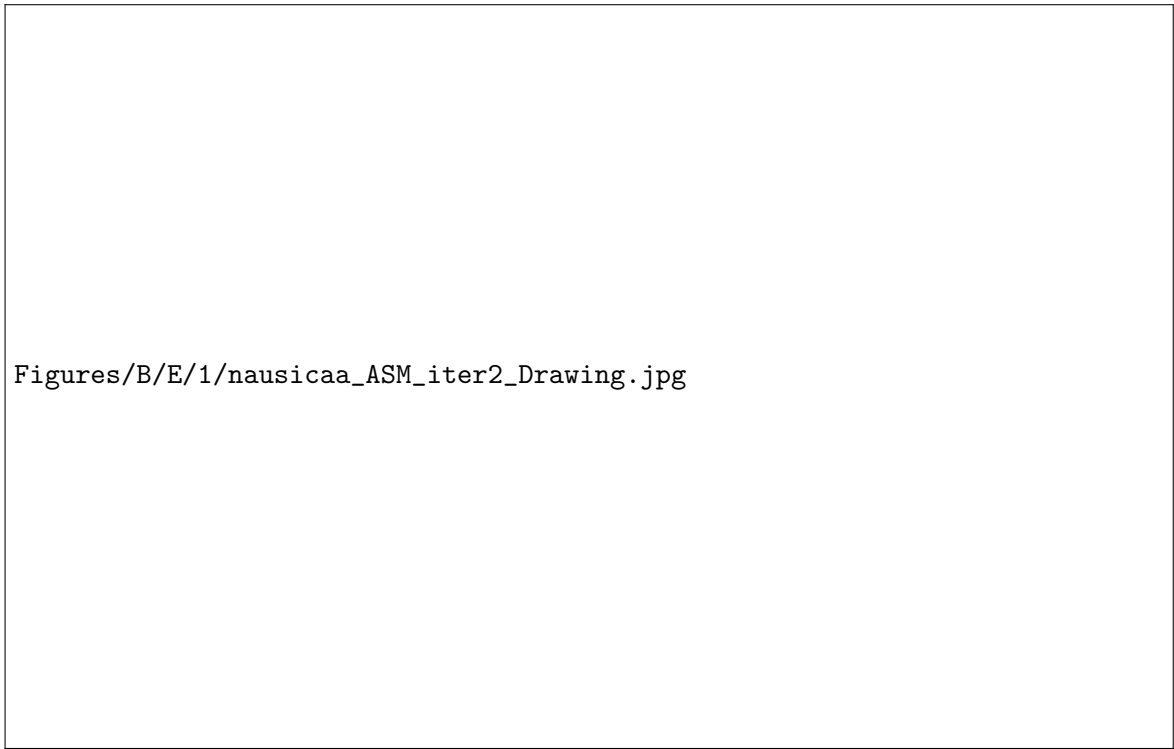


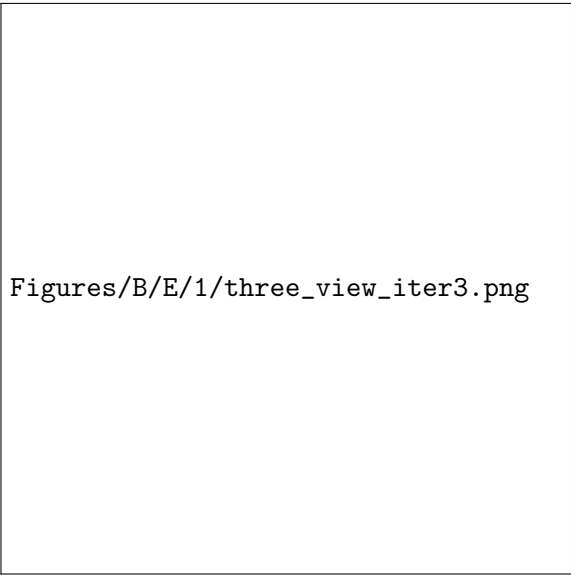
Fig. E.2. Second iteration optimised glider configuration.



Figures/B/E/1/nausicaa_ASM_iter2_Drawing.jpg

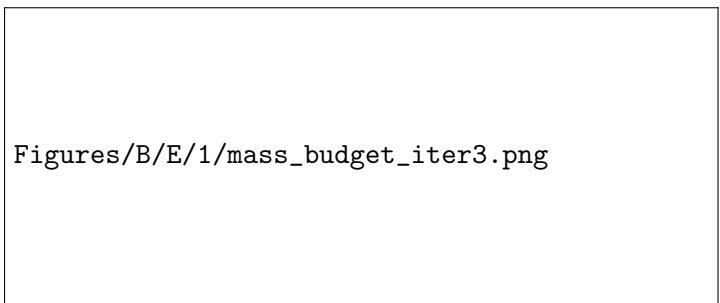
Fig. E.3. Detail engineering drawing of the second iteration glider assembly, corresponding to a half-span of 0.428 m and chord length of 0.207 m. Key 3D-printed mounting and skid components are shown; avionics, wiring, and tape reinforcements are omitted for clarity.

E.2.3 Third Iteration



Figures/B/E/1/three_view_iter3.png

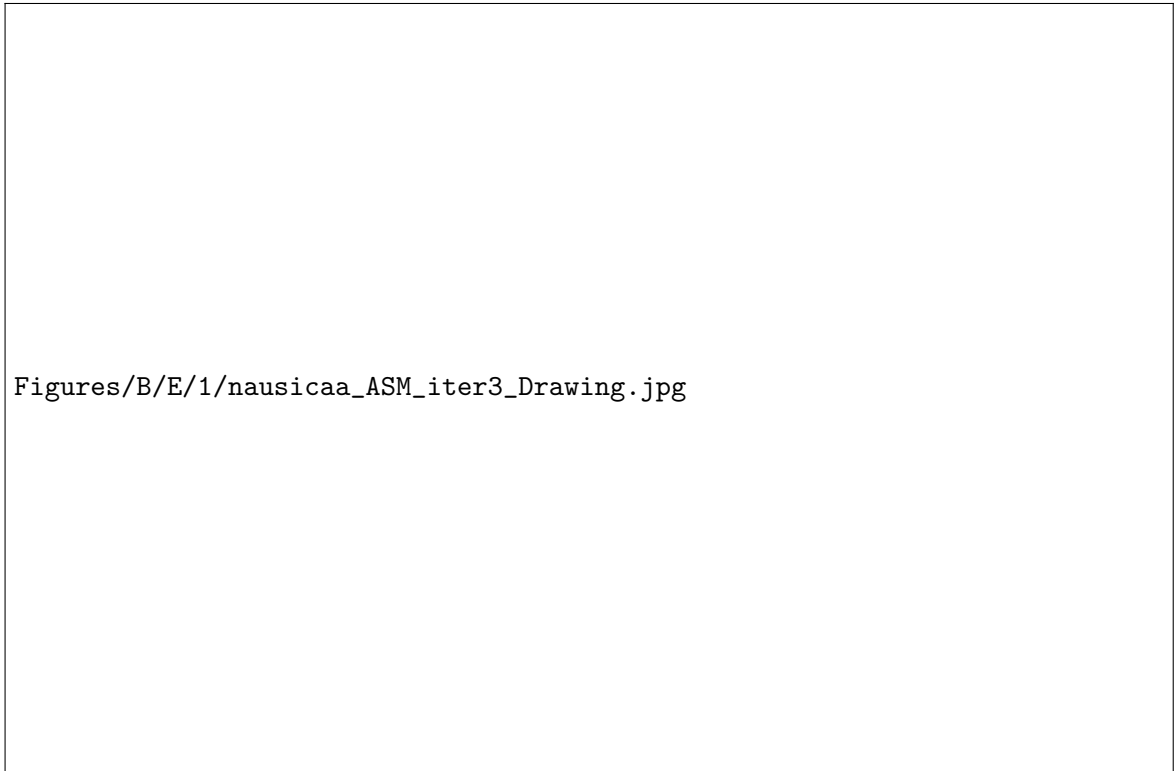
(a) Three-view geometry.



Figures/B/E/1/mass_budget_iter3.png

(b) Mass breakdown.

Fig. E.4. Third iteration optimised glider configuration.



Figures/B/E/1/nausicaa_ASM_iter3_Drawing.jpg

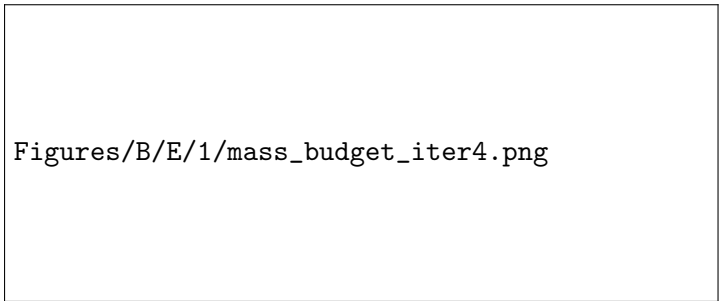
Fig. E.5. Detail engineering drawing of the third iteration glider assembly, corresponding to a half-span of 0.381 m and chord length of 0.182 m. Key 3D-printed mounting and skid components are shown; avionics, wiring, and tape reinforcements are omitted for clarity.

E.2.4 Fourth Iteration



Figures/B/E/1/three_view_iter4.png

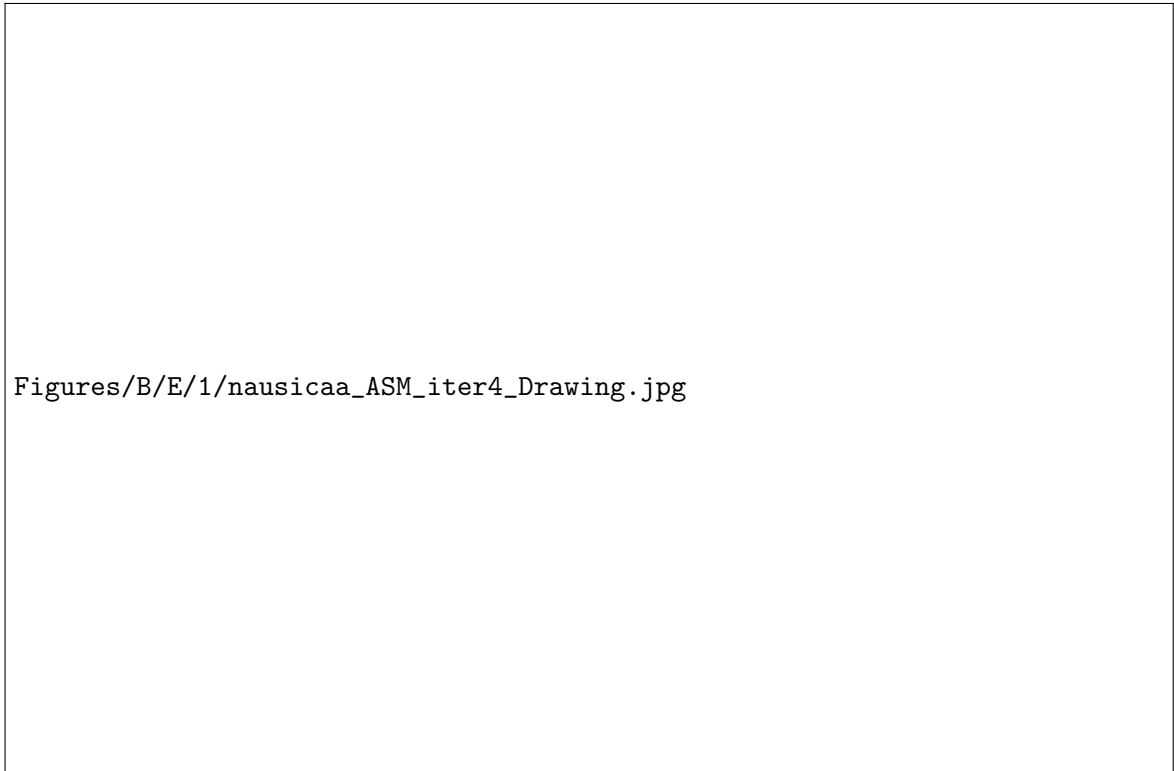
(a) Three-view geometry.



Figures/B/E/1/mass_budget_iter4.png

(b) Mass breakdown.

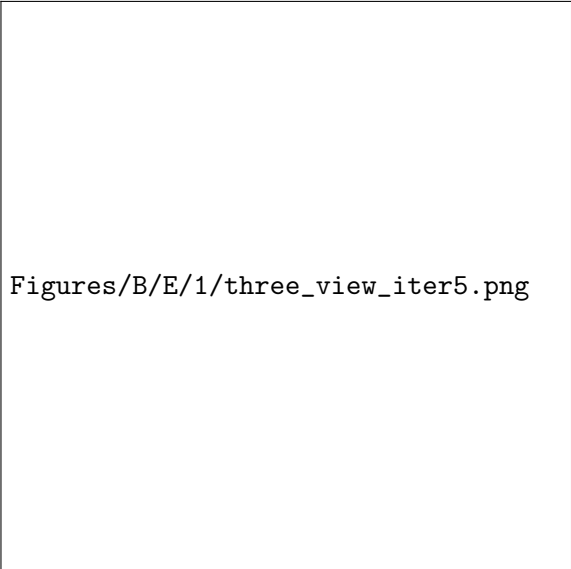
Fig. E.6. Fourth iteration optimised glider configuration.



Figures/B/E/1/nausicaa_ASM_iter4_Drawing.jpg

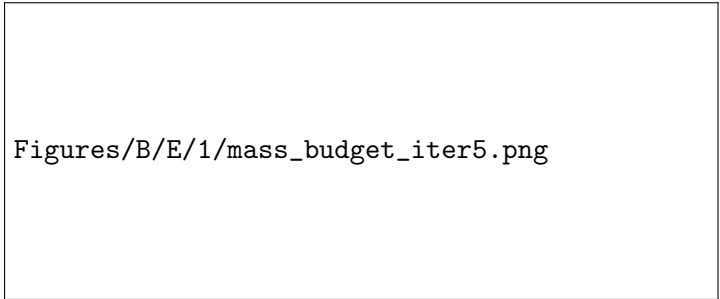
Fig. E.7. Detail engineering drawing of the fourth iteration glider assembly, corresponding to a half-span of 0.381 m and chord length of 0.160 m. Key 3D-printed mounting and skid components are shown; avionics, wiring, and tape reinforcements are omitted for clarity.

E.2.5 Fifth Iteration



Figures/B/E/1/three_view_iter5.png

(a) Three-view geometry.



Figures/B/E/1/mass_budget_iter5.png

(b) Mass breakdown.

Fig. E.8. Fifth iteration optimised glider configuration.

E.3 Glider Optimisation Model and Selected Design

E.3.1 Model Assumptions

Table E.7 Mission, arena, and aerodynamic settings used by the optimiser.

Quantity	Unit	Value
Nominal design speed	m s^{-1}	6.50
Turn diagnostic speed	m s^{-1}	4.00
Commanded bank angle	deg	50.0
Bank entry time	s	0.20
Arena length	m	8.40
Arena width	m	4.80
Arena height	m	3.50
Wall clearance allowance	m	0.20
Airfoil section	–	NACA0002
Wing dihedral	deg	10.0

Table E.8 Planform and control surface assumptions used by the optimiser.

Quantity	Unit	Value
Horizontal tail aspect ratio	-	4.0
Vertical tail aspect ratio	-	2.0
Wing section count	count	7
Tail section count	count	5
Aileron span stations	semispan	[0.30, 0.85]
Aileron chord ratio	-	0.30
Elevator chord ratio	-	0.30
Rudder chord ratio	-	0.35
Nose station	m	-0.10
Fuselage radius	m	0.002

Table E.9 Material and reinforcement assumptions used in the mass and stiffness model.

Quantity	Unit	Value
Wing foam density	kg m^{-3}	33.0
Tail foam density	kg m^{-3}	40.0
Wing foam thickness	mm	6.00
Tail foam thickness	mm	3.00
Wing secant modulus	Pa	1.5×10^6
Tail secant modulus	Pa	1.0×10^6
Wing spar outer diameter	mm	4.00
Wing spar inner diameter	mm	3.00
Spar modulus	Pa	25×10^9
Tape width	mm	10.0
Tape thickness	mm	0.13
Tape areal density	kg m^{-2}	0.175

Table E.10 Onboard mass and actuation assumptions used in the design model.

Quantity	Unit	Value
Centre module mass	g	28.0
Tail module mass	g	4.0
Battery mass	g	5.0
Receiver mass	g	6.0
Regulator mass	g	1.0
Servo mass, each	g	4.0
Servo count	count	4
Control deflection limits	deg	± 30
Servo rate limit	deg s^{-1}	400
Servo torque rating	N m	0.009

E.3.2 Selected Design

Table E.11 Geometry and mass properties.

Quantity	Symbol	Unit	Value
Wing span	b	m	0.7616
Wing chord	c	m	0.1646
Wing area	S	m ²	0.1254
Wing aspect ratio	AR	-	4.626
Tail moment arm	l_t	m	0.4000
Horizontal tail span	b_H	m	0.3656
Vertical tail height	b_V	m	0.1188
Aircraft mass	m	g	117.5
Longitudinal CG	x_{CG}	m	0.1137
Lateral CG	y_{CG}	m	0
Vertical CG	z_{CG}	m	0.0158

Table E.13 Grouped mass budget.

Group	Share	Mass
Wing and tail surfaces	25.3%	29.68 g
Spar, tape, and slot	8.3%	9.70 g
Mounts and joiners	15.5%	18.22 g
Boom and tail hardware	8.7%	10.24 g
Avionics and servos	34.9%	40.99 g
Glue allowance	7.4%	8.71 g
Total	100%	117.5 g

Table E.15 Turn footprint and roll response.

Quantity	Symbol	Unit	Value
Turn lift coefficient	$C_{L,turn}$	-	0.355
Allowable turn radius	R_{allow}	m	1.825
Predicted turn radius	R_{ach}	m	2.129
Predicted turn footprint	d_{turn}	m	2.704
Required lateral acceleration	$a_{y,req}$	m s ⁻²	8.767
Predicted lateral acceleration	$a_{y,ach}$	m s ⁻²	7.515
Lateral acceleration margin	Δa_y	m s ⁻²	-1.252
Steady roll rate	p_{ss}	rad s ⁻¹	5.240
Roll time constant	τ_p	s	0.0229
Bank angle in 0.2 s	$\phi(t_{be})$	deg	25.60

Table E.12 Trim and performance.

Quantity	Symbol	Unit	Value
Trim angle of attack	α_{trim}	deg	3.808
Trim elevator deflection	δ_e	deg	-0.114
Sink speed	V_z	m s ⁻¹	0.6703
Glide ratio	L/D	-	9.697
Wing loading	W/S	N m ⁻²	9.195
Wing Reynolds number	Re_c	-	7.33×10^4
Static margin	SM	MAC	0.100
Horizontal tail volume	V_H	-	0.647
Vertical tail volume	V_V	-	0.030
Lift coefficients	C_L	-	0.355
Drag coefficients	C_D	-	0.0366

Table E.14 Objective decomposition.

Objective term	Value
Total objective, J	2.166
Sink speed term	0.670
Mass term	0.0552
Trim effort term	1.72×10^{-6}
Wing deflection term	1.311
Horizontal tail deflection term	0.117
Roll time constant term	0.0126

Table E.16 Structural and stability.

Quantity	Symbol	Unit	Value
Wing deflection ratio	$\delta_w/\delta_{w,max}$	-	0.272
Horizontal tail deflection ratio	$\delta_h/\delta_{h,max}$	-	0.140
Maximum servo torque utilisation	u_s	-	0.00481
Lift curve slope	C_{L_α}	rad ⁻¹	5.648
Pitch stability	C_{m_α}	rad ⁻¹	-0.565
Dihedral effect	C_{ℓ_β}	rad ⁻¹	-0.199
Directional stability	C_{n_β}	rad ⁻¹	0.0257
Roll damping	C_{ℓ_p}	-	-0.779
Pitch damping	C_{m_q}	-	-11.47
Aileron effectiveness	$C_{\ell_{\delta_a}}$	rad ⁻¹	-0.827

E.4 Additive Manufacturing and Print Settings

Table E.17 Printer, material, and geometric print settings used for 3D-printed components.

Setting	Option
Printer	Bambu Lab X1 Carbon
Nozzle diameter	0.4 mm
Filament	PLA-CF @BBL X1C PAHT-CF @BBL X1C
Process preset	0.20 mm Strength @BBL X1C
Layer height	0.20 mm
Initial layer height	0.20 mm
Default line width	0.42 mm
Initial layer line width	0.50 mm
Inner wall line width	0.45 mm
Sparse infill line width	0.45 mm
Elephant foot compensation	0.15 mm
X-Y hole and contour compensation	0 mm

Table E.19 Speed and acceleration settings used for 3D-printed components.

Setting	Option
Initial layer speed	50 mm s ⁻¹
Initial layer infill speed	105 mm s ⁻¹
Outer wall speed	60 mm s ⁻¹
Inner wall speed	300 mm s ⁻¹
Sparse infill speed	270 mm s ⁻¹
Internal solid infill speed	250 mm s ⁻¹
Top surface speed	200 mm s ⁻¹
Bridge speed	50 mm s ⁻¹
Gap infill speed	250 mm s ⁻¹
Travel speed	500 mm s ⁻¹
Normal printing acceleration	10000 mm s ⁻²
Travel acceleration	10000 mm s ⁻²
Initial layer travel acceleration	6000 mm s ⁻²
Slow down for overhangs	On

Table E.18 Shell and infill settings used for 3D-printed components.

Setting	Option
Wall generator	Classic
Wall loops	4
Wall order	Internal perimeter first
Top surface pattern	Monotonic
Top shell layers	5
Top shell thickness	1 mm
Bottom surface pattern	Monotonic
Bottom shell layers	3
Internal solid infill pattern	Rectilinear
Sparse infill density	15%
Sparse infill pattern	Gyroid
Infill/wall overlap	15%
Infill direction	45°
Bridge direction	0°

Table E.20 Support and bed-adhesion settings used for 3D-printed components.

Setting	Option
Enable support	On
Support type	Tree
Threshold angle	30°
On build plate only	On
Remove small overhangs	On
Raft layers	0
Top and bottom Z distance	0.20 mm
Top interface layers	2
Top interface spacing	0.5 mm
Support/object X-Y distance	0.35 mm
Support/object first layer gap	0.20 mm
Brim type	Outer brim
Brim width	5 mm
Brim-object gap	0.24 mm

F Control Architecture (TBD)

F.1 Glider Controller Model (TBD)

G Simulation Evidence (TBD)

G.1 R5 (TBD)

G.2 R7 (TBD)

G.3 R8 (TBD)

G.4 R10 (TBD)

G.5 R11 (TBD)

H Experiment Results (TBD)